Choosing the Dictionary and Penalty for IV-LASSO*

Yukun Ma† Manu Navjeevan‡ Bohdan Salahub§

October 22, 2026

Abstract

Estimating an instrumental variables (IV) first stage with the least absolute shrinkage and selection operator (LASSO) requires choosing a dictionary and penalty. Existing rules target first-stage objectives, although these choices affect structural estimation. We derive a feasible approximate mean squared error (AMSE) criterion for full-sample IV-LASSO with one endogenous regressor and homoskedastic Gaussian errors. For a prespecified list of dictionary–penalty pairs, the criterion trades off approximation of the optimal instrument against a complexity cost that depends on the first two moments of the LASSO effective dimension and on the degree of endogeneity. The selected pair attains the smallest candidate-specific AMSE component up to vanishing relative error. In simulations calibrated to Paravisini et al. (2015), the criterion improves substantially on the plug-in rule at low endogeneity and on cross-validation at high endogeneity, with nearly identical results under non-Gaussian errors. A supplementary appendix formally extends analysis to a class of non-Gaussian errors.

KEYWORDS: Instrumental Variables, LASSO, Model Selection, Second-Order Asymptotics

JEL CODES: C26, C36, C52

1 Introduction

When a large set of technical instruments is available, LASSO makes estimation of an IV first stage feasible but leaves applied researchers to choose how heavily to penalize the fit and which dictionary of technical instruments to use.[1] Under standard sparse-first-stage conditions, Belloni et al. (2012) show that all sufficiently accurate LASSO fits

*An `R` package implementing IV-LASSO with the first-stage implementation guided by this paper can be found on GitHub. This package, along with the proofs of several technical lemmas in this paper, were developed with the help of Claude Fable. We take responsibility for all errors.

†Yukun Ma: yma69@ur.rochester.edu. University of Rochester

‡Manu Navjeevan: mnavjeevan@tamu.edu. Texas A&M University

§Bohdan Salahub: bsalahub@g.ucla.edu. University of California, Los Angeles

[1]See Paravisini et al. (2015), Gilchrist and Sands (2016), Reimers and Waldfogel (2021), Bald et al. (2022), Derenoncourt (2022), Agan et al. (2023), and Beraja et al. (2023) for prominent examples of IV-LASSO and IV-post-LASSO methods being applied in empirical economics.

yield IV estimators with the same first-order limiting distribution. Standard asymptotic theory therefore does not rank admissible implementations, even though their finite-sample performance can differ. We study how to choose among these implementations when the objective is accurate estimation of the structural parameter.

Figure 1 illustrates this issue in the eminent-domain application of Belloni et al. (2012). Across three nested first-stage dictionaries, the IV estimate changes materially along the penalty path. Moreover, different methods of selecting the first-stage penalty parameter can change whether the conventional pointwise 95% confidence interval for the resulting structural parameter estimate contains zero. For example, for the Case–Shiller home price index outcome, using cross-validation to select the penalty yields a significant result on the Restricted dictionary while using Stein's Unbiased Risk Estimate (SURE) does not. Given this variation, how should an applied researcher choose among possible first-stage implementations?

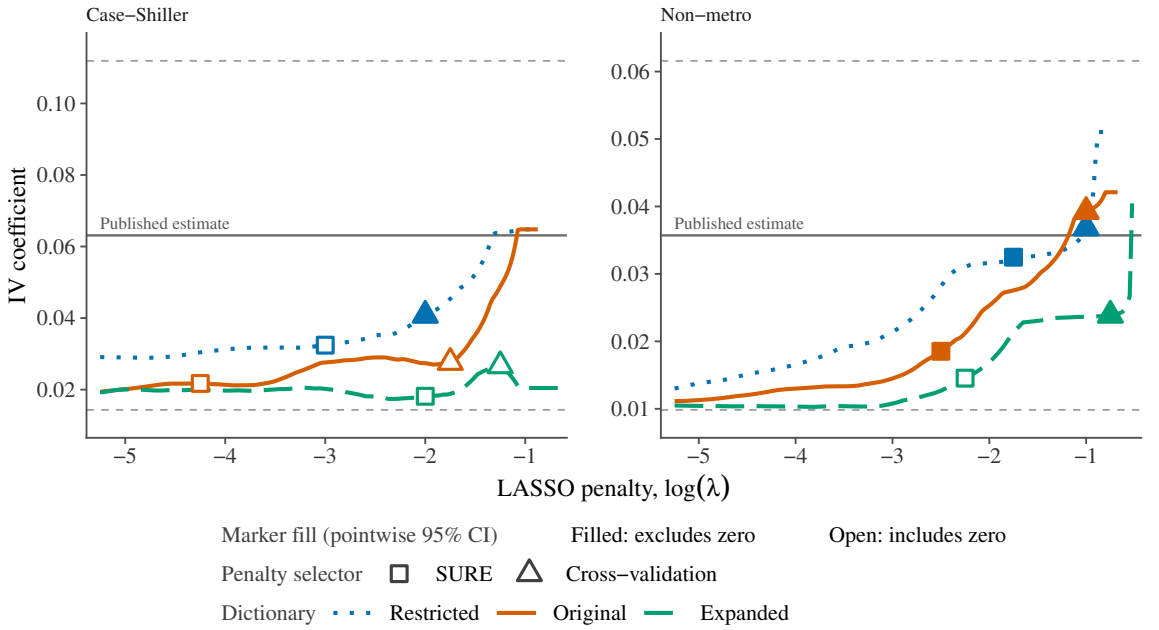


Figure 1: IV Estimates and Conventional Significance across First-Stage Implementations

Notes: Lines trace IV-LASSO estimates over LASSO penalties within three nested dictionaries. Original is the published dictionary; Restricted removes cross-products, and Expanded adds all nondegenerate second-order products. Filled markers denote pointwise 95% heteroskedasticity robust confidence intervals that exclude zero. The solid gray line and dashed bounds show the published IV-post-LASSO estimate and confidence interval. The panels report the two outcomes for which that interval excludes zero (183 Case–Shiller and 110 non-metro circuit-year observations).

Existing first-stage tuning rules are not designed to target accuracy of the resulting structural parameter estimates. The calibrated penalty rule of Belloni et al. (2012) and the

bootstrap-after-cross-validation method of Chetverikov and Sørensen (2025) are designed to control first-stage estimation error, while SURE and ordinary cross-validation target first-stage prediction (Tibshirani and Taylor, 2012; Chetverikov et al., 2021). A LASSO fit with a lighter penalty or richer dictionary may better approximate the true first stage, but it can also fit more of the first-stage noise. When the first-stage noise is correlated with the structural error, this in turn introduces a bias into the resulting structural parameter estiamte whose size grows with the degree of endogeneity. The optimal first-stage implementation therefore depends not only on goodness of first-stage fit and instrument strength, but also on the level of endogeneity.

This paper develops second-order asymptotic theory for the full-sample IV-LASSO estimator and uses it to provide a principled method for choosing the first-stage implementation. We focus on the leading case of a single endogenous regressor and work in a homoskedastic Gaussian model. As in Donald and Newey (2001), who build on the higher-order expansions of Nagar (1959) to choose among instrument sets, we take a finite, prespecified list of first-stage LASSO candidates, each consisting of a dictionary and a penalty, and rank them by the approximate mean squared error (AMSE) of the IV-LASSO estimators they produce. The candidate specific part of the AMSE can be decomposed into a first-stage approximation cost and a complexity cost, with complexity measured by a collinearity-robust analogue of the number of instruments selected by the LASSO. The size of this complexity cost rises with the squared covariance between the first-stage and structural errors, reflecting the discussion above.

This characterization leads to a feasible criterion that can be computed from each candidate's fitted first stage and its effective dimension. We show that, under our conditions, the pair that minimizes this criterion has a candidate-specific AMSE component as small as that of the best candidate in the list, up to vanishing relative error. This guarantee is stated relative to the second-order components that distinguish the candidates, rather than the first-order risk that all of them share. In a supplementary appendix, we extend these results to a class of non-Gaussian error distributions under some further regularity conditions.

To derive the AMSE of the IV-LASSO estimator, we need to account for both the shrinkage and the variable selection performed by the first-stage LASSO. Although the LASSO fitted values are continuous in the endogenous variable, they need not be differentiable at points where the selected set of instruments changes. This nonsmoothness complicates higher-order analysis, which typically relies on a smooth expansion of the first-stage estimator. Moreover, since our assumptions allow for coefficients near the selection threshold, we cannot avoid this difficulty by assuming that the selected model is eventually fixed.

We address this difficulty by using the geometric results of Tibshirani and Taylor (2012) and the second-order Stein identities of Bellec and Zhang (2021) to characterize moments involving the nonsmooth LASSO fit and structural errors. The fit is Lipschitz continuous and thus differentiable almost everywhere, allowing these calculations despite its nonsmoothness. The complexity cost depends on the average number of linearly independent instruments selected and how this number varies across samples. Estimating this cost requires fluctuations in this count to be small relative to its average. We obtain this stability under a partial beta-min condition on a growing group of coefficients large enough for their instruments to be selected with probability approaching one. This group must account for a nonvanishing share of the nonzero coefficients used to approximate the first stage. The condition also makes the AMSE approximation error small relative to the terms used to compare candidates.

Constructing a feasible criterion presents a further difficulty. Estimation error in the feasible criterion can asymptotically dominate the candidate-specific AMSE terms. We show that the leading part of this error is common to all candidates and therefore cancels when they are compared. After this common centering, the remaining estimation error is uniformly negligible relative to the candidate-specific AMSE components.

To evaluate the performance of the proposed criterion, we conduct a simulation study calibrated to the credit-supply and export data of Paravisini et al. (2015). We hold the original instruments fixed and vary the strength of identification and degree of endogeneity. At low levels of endogeneity, cross-validation and the proposed criterion perform comparably. As endogeneity rises, however, the criterion selects less complex first stages and achieves up to 50% lower mean squared error than cross-validation in designs with the highest levels of endogeneity. Compared to the cross-fit IV-LASSO estimator, the proposed criterion has substantially lower mean squared error when endogeneity is low and performs comparably when it is high and identification is strong. Interestingly, the relative performance of the criterion is nearly identical in the Gaussian designs and in the two non-Gaussian designs we consider.

Our results extend the literature on selecting and regularizing instruments according to the AMSE of the structural parameter estimator (Donald and Newey, 2001; Kuersteiner and Okui, 2010; Okui, 2011; Carrasco, 2012) to high-dimensional LASSO first stages, where each dictionary–penalty candidate itself selects instruments using the observed endogenous variable. Unlike a least-squares first stage on a prespecified instrument basis, fixing the dictionary and penalty does not fix the selected model. Our analysis accounts for this selection uncertainty as well as the shrinkage introduced by the penalty.

Our analysis also relates to machine-learning methods in econometrics (Belloni et al., 2014; Chernozhukov et al., 2018; Wager and Athey, 2018; Farrell et al., 2021) and higher-order asymptotics (Sargan, 1976; Rothenberg, 1984; Newey and Smith, 2004; Hahn et al., 2004). Velez (2026) develops second-order approximations for double/debiased machine learning estimators to study the choice of the number of cross-fitting folds. His higher-order analysis assumes a stochastic linear expansion of the first-stage estimators. We instead account directly for LASSO shrinkage and variable selection to choose the dictionary and penalty.

Section 2 presents the model and candidate first stages. Section 3 states the main results, and Section 4 reports the simulations. Proofs are in the main appendix, while a supplementary appendix extends our analysis to a class of non-Gaussian errors.

Notation. For $m \in \mathbb{N}$, let $[m] := \{1, \dots, m\}$. For an array $\{a_i\}_{i\in[n]}$, define

$$\mathbb{E}_n[a_i] := \frac{1}{n}\sum_{i\in[n]} a_i, \qquad \overline{\mathbb{E}}[a_i] := \mathbb{E}_n[\mathbb{E}[a_i]].$$

2 Model and Setup

To simplify exposition, we present the analysis assuming that any exogenous regressors have already been partialled out of both the outcome and the endogenous regressor. Our analysis assumes there is only a single endogenous variable. [2] The structural equation and first stage are given

$$\begin{aligned} y_i &= \beta x_i + \varepsilon_i \\ x_i &= \Pi_i + v_i \end{aligned} \tag{2.1}$$

We assume the researcher observes n independent observations (y_i, x_i, z_i), where $y_i, x_i \in \mathbb{R}$ and $z_i \in \mathbb{R}^p$. In (2.1), we let $\Pi_i = \mathbb{E}[x_i \mid z_i]$ denote the conditional expectation of the endogenous variable given the instruments and assume that the structural error is conditional-mean independent of the instruments, so that $\mathbb{E}[\varepsilon_i \mid z_i] = \mathbb{E}[v_i \mid z_i] = 0$. Throughout the theoretical analysis, we condition on the realized instruments and write $x, \Pi, \varepsilon, v \in \mathbb{R}^n$ for the corresponding stacked vectors.[3] All expectations and probability statements below are therefore understood as conditional on this realization, i.e., $\Pi_i = \mathbb{E}[x_i]$.

Assumption 1 (Conditional Gaussian Sampling)**.** *Assume*

(i) The errors $\{(\varepsilon_i, v_i)\}_{i=1}^n$ are independent of the instruments and independently and

[2] Appendices providing a formal treatment of the model with controls as well as an extension to multiple endogenous variables are available on request.

[3] I.e, $x = (x_1, \dots, x_n) \in \mathbb{R}^n$

identically distributed according to

$$\begin{pmatrix} \varepsilon_i \\ v_i \end{pmatrix} \sim N(0, \Omega), \qquad \Omega := \begin{pmatrix} \sigma_\varepsilon^2 & \sigma_{\varepsilon v} \\ \sigma_{\varepsilon v} & \sigma_v^2 \end{pmatrix},$$

where Ω is a fixed nonsingular matrix.

(ii) For constants $0 < \underline{H} < \overline{H} < \infty$,

$$\underline{H} \leq H := \mathbb{E}_n[\Pi_i^2] \leq \overline{H}, \qquad \sigma_{\varepsilon v} \neq 0. \tag{2.2}$$

Assumption 1(i) encodes the assumption of conditionally homoskedastic Gaussian errors, while Assumption 1(ii) encodes strong identification and assumes that there is an endogeneity problem to begin with.

Under conditional homoskedasticity, the conditional mean Π_i is the optimal instrument, since among IV estimators based on a nonrandom instrument, using Π_i itself minimizes the variance of the leading term.[4] Since Π_i is unknown, feasible estimation replaces it with a first-stage estimate. Any IV estimator $\widehat{\beta}$ based on a sufficiently well-estimated first stage (Section 3 states sufficient conditions) admits the same expansion,

$$\sqrt{n}(\widehat{\beta} - \beta) = \frac{\sqrt{n}\,\mathbb{E}_n[\Pi_i \varepsilon_i]}{H} + o_p(1), \tag{2.3}$$

whose leading term does not involve the first-stage estimate and, conditional on the instruments, is distributed exactly as $N(0, \sigma_\varepsilon^2/H)$ at every sample size. Comparing first-stage estimators thus requires analyzing terms hidden in the remainder.

2.1 Candidate First Stages and the IV Estimator

We select from a prespecified list of J candidates, where J is fixed. Each $c \in [J]$ consists of a dictionary of technical instruments and a nonrandom penalty $\lambda_c > 0$. The dictionary maps z_i to $z_{c,i} := (z_{c,i1}, \ldots, z_{c,ip_c})' \in \mathbb{R}^{p_c}$, whose entries are fixed functions of z_i. The dictionary dimension p_c may exceed n, and both the dictionary and the penalty may depend on n. Write

$$Z_c := (z_{c,1}, \ldots, z_{c,n})' \in \mathbb{R}^{n \times p_c}, \qquad Z_{c,B} := (Z_c)_{\cdot,B} \in \mathbb{R}^{n \times |B|} \quad \text{for } B \subseteq [p_c].$$

For candidate c and an arbitrary response vector $w := (w_1, \ldots, w_n) \in \mathbb{R}^n$, define the LASSO

[4] For nonrandom $g \in \mathbb{R}^n$ with $\mathbb{E}_n[g_i^2]$ bounded and $\mathbb{E}_n[g_i \Pi_i]$ bounded away from zero, the leading variance is $\sigma_\varepsilon^2 \mathbb{E}_n[g_i^2]/\mathbb{E}_n[g_i \Pi_i]^2 \geq \sigma_\varepsilon^2/H$, with equality when g is proportional to Π. The bound σ_ε^2/H is the fixed-design counterpart of the usual optimal-IV efficiency bound; see Chamberlain (1987) and Newey (1990).

coefficient estimate and fitted vector (Tibshirani, 1996) by

$$\widehat{\pi}_c(w) \in \underset{\pi \in \mathbb{R}^{p_c}}{\arg\min} \left\{ \frac{1}{2}\mathbb{E}_n[(w_i - z_{c,i}'\pi)^2] + \lambda_c \|\pi\|_1 \right\}, \qquad \mu_c(w) := Z_c \widehat{\pi}_c(w). \tag{2.4}$$

When the columns of Z_c are linearly dependent, the coefficient minimizer need not be unique, but the fitted vector $\mu_c(w)$ remains unique (Tibshirani and Taylor, 2012; Tibshirani, 2013). Applying this procedure to the endogenous variable x, define $\widehat{\pi}_c := \widehat{\pi}_c(x)$, $\widehat{\Pi}_c := \mu_c(x)$, and $\widehat{\Pi}_{c,i} := z_{c,i}'\widehat{\pi}_c$. The corresponding second-stage IV estimator is

$$\widehat{\beta}_c := \begin{cases} \dfrac{\mathbb{E}_n[\widehat{\Pi}_{c,i} y_i]}{\mathbb{E}_n[\widehat{\Pi}_{c,i} x_i]}, & \mathbb{E}_n[\widehat{\Pi}_{c,i}^2] > 0, \\ 0, & \mathbb{E}_n[\widehat{\Pi}_{c,i}^2] = 0. \end{cases} \tag{2.5}$$

A natural criterion for comparing the candidate estimators $\widehat{\beta}_c$ is their scaled mean squared error (MSE), $\mathbb{E}[n(\widehat{\beta}_c - \beta)^2]$. The exact MSE of an IV estimator, however, need not exist in finite samples. To see this, consider the infeasible estimator that uses the optimal instrument Π_i in place of $\widehat{\Pi}_{c,i}$ in (2.5). This estimator satisfies

$$\widehat{\beta} - \beta = \frac{\mathbb{E}_n[\Pi_i \varepsilon_i]}{\mathbb{E}_n[\Pi_i x_i]}.$$

Under Assumption 1, the numerator and denominator on the right-hand side are jointly Gaussian. Because Ω is nonsingular, the numerator is not an exact linear function of the denominator, and their ratio does not have a finite second moment. Thus, even the infeasible estimator based on the optimal instrument has no exact MSE. More generally, even when the exact MSE exists, it may be neither tractable nor consistently estimable.

Given this restriction, we instead follow the approach of Donald and Newey (2001). Under suitable conditions, we show that the scaled squared error of each estimator $\widehat{\beta}_c$ admits the decomposition

$$n(\widehat{\beta}_c - \beta)^2 = C_0 + Q_c + r_c. \tag{2.6}$$

Here, C_0 is common to all candidates and captures the component of squared error corresponding to the leading term in (2.3). By contrast, Q_c captures the second-order contribution of candidate c's first-stage procedure, while r_c is asymptotically negligible in the sense that $\max_{c \in [J]} |r_c| / \mathbb{E}[Q_c] \to_p 0$.

We define $\mathbb{E}[C_0 + Q_c]$ as the AMSE of $\sqrt{n}(\widehat{\beta}_c - \beta)$. For the infeasible estimator based on the optimal instrument, the expectation of the candidate-specific component is zero. Hence, $\bar{Q}_c := \mathbb{E}[Q_c]$ measures the excess AMSE of candidate c relative to that benchmark.

Because C_0 is common across candidates, ranking candidates by AMSE is equivalent to ranking them by $\bar{Q}_c$. In Section 3, we derive a closed-form asymptotic approximation to $\bar{Q}_c$, construct a consistent feasible counterpart, and use it to select among the first-order-equivalent candidate estimators.

2.2 The Effective Dimension

For the LASSO, the number of selected instruments is not always well defined. With collinear dictionary columns, LASSO coefficient solutions can have different supports even though their fitted values are identical. We therefore measure first-stage complexity using the rank of the equicorrelation set.

Let $u_c(w) := w - \mu_c(w)$ denote the LASSO residual and define the equicorrelation set by

$$E_c(w) := \left\{ j \in [p_c] : \left| \mathbb{E}_n[z_{c,ij} u_{c,i}(w)] \right| = \lambda_c \right\}, \tag{2.7}$$

This is the set of dictionary columns for which the LASSO Karush–Kuhn–Tucker (KKT) inequality binds. Because the residual is unique, $E_c(w)$ does not depend on which coefficient solution is used, and it contains the support of every solution. We define the effective dimension of the first-stage fit as

$$d_c := \operatorname{rank}(Z_{c,E_c(x)}). \tag{2.8}$$

The quantity d_c is observable and invariant to the choice of coefficient solution. When the LASSO solution is unique, d_c agrees with the number of selected instruments almost surely under our sampling assumption.[5] As shown by Tibshirani and Taylor (2012), the degrees of freedom of the LASSO fit are given by $\mathbb{E}[d_c]$.

3 Main Results

After stating the conditions required for our analysis, we characterize the candidate-specific AMSE components (Theorem 1) and show that the selected candidate attains the smallest of them up to vanishing relative error (Theorem 2).

3.1 Conditions on the Candidate First Stages

Along with the Gaussian sampling and strong identification conditions in Assumption 1, we impose conditions for first-stage prediction consistency and a partial beta-min condition that is used to establish concentration of the LASSO effective dimension but does not require exact support recovery.

[5] A sufficient condition for uniqueness is that the dictionary columns are in general position, which holds with probability one when the dictionary matrix has a joint continuous distribution (Tibshirani, 2013).

We use the approximately sparse representation of Belloni et al. (2012), in which for each candidate dictionary Z_c there is a nonrandom coefficient vector $\pi_c^0 \in \mathbb{R}^{p_c}$ and a prediction error $\xi_{c,i}$ such that

$$\Pi_i = z_{c,i}'\pi_c^0 + \xi_{c,i}, \qquad T_c := \operatorname{supp}(\pi_c^0), \qquad s_c := |T_c|, \tag{3.1}$$

where $\operatorname{supp}(b) := \{j : b_j \neq 0\}$, $s_c \geq 1$, and $\xi_c = (\xi_{c,1}, \ldots, \xi_{c,n})'$. The rate restrictions on s_c and ξ_c are stated below.

For a vector δ and an index set T, write $\delta_T := (\delta_j)_{j\in T}$ and $\|\delta\|_0 := |\operatorname{supp}(\delta)|$. Fix a constant $c_\lambda > 1$, common to all candidates, and set $L_\lambda := (c_\lambda + 1)/(c_\lambda - 1)$. Following Belloni et al. (2012), define the restricted eigenvalue (Bickel et al., 2009; van de Geer and Bühlmann, 2009)

$$\kappa_c := \min_{\substack{\delta \neq 0 \\ \|\delta_{T_c^c}\|_1 \leq L_\lambda \|\delta_{T_c}\|_1}} \frac{\sqrt{s_c \mathbb{E}_n[(z_{c,i}'\delta)^2]}}{\|\delta_{T_c}\|_1}, \tag{3.2}$$

and the upper and lower m-sparse eigenvalues

$$\phi_{\max,c}(m) := \max_{\substack{\delta \neq 0 \\ \|\delta\|_0 \leq m}} \frac{\mathbb{E}_n[(z_{c,i}'\delta)^2]}{\|\delta\|_2^2}, \qquad \phi_{\min,c}(m) := \min_{\substack{\delta \neq 0 \\ 1 \leq \|\delta\|_0 \leq m}} \frac{\mathbb{E}_n[(z_{c,i}'\delta)^2]}{\|\delta\|_2^2}. \tag{3.3}$$

These conditions control the design on sparse sets and the cone in (3.2), so they do not require the full dictionary to have linearly independent columns and do not rule out cases where $p_c \gg n$.

Assumption 2 (First-Stage Convergence)**.** *There are constants $0 \leq C_a < \infty$, $\alpha > 0$, $0 < \kappa_0 \leq 1$, $0 < \underline{\phi} \leq 1 \leq \overline{\phi} < \infty$, and $M > 4L_\lambda^2 \overline{\phi}/\kappa_0^2$, common to all candidates, such that the following hold for each $c \in [J]$:*

(i) The dictionary columns are normalized such that $\mathbb{E}_n[z_{c,ij}^2] = 1$ for each $j \in [p_c]$.

(ii) The approximation error satisfies

$$\mathbb{E}_n[\xi_{c,i}^2]^{1/2} \leq C_a \sqrt{\frac{s_c}{n}}.$$

(iii) The penalty satisfies

$$\lambda_c \geq c_\lambda \sqrt{2}\sigma_v \sqrt{\frac{\log(2p_c) + \alpha \log\log(p_c \vee n)}{n}}.$$

(iv) The design satisfies

$$\kappa_c \geq \kappa_0, \qquad \phi_{\max,c}(\lceil Ms_c \rceil \wedge p_c) \leq \overline{\phi}, \qquad \phi_{\min,c}(\lceil (M+1)s_c \rceil \wedge p_c) \geq \underline{\phi}.$$

(v) The sparsity index, s_c, and penalty λ_c satisfy

$$\frac{s_c}{\sqrt{n}} \to 0, \qquad \sqrt{s_c}\lambda_c \to 0.$$

Versions of each condition in Assumption 2 appear in Belloni et al. (2012), who also use them to establish convergence of first-stage LASSO estimates. Together with parts (i)–(iv) of Assumption 2, $\sqrt{s_c}\lambda_c \to 0$ yields first-stage prediction consistency, while $s_c/\sqrt{n} \to 0$ controls terms that arise in the higher-order expansion. Combined with Assumption 1, these conditions ensure that (2.3) holds for every candidate. Beyond guaranteeing suitable first-stage convergence of the candidate first-stage estimators, these conditions do little to restrict the candidate list, so dictionaries and penalties may vary across candidates (Bickel et al., 2009; van de Geer and Bühlmann, 2009; Belloni et al., 2012; Chetverikov and Sørensen, 2025).

The next condition ensures that the effective dimension grows and that its fluctuations become small relative to its mean. This allows us to estimate the complexity cost from the observed effective dimension, as described in Section 3.3, and to show that the terms omitted from the AMSE approximation are negligible relative to the candidate-specific component.

Assumption 3 (Partial Beta-Min)**.** *For each candidate, let*

$$\underline{b}_{n,c} := \frac{4}{\sqrt{\underline{\phi}}} \left\{ \frac{(1 + c_\lambda^{-1})\lambda_c \sqrt{s_c}}{\kappa_0} + C_a \sqrt{\frac{s_c}{n}} \right\},$$

where κ_0 and $\underline{\phi}$ are as in Assumption 2(iv). Define the set of strong coordinates by $A_c := \{j \in T_c : |\pi^0_{c,j}| \geq \underline{b}_{n,c}\}$ with $k_c := |A_c|$. There is a fixed constant $C_s \geq 1$ such that, uniformly over $c \in [J]$,

$$s_c \leq C_s k_c, \qquad k_c \to \infty, \qquad \frac{\log(p_c/k_c)}{k_c} \to 0. \tag{3.4}$$

The strong coordinates A_c are selected with probability approaching one. Unlike full beta-min conditions for support recovery (Zhao and Yu, 2006; Wainwright, 2009), Assumption 3 does not impose a lower bound on the coefficients in $T_c \setminus A_c$, nor does it require recovery of the entire support. The requirement that $k_c \to \infty$ parallels the growing-basis condition in Donald and Newey (2001) and, together with the bound $s_c \leq C_s k_c$, underlies the relative

concentration of d_c in Proposition 1. Since $\underline{b}_{n,c} \to 0$, the coefficients in the strong core may shrink with n. Our results require a growing effective dimension and do not cover a fixed effective dimension, whose analysis would need to consider additional terms from estimating the IV denominator. This extension appears possible, but is not pursued here.

The partial beta-min condition may be particularly well motivated under strong identification. If all first-stage coefficients are close to zero, the instruments may not induce enough exogenous variation in the endogenous variable to consistently estimate the structural parameter.

3.2 The AMSE Criterion

This subsection formally establishes the decomposition in (2.6) and presents the infeasible criterion, a closed-form asymptotic approximation to the expectation of the candidate-specific term. Exact expressions for the terms in (2.6) are given in the appendix but are not needed for the characterization below.

Two population quantities enter the infeasible criterion. The first is a measure of first-stage strength,

$$h_c := \overline{\mathbb{E}}[\widehat{\Pi}_{c,i}\Pi_i], \tag{3.5}$$

the population cross-moment between the fitted values and the optimal instrument. This is the population counterpart of the denominator of the IV estimator in (2.5). Under the conditions of Section 3.1, h_c converges to the second moment of the optimal instrument, H, and so is bounded away from zero in large samples.

The second quantity measures how well the fitted instrument approximates the optimal instrument. Because the IV estimator is unchanged when its instrument is multiplied by a nonzero constant, a fitted instrument proportional to the optimal instrument should register no error. We therefore measure the first-stage approximation error against the best rescaling of the optimal instrument,

$$\mathcal{A}_c := \min_{a\in\mathbb{R}} \overline{\mathbb{E}}\Big[(\widehat{\Pi}_{c,i} - a\Pi_i)^2\Big] = \overline{\mathbb{E}}[\widehat{\Pi}^2_{c,i}] - \frac{h_c^2}{H}. \tag{3.6}$$

The minimized value $\mathcal{A}_c$ is the mean squared residual from a population regression of the fitted instrument on Π. The ratio $\mathcal{A}_c/h_c^2$ appearing in the infeasible criterion below is invariant to rescaling of the fitted instrument.

The closed-form approximation to the expectation of the candidate-specific term combines these two quantities with certain error moments and moments of the effective dimension of

the LASSO fit defined in Section 2.2,

$$S_c := \frac{\sigma_\varepsilon^2 \mathcal{A}_c}{h_c^2} + \frac{\sigma_{\varepsilon v}^2 \, \mathbb{E}[d_c^2 + d_c]}{n h_c^2}. \tag{3.7}$$

We refer to S_c as the infeasible criterion. Each of its components is a population quantity, so the infeasible criterion cannot be computed directly from the data. Section 3.3 constructs a consistent feasible counterpart used to select a candidate.

Theorem 1 (Infeasible AMSE Criterion)**.** *Suppose Assumptions 1, 2, and 3 hold, with $J < \infty$ fixed. Then there exist random variables C_0, $\{Q_c\}_{c\in[J]}$, and $\{r_c\}_{c\in[J]}$ such that*

(i) For every candidate,

$$n(\widehat{\beta}_c - \beta)^2 = C_0 + Q_c + r_c, \tag{3.8}$$

where C_0 does not depend on the candidate, $\mathbb{E}[C_0] = \sigma_\varepsilon^2/H + O(n^{-1})$, and the remainder satisfies $\max_{c\in[J]} |r_c|/\mathbb{E}[Q_c] \longrightarrow_p 0$.

(ii) Uniformly over candidates, $\mathbb{E}[Q_c] > 0$ and $\mathbb{E}[Q_c] = S_c \left(1 + o(1)\right)$.

Theorem 1(i) establishes the validity of the decomposition in (2.6), while Theorem 1(ii) shows that S_c approximates the candidate-specific AMSE component $\bar{Q}_c = \mathbb{E}[Q_c]$ with vanishing relative error. Together, these results justify ranking candidates by S_c: minimizing S_c is asymptotically equivalent to minimizing $\bar{Q}_c$, and hence AMSE, over the candidate list. The remainder in part (i) is controlled in probability, so this result does not establish a ranking by exact finite-sample MSE.

The first term measures how well the candidate LASSO approximates the optimal instrument, while the second depends on the first two moments of the effective dimension. This term originates from the second moment of $\varepsilon'\widehat{\Pi}_c$, which we compute using results of Tibshirani and Taylor (2012) and Bellec and Zhang (2021).

When the first stage is least squares on a prespecified set of K linearly independent instruments ($\lambda = 0$), the effective dimension always equals K, and S_c reduces, up to lower-order terms, to the criterion of Donald and Newey (2001). With LASSO, however, the effective dimension depends on the realization of the sample, so the complexity cost depends on both its mean and its variance.[6]

A richer dictionary or a lighter penalty may reduce $\mathcal{A}_c$ while increasing the effective dimension. To examine this tradeoff, write $\rho = \mathrm{Corr}(\varepsilon_i, v_i)$, so that $\sigma_{\varepsilon v} = \rho\sigma_\varepsilon\sigma_v$, and rewrite (3.7)

[6]Under the partial beta-min condition in Assumption 3, the effective dimension grows, so that $\mathbb{E}[d_c^2]$ asymptotically dominates $\mathbb{E}[d_c]$. However, both terms appear in the expression for $\mathbb{E}[(\varepsilon'\widehat{\Pi}_c)^2]$, and we keep them since they may both be relevant in finite samples.

as

$$S_c = \frac{\sigma_\varepsilon^2}{h_c^2}\left\{\mathcal{A}_c + \rho^2\sigma_v^2\,\frac{\mathbb{E}[d_c^2 + d_c]}{n}\right\}. \tag{3.9}$$

In this form the second term is weighted by $\rho^2\sigma_v^2$, which is proportional to the squared covariance between the structural and first-stage errors. When the regressor is close to exogenous, the complexity term receives little weight.[7] When the regressor is highly endogenous, the same complexity produces a larger bias. This is the LASSO analogue of the many-instrument bias examined in Bekker (1994) and Donald and Newey (2001).

Cross-validation targets first-stage prediction error rather than the structural AMSE criterion in (3.7). Its rankings may therefore differ even when the complexity term receives little weight, because $\mathcal{A}_c$ allows rescaling and the criterion also depends on h_c. As endogeneity strengthens, the AMSE criterion prefers models with lower complexity, while cross-validation does not take this into consideration. The plug-in penalty rules of Belloni et al. (2012) and Chetverikov and Sørensen (2025) are designed to control first-stage noise as in Assumption 2(iii) and likewise do not account for endogeneity.

3.3 Feasible Candidate Selection

Since S_c depends on population quantities and is thus infeasible, we construct an observable criterion that estimates it up to a candidate-independent centering term and therefore preserves its ranking asymptotically. The only additional inputs not obtained from the candidate fits are estimates of the structural error moments σ_ε^2 and $\sigma_{\varepsilon v}$. We obtain these estimates using residuals from a pilot estimate $\check{\beta}$ of β.

For each candidate, the observed IV denominator

$$\widehat{h}_c := \mathbb{E}_n[\widehat{\Pi}_{c,i}x_i] \tag{3.10}$$

will serve as an estimator for first-stage cross-moment h_c. The KKT conditions for the LASSO optimization problem in (2.4) yield $\widehat{h}_c = \mathbb{E}_n[\widehat{\Pi}_{c,i}^2] + \lambda_c\|\widehat{\pi}_c\|_1 \geq 0$, so $\widehat{h}_c = 0$ if and only if $\widehat{\pi}_c = 0$. The pilot $\check{\beta}$ can be any estimator with $\check{\beta} - \beta = O_p(n^{-1/2})$, such as the IV-LASSO estimate $\widehat{\beta}_{\check{c}}$ from a possibly data-dependent candidate $\check{c} \in [J]$. Given the pilot, let

$$\widehat{\sigma}_\varepsilon^2 := \mathbb{E}_n[(y_i - \check{\beta}x_i)^2] \qquad \text{and} \qquad \widehat{\sigma}_{\varepsilon v} := \mathbb{E}_n[(y_i - \check{\beta}x_i)x_i]. \tag{3.11}$$

We put together these estimates to construct the feasible criterion

$$\widehat{S}_c := \frac{\widehat{\sigma}_\varepsilon^2\mathbb{E}_n[\widehat{\Pi}_{c,i}^2] + n^{-1}\widehat{\sigma}_{\varepsilon v}^2(d_c^2 + d_c)}{\widehat{h}_c^2}, \tag{3.12}$$

[7] Exact exogeneity is excluded by Assumption 1(ii), and the case $\sigma_{\varepsilon v} = 0$ requires a different expansion.

with the convention $\widehat{S}_c := +\infty$ when $\widehat{h}_c = 0$. The selected candidate minimizes the feasible criterion,

$$\widehat{c} := \min\left(\arg\min_{c\in[J]} \widehat{S}_c\right),$$

with ties broken by the smallest index.

The feasible criterion $\widehat{S}_c$ replaces the moment $\mathbb{E}[d_c^2+d_c]$ in the infeasible S_c with the realized value $d_c^2 + d_c$. This replacement is justified by the following proposition.

Proposition 1 (Ratio Consistency of the Effective Dimension)**.** *Suppose Assumptions 1, 2, and 3 hold, with $J < \infty$ fixed. Uniformly over candidates,*

$$\frac{d_c}{\mathbb{E}[d_c]} \longrightarrow_p 1, \tag{3.13}$$

$$\frac{d_c^2 + d_c}{\mathbb{E}[d_c^2 + d_c]} \longrightarrow_p 1. \tag{3.14}$$

The growth conditions on the strong core in Assumption 3, together with a variance bound derived from the second-order Stein identities of Bellec and Zhang (2021), yield relative concentration of d_c and hence of $d_c^2 + d_c$. This does not require the support of the LASSO solution to contain the support T_c in (3.1).

The numerator in $\widehat{S}_c$ contains the fitted second moment $\mathbb{E}_n[\widehat{\Pi}_{c,i}^2]$ rather than a direct estimate of the approximation error $\mathcal{A}_c$. Applying the decomposition in (3.6) to sample moments, with $\widehat{\mathcal{A}}_c := \min_{a\in\mathbb{R}} \mathbb{E}_n[(\widehat{\Pi}_{c,i} - a\Pi_i)^2]$, splits the first term of $\widehat{S}_c$ into two parts,

$$\frac{\widehat{\sigma}_\varepsilon^2 \mathbb{E}_n[\widehat{\Pi}_{c,i}^2]}{\widehat{h}_c^2} = \frac{\widehat{\sigma}_\varepsilon^2 \widehat{\mathcal{A}}_c}{\widehat{h}_c^2} + \frac{\widehat{\sigma}_\varepsilon^2}{H}\left(\frac{\mathbb{E}_n[\widehat{\Pi}_{c,i}\Pi_i]}{\widehat{h}_c}\right)^2.$$

The first part estimates $\sigma_\varepsilon^2 \mathcal{A}_c / h_c^2$, the first term of S_c. Because each candidate first stage is consistent, we can expand the second part around its value for the optimal instrument itself, $\widehat{\sigma}_\varepsilon^2 H / \mathbb{E}_n[\Pi_i x_i]^2$, where $\mathbb{E}_n[\Pi_i x_i] = H + \mathbb{E}_n[\Pi_i v_i]$ and $\mathbb{E}_n[\Pi_i v_i]$ is of order $n^{-1/2}$. This gives

$$B_n := \widehat{\sigma}_\varepsilon^2 \left(\frac{1}{H} - \frac{2\mathbb{E}_n[\Pi_i v_i]}{H^2}\right) \tag{3.15}$$

up to terms that are negligible relative to $\bar{Q}_c$. Theorem 2 below shows that, uniformly over candidates,

$$\widehat{S}_c = B_n + \bar{Q}_c(1 + o_p(1)).$$

Although B_n may dominate the candidate-specific AMSE components $\bar{Q}_c$, it is common to all candidates and therefore does not affect their feasible-criterion ranking. Minimizing $\widehat{S}_c$ is

then asymptotically equivalent to minimizing $\bar{Q}_c$. This does not require B_n to be observed.

Theorem 2 (Feasible AMSE Selection)**.** *Suppose Assumptions 1, 2, and 3 hold, with $J < \infty$ fixed, and that the pilot satisfies $\check{\beta} - \beta = O_p(n^{-1/2})$.*

(i) Up to the common centering B_n, the feasible criterion consistently estimates the infeasible criterion,

$$\max_{c\in[J]} \frac{|\widehat{S}_c - B_n - S_c|}{S_c} \longrightarrow_p 0, \tag{3.16}$$

and thus the expectation of the candidate-specific term,

$$\max_{c\in[J]} \frac{|\widehat{S}_c - B_n - \bar{Q}_c|}{\bar{Q}_c} \longrightarrow_p 0. \tag{3.17}$$

(ii) The selected candidate $\widehat{c}$ satisfies

$$\frac{\bar{Q}_{\widehat{c}}}{\min_{c\in[J]} \bar{Q}_c} \longrightarrow_p 1. \tag{3.18}$$

The supplementary appendix establishes the same AMSE criterion and feasible selection result for a class of non-Gaussian errors. That extension requires the structural error to have a linear conditional mean and constant conditional variance given the first-stage error, along with additional regularity conditions. Under more general departures from Gaussianity or homoskedasticity, AMSE optimality is not established here, although the results in Section 4.2 suggest that the AMSE in more general models may still be well approximated by the AMSE in the Gaussian model. If, under alternative conditions, all candidates continue to admit the common first-order expansion in (2.3) uniformly over the fixed candidate list, then the selected estimator retains that expansion and first-order inference remains valid.

4 Simulation Study

We compare the proposed criterion to first-stage selection via cross-validation and the plug-in penalty of Belloni et al. (2012). We also consider a cross-fitted IV estimator, with the dictionary and penalty chosen within each training sample (Chernozhukov et al., 2018). The simulations are calibrated to the data of Paravisini et al. (2015), who study the effect of credit supply on exports. We first consider Gaussian errors in Section 4.1 and then repeat the exercise in Section 4.2 with common-scale Laplace and smooth-transformation errors.

4.1 Calibrated Gaussian Designs

We hold the observed exposure instruments fixed and generate the outcome and endogenous variable from the IV model described in (2.1) with $\beta = 1$. The true first stage Π is proportional to the fitted values from a least-squares regression of credit growth on 17 instruments selected by a cross-validated LASSO from the original instruments and their 45 pairwise products. We also consider a dense alternative to this main first stage, proportional to the least-squares fitted values on all 55 of these instruments. We rescale these fitted values to vary identification strength, targeting expected first-stage F statistics of 2.7, 4.4, 7.7, 14.5, or 28.0 at the original sample size of $n = 14{,}208$ firm–product–destination export flows. Holding the instruments fixed and only rescaling the two first stages lets us examine how each rule's performance changes with identification strength and endogeneity in an empirically plausible setting.

The errors are generated homoskedastic Gaussian, with $\varepsilon_i = \rho v_i + \sqrt{1-\rho^2}\, e_i$ for $v_i, e_i \overset{\text{iid}}{\sim} N(0,1)$. The parameter ρ controls the degree of endogeneity. We consider a value calibrated to the data, $\widehat{\rho} \approx -0.22$, along with its multiples $\rho = c\,\widehat{\rho}$ for $c = 0, 0.25, \ldots, 2.75$ to examine how the performance of each rule changes with the level of endogeneity.[8] Each of the 120 designs is simulated 1,000 times. We report the average of $n(\widehat{\beta} - \beta)^2$, which we refer to as either the risk or the mean squared error.

Each candidate first stage implementation uses one of two nested dictionaries: either the ten original exposure instruments (Original) or a richer dictionary of 114 instruments including their pairwise products and the authors' alternative exposure measures (Expanded). For each dictionary, we consider 37 multiples of the plug-in penalty of Belloni et al. (2012) ("BCCH"), giving 74 candidates in total. This plug-in penalty level uses the known standard deviation of the first-stage error in place of an estimate. The proposed criterion selects among these candidate dictionary-penalty pairs by minimizing $\widehat{S}_c$ in (3.12), with estimated error moments coming from a pilot estimator of β. Cross-validation chooses from the same candidates using ten-fold first-stage prediction error, keeping observations from the same firm in the same fold. The cross-fitting estimator ("Cross-fit") makes this choice separately for each of ten folds formed by firm, using only the other nine folds, and the resulting predictions for each held-out fold are used as instruments in a single IV estimate. BCCH uses its plug-in penalty on the Original dictionary. When it selects no instruments, we use the pilot estimate instead.

Figure 2 shows how the rules respond to endogeneity. Consistent with the increasing complexity cost seen in (3.9), the criterion selects less complex first stages as endogeneity rises,

[8] Assumption 1(ii) rules out exact exogeneity, so the designs with $\rho = 0$ lie outside our theory, and we include them only as a limiting case.

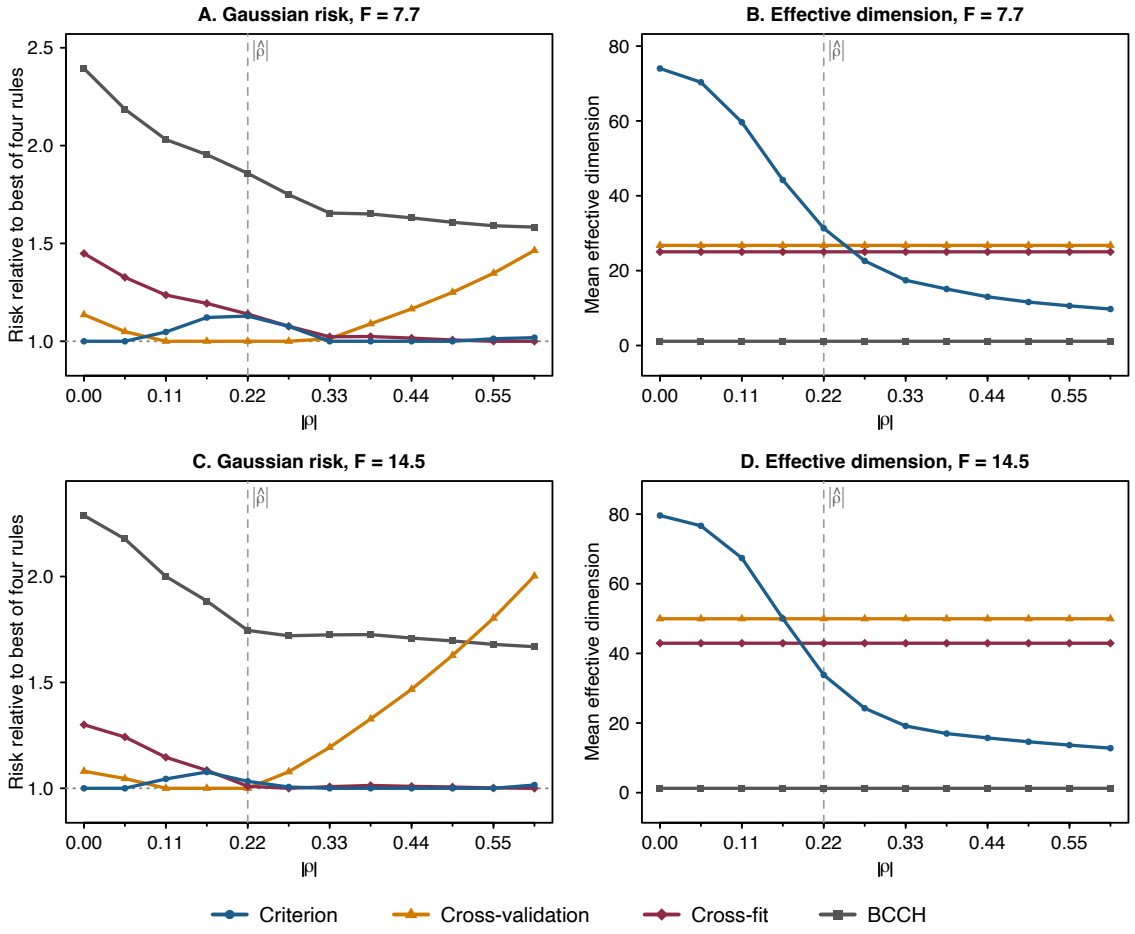


Figure 2: Relative risk and effective dimension under Gaussian errors.

Notes: Results use the main first stage, $n = 14{,}208$, and 1,000 replications. Panels A and C report risk relative to the lowest risk among the four rules; Panels B and D report mean effective dimension, averaged over folds for cross-fitting. Panels A–B use $F = 7.7$ and Panels C–D use $F = 14.5$. The horizontal axis is the absolute error correlation $|\rho|$, and the dashed line marks the calibrated correlation $|\widehat{\rho}| = 0.218$.

while the other three rules do not account for the level of endogeneity. At both $F = 7.7$ and $F = 14.5$, the criterion's mean effective dimension falls by more than six-fold. Its risk is more than 50% below that of BCCH at the lowest levels of endogeneity and between 30% and 50% below that of cross-validation at the highest level, although cross-validation performs better in some designs with intermediate levels of endogeneity.

Table 1 shows that the criterion remains close to the best feasible rule across the design grid, but does not dominate every design. With the main first stage, cross-validation has lower average relative risk than the criterion at $F = 2.7$ and near the calibrated correlation. As identification strengthens, the criterion and cross-fitting perform more similarly. The criterion's largest relative risk is about 20% above that of the best rule. The large BCCH

Table 1: Relative risk, Gaussian errors.

Design	Summary Measure	Feasible rules: Criterion	Feasible rules: Cross-validation	Feasible rules: Cross-fit	Feasible rules: BCCH	Infeasible: Known Moments
Panel A: Main signal, by first-stage strength						
$F = 2.7$	Average	1.072	1.034	1.589	14.529	0.857
	Maximum	1.199	1.143	2.045	21.117	0.976
$F = 4.4$	Average	1.051	1.074	1.240	1.813	0.924
	Maximum	1.173	1.291	1.615	2.459	0.987
$F = 7.7$	Average	1.034	1.126	1.124	1.824	0.962
	Maximum	1.129	1.464	1.449	2.394	1.023
$F = 14.5$	Average	1.015	1.302	1.068	1.835	0.975
	Maximum	1.077	2.002	1.300	2.288	1.009
$F = 28.0$	Average	1.036	1.301	1.038	1.908	1.014
	Maximum	1.074	2.009	1.163	2.170	1.044
Panel B: Main signal, by degree of endogeneity						
$0 \leq \lvert\rho\rvert \leq 0.109$	Average	1.009	1.048	1.422	3.490	0.865
	Maximum	1.047	1.143	2.045	9.883	1.000
$0.164 \leq \lvert\rho\rvert \leq 0.273$	Average	1.105	1.014	1.229	4.105	0.967
	Maximum	1.199	1.107	1.725	14.566	1.007
$0.327 \leq \lvert\rho\rvert < 0.6$	Average	1.026	1.304	1.098	4.966	0.976
	Maximum	1.183	2.009	1.534	21.117	1.044
Panel C: Dense signal, all designs						
All F and ρ	Average	1.035	1.145	1.374	4.339	0.926
	Maximum	1.175	2.017	2.474	23.204	1.002

Notes: Risk is the mean of $n(\widehat{\beta} - \beta)^2$ over 1,000 replications. All panels present risk relative to the best of the four feasible rules in each design. Panels A and B use the main first stage. Panel A averages or maximizes over the twelve values of ρ at each F. Panel B averages or maximizes over the five values of F and the values of $|\rho|$ in each range. Panel C averages or maximizes over all 60 designs with the dense first stage. Known Moments evaluates the feasible criterion at population error moments and is not one of the four feasible rules, so its relative risk can fall below one.

entries are driven by a few replications at $F = 2.7$ in which BCCH selects no instruments and the pilot estimate is used in its place.

Setting the error moments to their population values, $\sigma_\varepsilon^2 = 1$ and $\sigma_{\varepsilon v} = \rho$, lowers the criterion's average relative risk for the main first stage from 1.042 to 0.946. The Known Moments benchmark still uses estimated first-stage quantities to rank candidates and need not attain the lowest risk in every design. We do not include it among the four rules used

to normalize risk.

4.2 Robustness to Non-Gaussian Errors

We repeat the simulations of Section 4.1 under two departures from Gaussianity that differ in whether they preserve the constant conditional variance of the Gaussian design. The simulation details are otherwise left completely unchanged.

For the common-scale Laplace design, we draw mutually independent $W_i \sim \text{Exp}(1)$ and $a_i, b_i \sim N(0,1)$, independently across observations, and set

$$v_i = \sqrt{W_i}\, a_i, \qquad \varepsilon_i = \sqrt{W_i}\left(\rho a_i + \sqrt{1-\rho^2}\, b_i\right).$$

The scale is shared by the two errors within each observation, and both errors have variance-one Laplace marginals.

For the smooth-transformation design, we normalize the following transformation to unit variance. Its unnormalized form appears in Chetverikov et al. (2020, Figure 1):

$$Q(t) = \frac{t + t^3}{\nu(1+2t^2)}, \qquad \nu^2 = \mathbb{E}\left[\left(\frac{G+G^3}{1+2G^2}\right)^2\right], \quad G \sim N(0,1).$$

For independent standard Gaussian draws G_i, B_i, independent across observations, we set

$$v_i = Q(G_i), \qquad \varepsilon_i = \rho v_i + \sqrt{1-\rho^2}\, Q(B_i).$$

Both designs have mean-zero, unit-variance errors with correlation ρ. The smooth-transformation design also satisfies $\text{Var}(\varepsilon_i \mid v_i) = 1-\rho^2$, so it is covered by the non-Gaussian extension in the supplementary appendix. The common-scale Laplace design, in which this conditional variance depends on v_i, goes beyond the assumptions of that extension. As before, the designs with $\rho = 0$ serve as exogenous benchmarks.

Table 2 reports the criterion's relative risk under each error distribution, with Gaussian errors as the reference row. We use the same 60 designs for each first-stage signal and 1,000 replications per design. Average relative risks remain similar across error distributions, although the maximum for the dense signal reaches 1.221 under smooth-transformation errors. Under these errors, the criterion's unnormalized risk differs from its risk under Gaussian errors by only between -4.1% and 3.1%, and its mean effective dimension changes by at most 3.2%.

For the main signal at $F = 7.7$ under Laplace errors, the mean effective dimension of the first stage selected by the criterion still falls from 74.3 to 9.7 as $|\rho|$ increases. At

Table 2: Relative risk of the proposed criterion across error distributions.

	Main signal		Dense signal	
Error distribution	Average	Maximum	Average	Maximum
Gaussian	1.042	1.199	1.035	1.175
Common-scale Laplace	1.041	1.174	1.035	1.193
Smooth transformation	1.045	1.188	1.041	1.221

Notes: Each signal comprises 60 designs (five values of F and twelve values of ρ) with 1,000 replications per design. Relative risk divides the criterion's risk by the lowest risk among the four feasible rules in the same design and under the same error distribution. Known Moments is not part of this comparison. The Gaussian design is described in Section 4.1, and the two non-Gaussian constructions are defined in Section 4.2.

the highest level of endogeneity, the criterion's risk is about 28% below that of cross-validation but 5% above that of cross-fitting. Cross-validation retains an advantage in some designs with intermediate levels of endogeneity, and cross-fitting has lower average relative risk at $F = 28.0$. The largest BCCH risks under Laplace errors are again driven by a few replications in which BCCH selects no instruments. While AMSE optimality under the smooth-transformation design is established in the supplementary appendix, the results under Laplace errors suggest that the AMSE in more general models may be well approximated by the AMSE in the Gaussian model.

References

Agan, A., J. L. Doleac, and A. Harvey (2023). Misdemeanor prosecution. *The Quarterly Journal of Economics 138*(3), 1453–1505.

Bald, A., E. Chyn, J. Hastings, and M. Machelett (2022). The causal impact of removing children from abusive and neglectful homes. *Journal of Political Economy 130*(7), 1919–1962.

Bekker, P. A. (1994). Alternative approximations to the distributions of instrumental variable estimators. *Econometrica 62*(3), 657–681.

Bellec, P. C. and C.-H. Zhang (2021). Second-order Stein: SURE for SURE and other applications in high-dimensional inference. *The Annals of Statistics 49*(4), 1864–1903.

Belloni, A., D. Chen, V. Chernozhukov, and C. Hansen (2012). Sparse models and methods for optimal instruments with an application to eminent domain. *Econometrica 80*(6), 2369–2429.

Belloni, A., V. Chernozhukov, and C. Hansen (2014). Inference on treatment effects after

selection among high-dimensional controls. *The Review of Economic Studies 81*(2), 608–650.

Beraja, M., A. Kao, D. Y. Yang, and N. Yuchtman (2023). AI-tocracy. *The Quarterly Journal of Economics 138*(3), 1349–1402.

Bickel, P. J., Y. Ritov, and A. B. Tsybakov (2009). Simultaneous analysis of Lasso and Dantzig selector. *The Annals of Statistics 37*(4), 1705–1732.

Carrasco, M. (2012). A regularization approach to the many instruments problem. *Journal of Econometrics 170*(2), 383–398.

Chamberlain, G. (1987). Asymptotic efficiency in estimation with conditional moment restrictions. *Journal of Econometrics 34*(3), 305–334.

Chernozhukov, V., D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal 21*(1), C1–C68.

Chetverikov, D., Z. Liao, and V. Chernozhukov (2020). On cross-validated Lasso in high dimensions. arXiv:1605.02214v6, extended version.

Chetverikov, D., Z. Liao, and V. Chernozhukov (2021). On cross-validated Lasso in high dimensions. *The Annals of Statistics 49*(3), 1300–1317.

Chetverikov, D. and J. R.-V. Sørensen (2025). Selecting penalty parameters of high-dimensional M-estimators using bootstrapping after cross validation. *Journal of Political Economy 133*(10), 3208–3248.

Derenoncourt, E. (2022). Can you move to opportunity? evidence from the Great Migration. *American Economic Review 112*(2), 369–408.

Donald, S. G. and W. K. Newey (2001). Choosing the number of instruments. *Econometrica 69*(5), 1161–1191.

Farrell, M. H., T. Liang, and S. Misra (2021). Deep neural networks for estimation and inference. *Econometrica 89*(1), 181–213.

Gilchrist, D. S. and E. G. Sands (2016). Something to talk about: Social spillovers in movie consumption. *Journal of Political Economy 124*(5), 1339–1382.

Hahn, J., J. Hausman, and G. Kuersteiner (2004). Estimation with weak instruments: Accuracy of higher-order bias and MSE approximations. *The Econometrics Journal 7*(1), 272–306.

Isserlis, L. (1918). On a formula for the product-moment coefficient of any order of a normal frequency distribution in any number of variables. *Biometrika 12*(1–2), 134–139.

Kuersteiner, G. and R. Okui (2010). Constructing optimal instruments by first-stage prediction averaging. *Econometrica 78*(2), 697–718.

Laurent, B. and P. Massart (2000). Adaptive estimation of a quadratic functional by model selection. *The Annals of Statistics 28*(5), 1302–1338.

Nagar, A. L. (1959). The bias and moment matrix of the general k-class estimators of the parameters in simultaneous equations. *Econometrica 27*(4), 575–595.

Newey, W. K. (1990). Efficient instrumental variables estimation of nonlinear models. *Econometrica 58*(4), 809–837.

Newey, W. K. and R. J. Smith (2004). Higher order properties of GMM and generalized empirical likelihood estimators. *Econometrica 72*(1), 219–255.

Okui, R. (2011). Instrumental variable estimation in the presence of many moment conditions. *Journal of Econometrics 165*(1), 70–86.

Paravisini, D., V. Rappoport, P. Schnabl, and D. Wolfenzon (2015). Dissecting the effect of credit supply on trade: Evidence from matched credit-export data. *The Review of Economic Studies 82*(1), 333–359.

Reimers, I. and J. Waldfogel (2021). Digitization and pre-purchase information: The causal and welfare impacts of reviews and crowd ratings. *American Economic Review 111*(6), 1944–1971.

Rothenberg, T. J. (1984). Approximating the distributions of econometric estimators and test statistics. In Z. Griliches and M. D. Intriligator (Eds.), *Handbook of Econometrics*, Volume 2, Chapter 15, pp. 881–935. Elsevier.

Sargan, J. D. (1976). Econometric estimators and the Edgeworth approximation. *Econometrica 44*(3), 421–448.

Stein, C. M. (1981). Estimation of the mean of a multivariate normal distribution. *The Annals of Statistics 9*(6), 1135–1151.

Tibshirani, R. (1996). Regression shrinkage and selection via the Lasso. *Journal of the Royal Statistical Society: Series B 58*(1), 267–288.

Tibshirani, R. J. (2013). The Lasso problem and uniqueness. *Electronic Journal of Statistics 7*, 1456–1490.

Tibshirani, R. J. and J. Taylor (2012). Degrees of freedom in Lasso problems. *The Annals of Statistics 40*(2), 1198–1232.

van de Geer, S. A. and P. Bühlmann (2009). On the conditions used to prove oracle results for the Lasso. *Electronic Journal of Statistics 3*, 1360–1392.

Velez, A. (2026). Debiased machine learning with many cross-fitting folds. Working paper, Cornell University. August 2, 2026.

Wager, S. and S. Athey (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association 113*(523), 1228–1242.

Wainwright, M. J. (2009). Sharp thresholds for high-dimensional and noisy sparsity recovery using ℓ_1-constrained quadratic programming (Lasso). *IEEE Transactions on Information Theory 55*(5), 2183–2202.

Zhao, P. and B. Yu (2006). On model selection consistency of Lasso. *Journal of Machine Learning Research 7*, 2541–2563.

A Notation and Preliminary Reductions

Here we collect the notation used throughout the appendix. We write C for a positive constant that does not depend on n or on the candidate and whose value may change from line to line. Since J is fixed, a rate or limit established for each candidate holds uniformly over $c \in [J]$, and we use this without further comment.

LASSO notation. Lemma D.1, below, recalls certain facts about the LASSO fit from Tibshirani and Taylor (2012). In particular, it notes that the LASSO fitted-value map $\mu_c : \mathbb{R}^n \to \mathbb{R}^n$ is one-Lipschitz, and hence Lebesgue-almost-everywhere differentiable by Rademacher's theorem. At a differentiability point $w \in \mathbb{R}^n$, write

$$\dot{\mu}_c(w) := D\mu_c(w), \qquad \operatorname{div} \mu_c(w) := \operatorname{tr}\{\dot{\mu}_c(w)\}.$$

Since $\Pi + v$ has a Lebesgue density under Assumption 1(i), the fit is almost surely differentiable at the observed $x = \Pi + v$, and we write $P_c := \dot{\mu}_c(\Pi + v)$ for its derivative there. Then Lemma D.1 and (D.5) give, almost surely,

$$P_c = P_c' = P_c^2, \qquad \operatorname{tr}(P_c) = d_c, \qquad P_c \widehat{\Pi}_c = \widehat{\Pi}_c. \tag{A.1}$$

That is, P_c is the orthogonal projection onto the column space of $Z_{c,E_c(x)}$, the active columns indexed by the equicorrelation set in (2.7), so that both P_c and $I - P_c$ are contractions.

With d_c defined in (2.8), we will save notation later in the proofs and write

$$\bar{d}_c := \mathbb{E}[d_c], \qquad R_{d,c} := \mathbb{E}[d_c^2 + d_c] = \bar{d}_c^2 + \mathrm{Var}(d_c) + \bar{d}_c. \tag{A.2}$$

Population projection residual. The minimum in (3.6) is attained at $a = h_c/H$. We write

$$M_c := \widehat{\Pi}_c - \frac{h_c}{H}\Pi \tag{A.3}$$

for the residual from the population projection of $\widehat{\Pi}_c$ on Π, so that $\mathbb{E}\|M_c\|_2^2 = n\mathcal{A}_c$. With $b = (h_c/H)\Pi$ in Lemma E.2, the residual M of that lemma is M_c, and the identity (E.7) becomes

$$\mathbb{E}[(v'M_c - \sigma_v^2 d_c)^2] = \sigma_v^2 n\mathcal{A}_c + \sigma_v^4 \bar{d}_c. \tag{A.4}$$

Gaussian error decomposition. Under Assumption 1(i) set $e := \varepsilon - \frac{\sigma_{\varepsilon v}}{\sigma_v^2} v$. Then e is Gaussian with mean zero and $\mathrm{Cov}(e_i, v_i) = 0$, hence independent of v:

$$\varepsilon = \frac{\sigma_{\varepsilon v}}{\sigma_v^2} v + e, \qquad e \perp v, \qquad \mathrm{Var}(e_i) = \sigma_\varepsilon^2 - \frac{\sigma_{\varepsilon v}^2}{\sigma_v^2}. \tag{A.5}$$

Positive denominator event. Lemma F.2 gives a fixed constant $\underline{h} > 0$ with $\min_{c\in[J]} h_c \geq \underline{h}$ for all sufficiently large n. With $\widehat{h}_c$ the sample denominator in (3.10), define

$$\mathcal{H}_n := \left\{ \min_{c\in[J]} \widehat{h}_c \geq \frac{\underline{h}}{2} \right\}. \tag{A.6}$$

Since analyzing the properties of IV-LASSO is difficult when the denominator is close to zero, our proofs will often condition on the event $\mathcal{H}_n$ to establish stochastic bounds. Because $\mathbb{P}(\mathcal{H}_n) \to 1$ by Lemma B.1, these bounds also hold without restricting to $\mathcal{H}_n$, while all expectations below are unconditional.

B Proof of Theorem 1

We take n large enough that $\min_c h_c \geq \underline{h}$, as in (F.11), and set the values of C_0, Q_c, and r_c for the remaining finitely many n at the end of the proof. The three supporting lemmas are given in Appendix B.1, after the proof.

Proof. Begin by defining the numerator, scaled by the population denominator, and the denominator correction,

$$\Gamma_c := \frac{\sqrt{n}\,\mathbb{E}_n[\varepsilon_i \widehat{\Pi}_{c,i}]}{h_c}, \qquad \Delta_c := \frac{\widehat{h}_c - h_c}{h_c}. \tag{B.1}$$

On $\mathcal{H}_n$, Lemma B.1 shows that $\widehat{\beta}_c$ is the ratio in (2.5), and after substituting $y_i = \beta x_i + \varepsilon_i$ we can write

$$\sqrt{n}(\widehat{\beta}_c - \beta) = \frac{\sqrt{n}\,\mathbb{E}_n[\varepsilon_i \widehat{\Pi}_{c,i}]}{\widehat{h}_c} = \frac{\Gamma_c}{1+\Delta_c} = \Gamma_c(1-\Delta_c) + \frac{\Gamma_c \Delta_c^2}{1+\Delta_c}. \tag{B.2}$$

Squaring $\Gamma_c(1-\Delta_c)$ directly is difficult, since it requires moments such as $\mathbb{E}[\Gamma_c^2 \Delta_c]$, in which both factors depend on the first-stage fit. Instead, we replace Γ_c in the correction by the leading term in (2.3),

$$L_n := \frac{\sqrt{n}\,\mathbb{E}_n[\varepsilon_i \Pi_i]}{H}, \tag{B.3}$$

which does not depend on the candidate and is linear in the structural errors, so that its conditional second moment given v is a quadratic in $\Pi' v$ with nonrandom coefficients. After squaring, the correction then enters only through $\mathbb{E}[\Delta_c L_n^2]$ and $\mathbb{E}[\Delta_c^2 L_n^2]$, together with a cross term that Cauchy–Schwarz bounds using $\mathbb{E}[(\Gamma_c - L_n)^2]$. With $q := \mathbb{E}_n[\Pi_i v_i]/H$, we then define

$$\begin{aligned} \Lambda_c &:= \Gamma_c - \Delta_c L_n, \qquad & C_0 &:= (1-q)^2 L_n^2, \\ Q_c &:= \Lambda_c^2 - C_0, \qquad & r_c &:= n(\widehat{\beta}_c - \beta)^2 - \Lambda_c^2. \end{aligned} \tag{B.4}$$

These definitions make (3.8) an identity, and C_0 does not depend on the candidate. When the optimal instrument Π is used in place of $\widehat{\Pi}_c$, h_c becomes H and $\widehat{h}_c$ becomes $\mathbb{E}_n[\Pi_i x_i] = H(1+q)$, so that $\Gamma_c = L_n$, $\Delta_c = q$, $\Lambda_c = (1-q)L_n$, and $Q_c = 0$. We show in four steps that $\mathbb{E}[Q_c]$ is approximated by S_c and that r_c is negligible.

Step 1: $\sqrt{n}(\widehat{\beta}_c - \beta) = \Lambda_c + o_p(S_c)$. Subtracting Λ_c from (B.2), on $\mathcal{H}_n$ we have

$$\sqrt{n}(\widehat{\beta}_c - \beta) - \Lambda_c = -\Delta_c(\Gamma_c - L_n) + \frac{\Gamma_c \Delta_c^2}{1+\Delta_c}.$$

By Lemma B.1, $\Delta_c = O_p(n^{-1/2})$, while $1 + \Delta_c = \widehat{h}_c / h_c$ is bounded away from zero on $\mathcal{H}_n$ because $\widehat{h}_c \geq \underline{h}/2$ there and h_c is bounded above by (F.11). By Lemma B.2 and Markov's inequality, $\Gamma_c - L_n = O_p(\sqrt{S_c})$, and since L_n is centered Gaussian with $\mathbb{E}[L_n^2] = \sigma_\varepsilon^2 / H$ and $\max_c S_c \to 0$ by (F.14), we also have $\Gamma_c = O_p(1)$. Therefore

$$\sqrt{n}(\widehat{\beta}_c - \beta) - \Lambda_c = O_p\left(\sqrt{\frac{S_c}{n}} + \frac{1}{n}\right) = S_c \cdot O_p\left(\frac{1}{\sqrt{nS_c}} + \frac{1}{nS_c}\right),$$

which is $o_p(S_c)$ because $\min_c nS_c \to \infty$ by (F.14). Since $\mathbb{P}(\mathcal{H}_n) \to 1$, this holds without restricting to $\mathcal{H}_n$. The same bounds show that

$$\max_{c \in [J]} |\Lambda_c| = O_p(1), \qquad \max_{c \in [J]} |\sqrt{n}(\widehat{\beta}_c - \beta)| = O_p(1). \tag{B.5}$$

Step 2: $\mathbb{E}[\Gamma_c^2] = \sigma_\varepsilon^2/H + S_c$. The second-moment identity (E.8) in Lemma E.3, which follows from the Stein identities in Lemma E.1, gives

$$\mathbb{E}[\Gamma_c^2] = \frac{\mathbb{E}[(\varepsilon'\widehat{\Pi}_c)^2]}{nh_c^2} = \frac{\sigma_\varepsilon^2 \mathbb{E}\|\widehat{\Pi}_c\|_2^2}{nh_c^2} + \frac{\sigma_{\varepsilon v}^2 \mathbb{E}[d_c^2 + d_c]}{nh_c^2}.$$

By (3.6), $\mathbb{E}\|\widehat{\Pi}_c\|_2^2/n = \mathcal{A}_c + h_c^2/H$, and substituting this into the first term yields

$$\mathbb{E}[\Gamma_c^2] = \frac{\sigma_\varepsilon^2}{H} + \underbrace{\frac{\sigma_\varepsilon^2 \mathcal{A}_c}{h_c^2} + \frac{\sigma_{\varepsilon v}^2 \mathbb{E}[d_c^2 + d_c]}{nh_c^2}}_{S_c}.$$

Step 3: $\mathbb{E}[Q_c] = S_c\{1+o(1)\}$. Expanding $\Lambda_c^2 = (\Gamma_c - \Delta_c L_n)^2$ and writing $\Gamma_c = L_n + (\Gamma_c - L_n)$ in the cross term,

$$\mathbb{E}[\Lambda_c^2] - \mathbb{E}[\Gamma_c^2] = -2\mathbb{E}[\Delta_c L_n^2] - 2\mathbb{E}[\Delta_c L_n(\Gamma_c - L_n)] + \mathbb{E}[\Delta_c^2 L_n^2].$$

By Lemma B.3, $|\mathbb{E}[\Delta_c L_n^2]| \le C\{\sqrt{S_c/n} + 1/n\}$ and $\mathbb{E}[\Delta_c^2 L_n^2] \le C/n$. For the middle term, Cauchy–Schwarz and Lemmas B.2 and B.3 give

$$|\mathbb{E}[\Delta_c L_n(\Gamma_c - L_n)]| \le \{\mathbb{E}[\Delta_c^2 L_n^2]\,\mathbb{E}[(\Gamma_c - L_n)^2]\}^{1/2} \le C\sqrt{\frac{S_c}{n}}.$$

We next compute $\mathbb{E}[C_0]$. Under Assumption 1, $(\Pi'\varepsilon, \Pi' v)$ is centered jointly Gaussian with covariance matrix $\|\Pi\|_2^2 \Omega = nH\Omega$. Its odd moments vanish, and the product-moment formula of Isserlis (1918) states that $\mathbb{E}[X^2Y^2] = \mathbb{E}[X^2]\mathbb{E}[Y^2] + 2\{\mathbb{E}[XY]\}^2$ for a centered jointly Gaussian pair (X, Y). We therefore have

$$\mathbb{E}[L_n^2] = \frac{\sigma_\varepsilon^2}{H}, \qquad \mathbb{E}[qL_n^2] = 0, \qquad \mathbb{E}[q^2 L_n^2] = \frac{\sigma_\varepsilon^2 \sigma_v^2 + 2\sigma_{\varepsilon v}^2}{nH^2}.$$

Expanding $C_0 = (1 - 2q + q^2)L_n^2$, we obtain $\mathbb{E}[C_0] = \sigma_\varepsilon^2/H + O(n^{-1})$, which is the bound on $\mathbb{E}[C_0]$ in part (i). Combining these bounds with Step 2 and $\mathbb{E}[Q_c] = \mathbb{E}[\Lambda_c^2] - \mathbb{E}[C_0]$,

$$|\mathbb{E}[Q_c] - S_c| = \left|\mathbb{E}[\Lambda_c^2] - \mathbb{E}[\Gamma_c^2] + \frac{\sigma_\varepsilon^2}{H} - \mathbb{E}[C_0]\right| \le C\left\{\sqrt{\frac{S_c}{n}} + \frac{1}{n}\right\}.$$

Dividing by S_c and using $\min_c nS_c \to \infty$ once more, we have $\max_c |\mathbb{E}[Q_c]/S_c - 1| \to 0$, so that $\mathbb{E}[Q_c] > 0$ for all sufficiently large n and part (ii) follows.

Step 4: $\max_c |r_c|/\mathbb{E}[Q_c] \to_p 0$. Factoring the difference of squares in (B.4),

$$|r_c| = |\sqrt{n}(\widehat{\beta}_c - \beta) - \Lambda_c|\,|\sqrt{n}(\widehat{\beta}_c - \beta) + \Lambda_c|.$$

The first factor is $o_p(S_c)$ by Step 1, and the second is $O_p(1)$ by (B.5). Since $S_c \le 2\mathbb{E}[Q_c]$ for all sufficiently large n by Step 3, we have $\max_c |r_c|/\mathbb{E}[Q_c] = o_p(1)$, which proves part (i).

Finally, by (F.11) and Step 3 we can choose n_0 such that $\min_c h_c \ge \underline{h}$ and $\min_c \mathbb{E}[Q_c] > 0$ whenever $n \ge n_0$. For $n < n_0$, set $C_0 := 0$, $Q_c := 1$, and $r_c := n(\widehat{\beta}_c - \beta)^2 - 1$. These choices preserve the decomposition and the positivity of $\mathbb{E}[Q_c]$ for every n, and they do not affect any of the asymptotic statements. □

B.1 Supporting Lemmas

Lemma B.1 (The Sample Denominator)**.** *(i) Under Assumption 1, for each candidate,*

$$\mathbb{E}[\widehat{h}_c] = h_c + \frac{\sigma_v^2 \bar{d}_c}{n}, \qquad \mathrm{Var}(\widehat{h}_c) \le \frac{4\sigma_v^2(H + \sigma_v^2)}{n}. \tag{B.6}$$

(ii) If Assumptions 2 and 3 also hold, with $J < \infty$ fixed and Δ_c as in (B.1), then $\mathbb{E}[(\widehat{h}_c - h_c)^2] \le C/n$ for every candidate and

$$\max_{c\in[J]} |\widehat{h}_c - h_c| = O_p(n^{-1/2}), \qquad \max_{c\in[J]} |\Delta_c| = O_p(n^{-1/2}). \tag{B.7}$$

(iii) Under the hypotheses of (ii), $\mathbb{P}(\mathcal{H}_n) \to 1$, and on $\mathcal{H}_n$ every candidate satisfies $\mathbb{E}_n[\widehat{\Pi}_{c,i}^2] > 0$ and $\widehat{\beta}_c = \mathbb{E}_n[\widehat{\Pi}_{c,i} y_i]/\widehat{h}_c$.

Proof. Since $n\widehat{h}_c = x'\widehat{\Pi}_c = \Pi'\widehat{\Pi}_c + v'\widehat{\Pi}_c$, we have $\mathbb{E}[\widehat{h}_c] = h_c + \sigma_v^2 \bar{d}_c/n$ by the definition of h_c and the first identity in (E.9). For the variance, write $x'\widehat{\Pi}_c = x'\mu_c(x)$ with $x = \Pi + v$. The product rule and $P_c' = P_c$ show that its gradient with respect to v is $\widehat{\Pi}_c + P_c x$, whose norm is at most $2\|x\|_2$ because P_c is an orthogonal projection and $\|\widehat{\Pi}_c\|_2 \le \|x\|_2$ by (D.4). The Gaussian Poincaré inequality (E.1) and $\mathbb{E}\|x\|_2^2 = n(H + \sigma_v^2)$ then give

$$\mathrm{Var}(\widehat{h}_c) = \frac{\mathrm{Var}(x'\widehat{\Pi}_c)}{n^2} \le \frac{4\sigma_v^2 \mathbb{E}\|x\|_2^2}{n^2} = \frac{4\sigma_v^2(H + \sigma_v^2)}{n},$$

which proves (i). Under the additional assumptions in (ii), we have $\bar{d}_c = o(\sqrt{n})$ by (F.13), so that $\mathbb{E}[(\widehat{h}_c - h_c)^2] = \mathrm{Var}(\widehat{h}_c) + \sigma_v^4 \bar{d}_c^2/n^2 \le C/n$. The first bound in (B.7) then follows from Markov's inequality, and the second because $h_c \ge \underline{h}$ for all sufficiently large n by (F.11). By the same lower bound, the complement of $\mathcal{H}_n$ is contained in $\{\max_c |\widehat{h}_c - h_c| > \underline{h}/2\}$, whose probability tends to zero by (B.7), so that $\mathbb{P}(\mathcal{H}_n) \to 1$. Finally, if $\widehat{\Pi}_c = 0$ then $\widehat{h}_c = 0$, so

that on $\mathcal{H}_n$ every $\widehat{\Pi}_c$ is nonzero and the estimator is the ratio in (2.5). □

Lemma B.2. *Suppose Assumptions 1 and 2 hold, with $J < \infty$ fixed, and let Γ_c and L_n be as in (B.1) and (B.3). Then, for each candidate and all sufficiently large n,*

$$\mathbb{E}[(\Gamma_c - L_n)^2] \le 2S_c. \tag{B.8}$$

Proof. Fix a candidate and suppress its subscript. With M the population projection residual in (A.3), we have $\Gamma - L_n = \varepsilon'\{\widehat{\Pi} - (h/H)\Pi\}/(\sqrt{n}\,h) = \varepsilon' M/(\sqrt{n}\,h)$. Using the Gaussian error decomposition (A.5), we can write

$$\varepsilon' M - \sigma_{\varepsilon v} d = \frac{\sigma_{\varepsilon v}}{\sigma_v^2}(v' M - \sigma_v^2 d) + e' M.$$

Conditional on v, the vector M and the rank d are fixed, while $e'M$ has mean zero and variance $(\sigma_\varepsilon^2 - \sigma_{\varepsilon v}^2/\sigma_v^2)\|M\|_2^2$. The cross term therefore has expectation zero, and (A.4) together with $\mathbb{E}\|M\|_2^2 = n\mathcal{A}$ yields

$$\mathbb{E}[(\varepsilon' M - \sigma_{\varepsilon v} d)^2] = \frac{\sigma_{\varepsilon v}^2}{\sigma_v^4}(\sigma_v^2 n\mathcal{A} + \sigma_v^4 \bar{d}) + \left(\sigma_\varepsilon^2 - \frac{\sigma_{\varepsilon v}^2}{\sigma_v^2}\right) n\mathcal{A} = \sigma_\varepsilon^2 n\mathcal{A} + \sigma_{\varepsilon v}^2 \bar{d}.$$

Writing $\varepsilon' M = (\varepsilon' M - \sigma_{\varepsilon v} d) + \sigma_{\varepsilon v} d$ and using $(a+b)^2 \le 2a^2 + 2b^2$, we conclude that

$$\mathbb{E}[(\Gamma - L_n)^2] = \frac{\mathbb{E}[(\varepsilon' M)^2]}{nh^2} \le \frac{2}{nh^2}\left\{\sigma_\varepsilon^2 n\mathcal{A} + \sigma_{\varepsilon v}^2 \mathbb{E}[d^2 + d]\right\} = 2S.$$

□

Lemma B.3 (Moments of the Denominator Correction)**.** *Suppose Assumptions 1, 2, and 3 hold, with $J < \infty$ fixed, and let Δ_c and L_n be as in (B.1) and (B.3). There is a constant $C < \infty$ such that, for all sufficiently large n and every candidate,*

$$\mathbb{E}[\Delta_c^2 L_n^2] \le \frac{C}{n}, \tag{B.9}$$

$$|\mathbb{E}[\Delta_c L_n^2]| \le C\left\{\sqrt{\frac{S_c}{n}} + \frac{1}{n}\right\}. \tag{B.10}$$

Proof. Fix a candidate and suppress its subscript. The quantities H, h, and their reciprocals are bounded by (2.2) and (F.11). Both bounds rest on the Stein identity

$$\mathbb{E}[(\Pi' v) g(v)] = \sigma_v^2 \mathbb{E}[\Pi' \nabla g(v)], \tag{B.11}$$

which follows from Stein's lemma (Stein, 1981, Lemma 2) and holds for locally Lipschitz g whose value and gradient grow at most polynomially. Every g used below has this property. Since $\|\widehat{\Pi}\|_2 \le \|x\|_2$ by Lemma D.1, Cauchy–Schwarz bounds $|\Delta|$ by $1 + \|x\|_2^2/(nh)$, and since the Jacobian of the fit is also a projection by that lemma, the gradients of Δ and $\Pi'\widehat{\Pi}$ grow at most linearly in $\|x\|_2 \le \|\Pi\|_2 + \|v\|_2$.

Before proceeding, we collect three facts. First, by the proof of Lemma B.1, the gradient of Δ with respect to v is

$$\nabla\Delta = \frac{\widehat{\Pi} + Px}{nh},$$

so that $\mathbb{E}\|\nabla\Delta\|_2^2 \le 4\mathbb{E}\|x\|_2^2/(nh)^2 = 4(H + \sigma_v^2)/(nh^2) \le C/n$. Second, by Lemma B.1, $\mathbb{E}[\Delta] = \sigma_v^2\bar{d}/(nh)$ and $\mathbb{E}[\Delta^2] = \mathbb{E}[(\widehat{h} - h)^2]/h^2 \le C/n$. Third, by the error decomposition (A.5),

$$L_n = \frac{\sigma_{\varepsilon v}}{\sigma_v^2\sqrt{n}\,H}\Pi'v + \frac{\Pi'e}{\sqrt{n}\,H},$$

where the second term is independent of v, with mean zero and variance $(\sigma_\varepsilon^2 - \sigma_{\varepsilon v}^2/\sigma_v^2)/H$, so that

$$\mathbb{E}[L_n^2 \mid v] = \frac{\sigma_\varepsilon^2 - \sigma_{\varepsilon v}^2/\sigma_v^2}{H} + \frac{\sigma_{\varepsilon v}^2}{\sigma_v^4 nH^2}(\Pi'v)^2. \tag{B.12}$$

Step 1: $\mathbb{E}[\Delta^2 L_n^2] \le C/n$. Since Δ depends only on v, multiplying (B.12) by Δ^2 and taking expectations yields $\mathbb{E}[\Delta^2 L_n^2] \le C\mathbb{E}[\Delta^2] + (C/n)\mathbb{E}[(\Pi'v)^2\Delta^2]$, so it suffices to show that $\mathbb{E}[(\Pi'v)^2\Delta^2] \le C$. Apply (B.11) with $g(v) = (\Pi'v)\Delta^2$. Since $\nabla(\Pi'v) = \Pi$ and $\|\Pi\|_2^2 = nH$, the product rule gives

$$\begin{aligned}\mathbb{E}[(\Pi'v)^2\Delta^2] &= \sigma_v^2 nH\mathbb{E}[\Delta^2] + 2\sigma_v^2\mathbb{E}[(\Pi'v)\Delta\cdot\Pi'\nabla\Delta] \\ &\le \sigma_v^2 nH\mathbb{E}[\Delta^2] + \tfrac{1}{2}\mathbb{E}[(\Pi'v)^2\Delta^2] + 2\sigma_v^4\mathbb{E}[(\Pi'\nabla\Delta)^2],\end{aligned}$$

where the inequality comes from Young's inequality.[9] By Cauchy–Schwarz, the last term is at most $2\sigma_v^4 nH\mathbb{E}\|\nabla\Delta\|_2^2 \le C$, and the first is at most C because $\mathbb{E}[\Delta^2] \le C/n$. Since $|\Delta| \le 1 + \|x\|_2^2/(nh)$, the term $\frac{1}{2}\mathbb{E}[(\Pi'v)^2\Delta^2]$ is finite, and moving it to the left-hand side we obtain $\mathbb{E}[(\Pi'v)^2\Delta^2] \le C$.

Step 2: $|\mathbb{E}[\Delta L_n^2]| \le C\{\sqrt{S/n} + 1/n\}$. Cauchy–Schwarz and Step 1 bound $|\mathbb{E}[\Delta L_n^2]|$ only by a multiple of $n^{-1/2}$, which need not be small relative to S. However, for the optimal instrument the denominator correction is q, and $\mathbb{E}[qL_n^2] = 0$ by Gaussian symmetry, so a signed calculation can do better. To this end, multiply (B.12) by Δ and take expectations. Applying (B.11) with $g(v) = (\Pi'v)\Delta$, we have $\mathbb{E}[(\Pi'v)^2\Delta] = \sigma_v^2 nH\mathbb{E}[\Delta] + \sigma_v^2\mathbb{E}[(\Pi'v)\Pi'\nabla\Delta]$.

[9] Young's inequality states that $2ab \le \eta a^2 + b^2/\eta$ for real a, b and any $\eta > 0$, which follows from expanding $(\sqrt{\eta}\,a - b/\sqrt{\eta})^2 \ge 0$. Here we apply it with $a = (\Pi'v)\Delta$, $b = \sigma_v^2\Pi'\nabla\Delta$, and $\eta = 1/2$.

After this substitution, the two multiples of $\mathbb{E}[\Delta]$ add up to $(\sigma_\varepsilon^2 - \sigma_{\varepsilon v}^2/\sigma_v^2)/H + \sigma_{\varepsilon v}^2/(\sigma_v^2 H) = \sigma_\varepsilon^2/H$, so that

$$\mathbb{E}[\Delta L_n^2] = \frac{\sigma_\varepsilon^2}{H}\mathbb{E}[\Delta] + \frac{\sigma_{\varepsilon v}^2}{\sigma_v^2 n H^2}\mathbb{E}[(\Pi' v)\Pi'\nabla\Delta]. \tag{B.13}$$

The first term is at most $C\bar{d}/n$. For the optimal instrument, $\Delta = q$ has $\mathbb{E}[q] = 0$ and $\Pi'\nabla q = 1$, so that both terms vanish. For the LASSO fit, we show that the second term is at most $C\{1/n + \sqrt{\mathcal{A}/n}\}$ rather than of order $n^{-1/2}$. Since $P\widehat{\Pi} = \widehat{\Pi}$, we have $(I - P)\Pi = -(H/h)(I - P)M$, so that the part of the optimal instrument outside the column space of the active columns satisfies

$$\mathbb{E}\|(I - P)\Pi\|_2^2 \le \frac{H^2}{h^2}\mathbb{E}\|M\|_2^2 = \frac{H^2}{h^2} n\mathcal{A}.$$

Since P is symmetric, the formula for $\nabla\Delta$ above shows that $nh\,\Pi'\nabla\Delta = \Pi'\widehat{\Pi} + x'P\Pi$, and writing $x'P\Pi = x'\Pi - x'(I - P)\Pi$ with $x'\Pi = nH + \Pi' v$, we have

$$nh\,\mathbb{E}[(\Pi' v)\Pi'\nabla\Delta] = \underbrace{\mathbb{E}[(\Pi' v)\Pi'\widehat{\Pi}]}_{\mathrm{I}} + \underbrace{\mathbb{E}[(\Pi' v)(nH + \Pi' v)]}_{\mathrm{II}} - \underbrace{\mathbb{E}[(\Pi' v)x'(I - P)\Pi]}_{\mathrm{III}}.$$

Applying (B.11) with $g(v) = \Pi'\widehat{\Pi}$, whose gradient is $P\Pi$, we have

$$0 \le \mathrm{I} = \sigma_v^2\mathbb{E}[\Pi' P\Pi] \le \sigma_v^2 nH,$$

and $\mathrm{II} = \mathbb{E}[(\Pi' v)^2] = \sigma_v^2 nH$ because $\mathbb{E}[\Pi' v] = 0$. For III, since $\mathbb{E}[(\Pi' v)^4] = 3\sigma_v^4 (nH)^2$ and $\mathbb{E}\|x\|_2^4 \le 8\{(nH)^2 + \sigma_v^4 n(n + 2)\}$, Cauchy–Schwarz bounds $\mathbb{E}[(\Pi' v)^2\|x\|_2^2]$ by Cn^2, and a second application yields

$$|\mathrm{III}| \le \left\{\mathbb{E}[(\Pi' v)^2\|x\|_2^2]\,\mathbb{E}\|(I - P)\Pi\|_2^2\right\}^{1/2} \le Cn^{3/2}\sqrt{\mathcal{A}}.$$

Combining the bounds on I, II, and III and dividing by nh, we have $|\mathbb{E}[(\Pi' v)\Pi'\nabla\Delta]| \le C(1 + \sqrt{n\mathcal{A}})$, so that by (B.13),

$$|\mathbb{E}[\Delta L_n^2]| \le C\left\{\frac{\bar{d}}{n} + \sqrt{\frac{\mathcal{A}}{n}} + \frac{1}{n}\right\}.$$

Finally, the first bound in (F.15) gives $\bar{d}/n + \sqrt{\mathcal{A}/n} \le C\sqrt{S/n}$, which proves (B.10). □

C Proofs of Proposition 1 and Theorem 2

C.1 Proof of Proposition 1

Proof. The first conclusion (3.13) is part of (F.13) of Lemma F.2. For (3.14), write

$$\frac{d_c^2 + d_c}{R_{d,c}} = \frac{(d_c/\bar{d}_c)^2 + (d_c/\bar{d}_c)/\bar{d}_c}{R_{d,c}/\bar{d}_c^2}. \tag{C.1}$$

By (F.13), $d_c/\bar{d}_c \to_p 1$, $\bar{d}_c \to \infty$, and $R_{d,c}/\bar{d}_c^2 \to 1$, so the right-hand side of (C.1) converges to one in probability. Since $R_{d,c} = \mathbb{E}[d_c^2 + d_c]$, this is (3.14). □

C.2 Proof of Theorem 2

Proof. We work on $\mathcal{H}_n$, where every $\widehat{h}_c$ is bounded away from zero and every $\widehat{S}_c$ is finite. Since $\mathbb{P}(\mathcal{H}_n) \to 1$ by Lemma B.1, the stochastic orders below also hold without this restriction. We expand the two terms of $\widehat{S}_c$ in turn and show that the first estimates $\sigma_\varepsilon^2 \mathcal{A}_c / h_c^2$ up to the common centering B_n and that the second estimates $\sigma_{\varepsilon v}^2 \mathbb{E}[d_c^2 + d_c]/(n h_c^2)$. The two supporting lemmas are stated and proved in Appendix C.3.

Step 1: $\widehat{\sigma}_\varepsilon^2 \mathbb{E}_n[\widehat{\Pi}_{c,i}^2]/\widehat{h}_c^2 = \sigma_\varepsilon^2 \mathcal{A}_c / h_c^2 + B_n + o_p(S_c)$. Applying the decomposition in (3.6) to sample moments, and recalling that $H = \mathbb{E}_n[\Pi_i^2]$, we have $\mathbb{E}_n[\widehat{\Pi}_{c,i}^2] = \widehat{\mathcal{A}}_c + \{\mathbb{E}_n[\Pi_i \widehat{\Pi}_{c,i}]\}^2 / H$, with $\widehat{\mathcal{A}}_c$ as defined in Section 3.3. Since $x_i = \Pi_i + v_i$, we also have $\mathbb{E}_n[\Pi_i \widehat{\Pi}_{c,i}] = \widehat{h}_c - \mathbb{E}_n[v_i \widehat{\Pi}_{c,i}]$, so that, writing $\widehat{q}_c := \mathbb{E}_n[v_i \widehat{\Pi}_{c,i}]/\widehat{h}_c$,

$$\frac{\widehat{\sigma}_\varepsilon^2 \mathbb{E}_n[\widehat{\Pi}_{c,i}^2]}{\widehat{h}_c^2} = \frac{\widehat{\sigma}_\varepsilon^2 \widehat{\mathcal{A}}_c}{\widehat{h}_c^2} + \frac{\widehat{\sigma}_\varepsilon^2}{H}(1 - \widehat{q}_c)^2. \tag{C.2}$$

For the first term, $\widehat{\mathcal{A}}_c - \mathcal{A}_c = o_p(S_c)$ by Lemma C.1, while Lemmas B.1 and C.2 show that $\widehat{h}_c - h_c$ and $\widehat{\sigma}_\varepsilon^2 - \sigma_\varepsilon^2$ are both $O_p(n^{-1/2})$. Since $\mathcal{A}_c \le C S_c$ by (F.14),

$$\frac{\widehat{\sigma}_\varepsilon^2 \widehat{\mathcal{A}}_c}{\widehat{h}_c^2} = \frac{\sigma_\varepsilon^2 \mathcal{A}_c}{h_c^2} + o_p(S_c) + O_p(\mathcal{A}_c n^{-1/2}) = \frac{\sigma_\varepsilon^2 \mathcal{A}_c}{h_c^2} + o_p(S_c).$$

For the second term, with $q = \mathbb{E}_n[\Pi_i v_i]/H$ as in the proof of Theorem 1, Lemma C.1 shows that $\widehat{q}_c = q + o_p(S_c)$, and $q = O_p(n^{-1/2})$ because $\mathbb{E}[q^2] = \sigma_v^2/(nH)$. Factoring the difference of squares,

$$(1 - \widehat{q}_c)^2 - (1 - q)^2 = (q - \widehat{q}_c)(2 - q - \widehat{q}_c) = o_p(S_c),$$

while $(1 - q)^2 = 1 - 2q + q^2$ with $q^2 = O_p(n^{-1}) = o_p(S_c)$, since $\min_c n S_c \to \infty$. As $\widehat{\sigma}_\varepsilon^2 / H = O_p(1)$, the second term in (C.2) equals $\widehat{\sigma}_\varepsilon^2 (1 - 2q)/H + o_p(S_c) = B_n + o_p(S_c)$, with B_n the common centering in (3.15). We keep $\widehat{\sigma}_\varepsilon^2$ inside B_n, since its estimation error is of order $n^{-1/2}$ and need not be $o_p(S_c)$. Adding the two terms proves Step 1.

Step 2: $\widehat{\sigma}^2_{\varepsilon v}(d_c^2+d_c)/(n\widehat{h}_c^2) = \sigma^2_{\varepsilon v}\mathbb{E}[d_c^2+d_c]/(nh_c^2) + o_p(S_c)$. Write

$$\frac{\widehat{\sigma}^2_{\varepsilon v}(d_c^2+d_c)}{n\widehat{h}_c^2} = \frac{\sigma^2_{\varepsilon v}\mathbb{E}[d_c^2+d_c]}{nh_c^2}\cdot\frac{\widehat{\sigma}^2_{\varepsilon v}}{\sigma^2_{\varepsilon v}}\cdot\frac{d_c^2+d_c}{\mathbb{E}[d_c^2+d_c]}\cdot\frac{h_c^2}{\widehat{h}_c^2}. \tag{C.3}$$

Of the last three factors, the first converges to one in probability by Lemma C.2 and $\sigma_{\varepsilon v}\neq 0$, the second by Proposition 1, and the third by Lemma B.1. Since the leading factor is at most S_c, the product equals $\sigma^2_{\varepsilon v}\mathbb{E}[d_c^2+d_c]/(nh_c^2)+o_p(S_c)$.

Step 3: Part (i). Adding Steps 1 and 2 and using (3.12) and (3.7), we obtain $\widehat{S}_c = B_n + S_c + o_p(S_c)$, which is (3.16). Since $\bar{Q}_c/S_c \to 1$ by Theorem 1(ii),

$$\widehat{S}_c - B_n - \bar{Q}_c = (\widehat{S}_c - B_n - S_c) + (S_c - \bar{Q}_c) = o_p(S_c) = o_p(\bar{Q}_c),$$

which is (3.17).

Step 4: Part (ii). By Theorem 1(ii), every $\bar{Q}_c$ is positive. Let

$$\omega_n := \max_{c\in[J]}\frac{|\widehat{S}_c - B_n - \bar{Q}_c|}{\bar{Q}_c},$$

By Step 3, $\omega_n \to_p 0$. On $\{\omega_n<1\}$, since $\widehat{c}$ minimizes the feasible criterion, for every $c\in[J]$,

$$B_n + (1-\omega_n)\bar{Q}_{\widehat{c}} \le \widehat{S}_{\widehat{c}} \le \widehat{S}_c \le B_n + (1+\omega_n)\bar{Q}_c.$$

Cancelling B_n, minimizing the right-hand side over c, and dividing by $(1-\omega_n)\min_c \bar{Q}_c$, we obtain

$$0 \le \frac{\bar{Q}_{\widehat{c}}}{\min_c \bar{Q}_c} - 1 \le \frac{2\omega_n}{1-\omega_n}, \tag{C.4}$$

and the right-hand side converges to zero in probability. Since $\mathbb{P}(\omega_n<1)\to 1$, this proves (3.18). □

C.3 Supporting Lemmas

Lemma C.1 (Sample Counterparts of $\mathcal{A}_c$ and q)**.** *Suppose Assumptions 1 to 3 hold, with $J<\infty$ fixed, and let $q = \mathbb{E}_n[\Pi_i v_i]/H$ and $\widehat{q}_c = \mathbb{E}_n[v_i\widehat{\Pi}_{c,i}]/\widehat{h}_c$ be as in the proofs of Theorems 1 and 2. With $\widehat{\mathcal{A}}_c = \min_{a\in\mathbb{R}}\mathbb{E}_n[(\widehat{\Pi}_{c,i}-a\Pi_i)^2]$ as defined in Section 3.3, we have*

$$\widehat{\mathcal{A}}_c - \mathcal{A}_c = O_p\left(\sqrt{\frac{\mathcal{A}_c}{n}}+\frac{1}{n}\right) = o_p(S_c), \tag{C.5}$$

and, on $\mathcal{H}_n$,

$$\widehat{q}_c - q = O_p\left(\sqrt{\frac{\mathcal{A}_c}{n}} + \frac{\bar{d}_c}{n} + \frac{1}{n}\right) = o_p(S_c). \tag{C.6}$$

Proof. Fix a candidate and suppress its subscript.

Step 1: $\widehat{\mathcal{A}} - \mathcal{A} = O_p(\sqrt{\mathcal{A}/n} + 1/n)$. Applying the decomposition in (3.6) to sample moments, as in the proof of Theorem 2,

$$\widehat{\mathcal{A}} = \mathbb{E}_n[\widehat{\Pi}_i^2] - \frac{\{\mathbb{E}_n[\Pi_i\widehat{\Pi}_i]\}^2}{H} = \frac{1}{n}\left\|\left(I - \frac{\Pi\Pi'}{nH}\right)\widehat{\Pi}\right\|_2^2.$$

Since $\overline{\mathbb{E}}[\Pi_i\widehat{\Pi}_i] = h$, we have $\mathbb{E}[\{\mathbb{E}_n[\Pi_i\widehat{\Pi}_i]\}^2] = h^2 + \mathrm{Var}(\mathbb{E}_n[\Pi_i\widehat{\Pi}_i])$, and taking expectations and comparing with (3.6), we obtain $\mathcal{A} - \mathbb{E}[\widehat{\mathcal{A}}] = \mathrm{Var}(\mathbb{E}_n[\Pi_i\widehat{\Pi}_i])/H$. The gradient of $\mathbb{E}_n[\Pi_i\widehat{\Pi}_i]$ with respect to v is $P\Pi/n$, whose squared norm is at most $\|\Pi\|_2^2/n^2 = H/n$, so that by the Gaussian Poincaré inequality (E.1),

$$0 \le \mathcal{A} - \mathbb{E}[\widehat{\mathcal{A}}] \le \frac{\sigma_v^2}{n}. \tag{C.7}$$

Similarly, the gradient of $\widehat{\mathcal{A}}$ is $(2/n)P\{I - \Pi\Pi'/(nH)\}\widehat{\Pi}$, whose squared norm is at most $4\widehat{\mathcal{A}}/n$, so that

$$\mathrm{Var}(\widehat{\mathcal{A}}) \le \frac{4\sigma_v^2}{n}\mathbb{E}[\widehat{\mathcal{A}}] \le \frac{4\sigma_v^2\mathcal{A}}{n}.$$

The claim follows from Chebyshev's inequality and (C.7).

Step 2: $\widehat{q} - q = O_p(\sqrt{\mathcal{A}/n} + \bar{d}/n + 1/n)$ *on* $\mathcal{H}_n$. With M the population projection residual in (A.3), we have $\mathbb{E}_n[v_i\widehat{\Pi}_i] = v'M/n + hq$, and therefore

$$\widehat{q} - q = \frac{v'M/n}{\widehat{h}} - q\frac{\widehat{h} - h}{\widehat{h}}. \tag{C.8}$$

By (A.4) and Markov's inequality, $v'M = \sigma_v^2 d + O_p(\sqrt{n\mathcal{A}} + \sqrt{\bar{d}})$. Since $d = O_p(\bar{d})$ and $\bar{d} \ge 1$ for all sufficiently large n by (F.13), we have $v'M/n = O_p(\sqrt{\mathcal{A}/n} + \bar{d}/n)$. Moreover, $q = O_p(n^{-1/2})$ since $\mathbb{E}[q^2] = \sigma_v^2/(nH)$, while $\widehat{h} - h = O_p(n^{-1/2})$ by Lemma B.1 and $\widehat{h} \ge \underline{h}/2$ on $\mathcal{H}_n$. Substituting these bounds into (C.8) yields the claim.

Step 3: Both rates are $o_p(S)$. By the second bound in (F.15), $\sqrt{\mathcal{A}/n} + \bar{d}/n \le CS/\bar{d}$, which is $o(S)$ because $\bar{d} \to \infty$ by (F.13). The remaining term $1/n$ is also $o(S)$, since $nS \to \infty$ by (F.14). □

Lemma C.2 (Estimating the Error Moments)**.** *Suppose Assumptions 1 to 3 hold and $J < \infty$ is fixed. For any (possibly data-dependent) $[J]$-valued index $\check{c}$,*

$$\widehat{\beta}_{\check{c}} - \beta = O_p(n^{-1/2}). \tag{C.9}$$

If the pilot satisfies $\check{\beta} - \beta = O_p(n^{-1/2})$ and $\widehat{\sigma}^2_\varepsilon, \widehat{\sigma}_{\varepsilon v}$ are as in (3.11), then

$$\widehat{\sigma}^2_\varepsilon - \sigma^2_\varepsilon = O_p(n^{-1/2}), \qquad \widehat{\sigma}_{\varepsilon v} - \sigma_{\varepsilon v} = O_p(n^{-1/2}). \tag{C.10}$$

Proof. Since $\check{c} \in [J]$, (B.5) in the proof of Theorem 1 gives

$$|\sqrt{n}(\widehat{\beta}_{\check{c}} - \beta)| \le \max_{c \in [J]} |\sqrt{n}(\widehat{\beta}_c - \beta)| = O_p(1),$$

which proves (C.9).

By independence, Gaussian fourth moments, $H = \mathbb{E}_n[\Pi_i^2] = O(1)$, and Chebyshev's inequality,

$$\mathbb{E}_n[\varepsilon_i^2] = \sigma^2_\varepsilon + O_p(n^{-1/2}), \qquad \mathbb{E}_n[\varepsilon_i x_i] = \sigma_{\varepsilon v} + O_p(n^{-1/2}).$$

Also, $\overline{\mathbb{E}}[x_i^2] = H + \sigma_v^2 = O(1)$, so $\mathbb{E}_n[x_i^2] = O_p(1)$ by Markov's inequality. Since $y_i - \check{\beta} x_i = \varepsilon_i - \delta_\beta x_i$ for $\delta_\beta := \check{\beta} - \beta$, we have

$$\widehat{\sigma}^2_\varepsilon = \mathbb{E}_n[\varepsilon_i^2] - 2\delta_\beta \mathbb{E}_n[\varepsilon_i x_i] + \delta_\beta^2 \mathbb{E}_n[x_i^2], \quad \widehat{\sigma}_{\varepsilon v} = \mathbb{E}_n[\varepsilon_i x_i] - \delta_\beta \mathbb{E}_n[x_i^2]. \tag{C.11}$$

Substituting these moment bounds and $\delta_\beta = O_p(n^{-1/2})$ into the identities (C.11), which do not use Gaussian errors, proves (C.10). □

D LASSO Fitted-Value Geometry

Fix a candidate and suppress its subscript. The following geometry holds for fixed Z and $\lambda > 0$, without distributional restrictions on the response. Let $u(w) := w - \mu(w)$ and let $E(w)$ be as in (2.7); both are determined by the unique fitted vector. Define the closed convex set containing zero

$$\mathcal{K} := \{q \in \mathbb{R}^n : \|Z'q/n\|_\infty \le \lambda\}.$$

In Lemma D.1, $\widehat{\pi}(w)$ is any minimizer of (2.4); Lemma D.2 fixes a Borel choice.

Lemma D.1 (Geometry of the LASSO Fit)**.** *For every $w \in \mathbb{R}^n$ and every minimizer $\widehat{\pi}(w)$ of the problem in (2.4),*

$$|\mathbb{E}_n[z_{ij}u_i(w)]| \le \lambda, \qquad \widehat{\pi}_j(w)\mathbb{E}_n[z_{ij}u_i(w)] = \lambda|\widehat{\pi}_j(w)|, \quad j \in [p], \tag{D.1}$$

and hence $\operatorname{supp}\{\widehat{\pi}(w)\} \subseteq E(w)$ *and*

$$\mathbb{E}_n[\mu_i(w)w_i] = \mathbb{E}_n[\mu_i(w)^2] + \lambda\|\widehat{\pi}(w)\|_1. \tag{D.2}$$

Moreover,

$$u(w) = P_{\mathcal{K}}(w), \qquad \mu(w) = w - P_{\mathcal{K}}(w), \tag{D.3}$$

where $P_{\mathcal{K}}$ is Euclidean projection onto $\mathcal{K}$. The map μ is one-Lipschitz, $\mu(0) = 0$, and

$$\|\mu(w)\|_2 \le \|w\|_2. \tag{D.4}$$

For Lebesgue-almost every $w \in \mathbb{R}^n$,

$$\dot{\mu}(w) = P_{\operatorname{col}(Z_{E(w)})}, \qquad \dot{\mu}(w)' = \dot{\mu}(w) = \dot{\mu}(w)^2, \qquad \dot{\mu}(w)\mu(w) = \mu(w). \tag{D.5}$$

Proof. The KKT conditions give (D.1); multiplying by the coefficients and summing gives (D.2). Multiplying (2.4)'s objective by n gives penalty $n\lambda$. Lemmas 1 and 3 of Tibshirani and Taylor (2012) then give (D.3) and the one-Lipschitz property. Since $0 \in \mathcal{K}$, $\mu(0) = 0$, proving (D.4). Their equation (21) and Lemma 5 give $\dot{\mu}(w) = Z_{E(w)}Z^{+}_{E(w)} = P_{\operatorname{col}(Z_{E(w)})}$ for Lebesgue-almost every w, with $+$ denoting the Moore–Penrose pseudoinverse. The remaining identities in (D.5) follow because this is an orthogonal projection and $\mu(w) \in \operatorname{col}(Z_{E(w)})$ by (D.1). □

Lemma D.2 (A Measurable LASSO Solution)**.** *There exists a Borel-measurable LASSO solution $w \mapsto \widehat{\pi}(w)$ such that, for every $w \in \mathbb{R}^n$,*

$$\operatorname{rank}\Big(Z_{\operatorname{supp}\{\widehat{\pi}(w)\}}\Big) = \big|\operatorname{supp}\{\widehat{\pi}(w)\}\big| \le \operatorname{rank}(Z) \le n, \tag{D.6}$$

and $Z\widehat{\pi}(w)$ equals the unique fitted vector $\mu(w)$.

Proof. Lemma 14 of Tibshirani (2013), with penalty $n\lambda$, gives an independent-support minimizer for every w. Order the finite set $\mathcal{B} = \{B \subseteq [p] : \operatorname{rank}(Z_B) = |B|\}$, including $B = \varnothing$. For nonempty $B \in \mathcal{B}$, set

$$b^{(B)}_B(w) = (Z_B'Z_B)^{-1}Z_B'\mu(w), \qquad b^{(B)}_{B^c}(w) = 0,$$

and set $b^{(\varnothing)}(w) = 0$. Let $\mathcal{R}_B$ be the set of responses for which $b^{(B)}(w)$ satisfies the KKT conditions (D.1) with residual $w - Zb^{(B)}(w)$. Each $\mathcal{R}_B$ is closed: μ and $b^{(B)}$ are continuous by Lemma D.1, and the KKT conditions are closed. For any w, the independent-support minimizer above has support B and, by uniqueness of $\mu(w)$ and full column rank of Z_B, equals $b^{(B)}(w)$, including when $B = \varnothing$. Thus the $\mathcal{R}_B$ cover $\mathbb{R}^n$. Choose $b^{(B)}(w)$ for the first B with $w \in \mathcal{R}_B$. The first-choice regions are Borel and KKT suffices for optimality, so this is a Borel minimizer. Its support lies in independent B, proving (D.6), and its fitted vector is $\mu(w)$. □

E Gaussian Identities and Rank Variance

By Rademacher's theorem, any locally Lipschitz $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable almost everywhere. Define $\nabla f(w)$ to be its gradient where it is differentiable and zero otherwise, and set $\partial_i f(w) := \{\nabla f(w)\}_i$.

Lemma E.1 (Poincaré Inequality and Stein Identities)**.** *Let $W \sim N(0, \sigma^2 I_n)$, $\sigma > 0$. If $g : \mathbb{R}^n \to \mathbb{R}$ is locally Lipschitz with $\mathbb{E}[g(W)^2] < \infty$ and $\mathbb{E}[\|\nabla g(W)\|_2^2] < \infty$, then*

$$\operatorname{Var}\{g(W)\} \le \sigma^2 \mathbb{E}[\|\nabla g(W)\|_2^2]. \tag{E.1}$$

Separately, let $m : \mathbb{R}^n \to \mathbb{R}^n$ be Lipschitz, set $A(w) := Dm(w)$ where differentiable and zero otherwise, and define $\operatorname{div} m(w) := \operatorname{tr}\{A(w)\}$. Then

$$\mathbb{E}[W' m(W)] = \sigma^2 \mathbb{E}[\operatorname{div} m(W)], \tag{E.2}$$

$$\mathbb{E}[\{W' m(W) - \sigma^2 \operatorname{div} m(W)\}^2] = \sigma^2 \mathbb{E}[\|m(W)\|_2^2] + \sigma^4 \mathbb{E}[\operatorname{tr}\{A(W)^2\}]. \tag{E.3}$$

For this m, let g be a locally Lipschitz scalar function with $|g(w)| + \|\nabla g(w)\|_2 \le C(1 + \|w\|_2^r)$ almost everywhere for some $C < \infty$ and $r \ge 0$, then

$$\mathbb{E}[\{W' m(W) - \sigma^2 \operatorname{div} m(W)\} g(W)] = \sigma^2 \mathbb{E}[m(W)' \nabla g(W)]. \tag{E.4}$$

Proof. For (E.1), rescale by $G = W/\sigma$ and $h(z) = g(\sigma z)$. The assumptions put h in $W^{1,2}(N(0, I_n))$, so Theorem 2.2, equation (2.12), of Bellec and Zhang (2021) with zero vector field gives the result. A Lipschitz m has at most linear growth and an almost-everywhere bounded weak Jacobian, ensuring Gaussian integrability. The first-order Stein formula of Stein (1981) applied componentwise gives (E.2); applying it to $m_i g$ and summing the product rule gives (E.4). The growth condition on g ensures the latter application is integrable. Theorem 2.1(ii), equation (2.7), of Bellec and Zhang (2021) gives (E.3). Transposing their Jacobian convention leaves divergence and $\operatorname{tr}(A^2)$ unchanged. □

Lemma E.2 (Moments of a Recentered LASSO Fit)**.** *Suppose Assumption 1(i) holds. Fix a candidate, suppress its subscript, and let $b \in \mathbb{R}^n$ be nonrandom. Define $m(a) := \mu(\Pi + a) - b$ for $a \in \mathbb{R}^n$ and $M := m(v) = \widehat{\Pi} - b$. Let d be as in (2.8) and $\bar{d} := \mathbb{E}[d]$ as in (A.2). Then m is one-Lipschitz, $\mathbb{E}[\|M\|_2^r] < \infty$ for every fixed $r > 0$, and, for every nonrandom $c_0 \in \mathbb{R}^n$,*

$$\mathrm{Var}(\|M\|_2^2) \le 4\sigma_v^2 \mathbb{E}[\|M\|_2^2], \tag{E.5}$$

$$\mathrm{Var}\{(\Pi + v + c_0)'M\} \le 2\sigma_v^2 \left\{ \mathbb{E}[\|M\|_2^2] + \mathbb{E}[\|\Pi + v + c_0\|_2^2] \right\}, \tag{E.6}$$

$$\mathbb{E}[v'M] = \sigma_v^2 \bar{d},$$

$$\mathbb{E}[(v'M - \sigma_v^2 d)^2] = \sigma_v^2 \mathbb{E}[\|M\|_2^2] + \sigma_v^4 \bar{d}. \tag{E.7}$$

Proof. By Lemma D.1, m is one-Lipschitz, has at most linear growth, and its weak Jacobian $Dm(a)$ is almost everywhere an orthogonal projection. Thus all required Gaussian moments are finite. At almost every a,

$$\nabla \|m(a)\|_2^2 = 2Dm(a)'m(a),$$
$$\nabla\{(\Pi + a + c_0)'m(a)\} = m(a) + Dm(a)'(\Pi + a + c_0).$$

The contraction property of $Dm(a)$ and Lemma E.1 give the variance bounds (E.5) and (E.6). Since $\operatorname{div} m(v) = \mathrm{tr}\{Dm(v)^2\} = d$ almost surely, (E.2) and (E.3) give (E.7). □

Lemma E.3 (Second Moment of the Numerator)**.** *Under Assumption 1(i), fix a candidate and suppress its subscript. Then*

$$\mathbb{E}[(\varepsilon'\widehat{\Pi})^2] = \sigma_\varepsilon^2 \mathbb{E}[\|\widehat{\Pi}\|_2^2] + \sigma_{\varepsilon v}^2 \mathbb{E}[d^2 + d]. \tag{E.8}$$

Define

$$\zeta := v'\widehat{\Pi} - \sigma_v^2 d.$$

Then, with ζ so defined,[10]

$$\mathbb{E}[v'\widehat{\Pi}] = \sigma_v^2 \mathbb{E}[d], \qquad \mathbb{E}[\zeta^2] = \sigma_v^2 \mathbb{E}[\|\widehat{\Pi}\|_2^2] + \sigma_v^4 \mathbb{E}[d], \qquad \mathbb{E}[d\zeta] = 0. \tag{E.9}$$

Proof. Lemma E.2 at $b = 0$ gives the first two identities in (E.9). Write $m(a) := \mu(\Pi + a)$ and $P := Dm(v)$. By Lemma D.1, $P' = P = P^2$ and $P\widehat{\Pi} = \widehat{\Pi}$ almost surely. For $g(a) := a'm(a)$, $\nabla g(v) = \widehat{\Pi} + Pv$. The function g is locally Lipschitz, and g and its weak gradient have polynomial growth. Thus (E.4), followed by the first two identities in (E.9),

[10] All quantities in (E.8)–(E.9) depend on the LASSO only through $\widehat{\Pi}$ and d, so no selection among non-unique coefficient solutions is involved.

gives

$$\begin{aligned}\sigma_v^2\mathbb{E}[d\zeta] &= \mathbb{E}[(v'\widehat{\Pi})\zeta] - \mathbb{E}[\zeta^2] \\ &= \sigma_v^2\mathbb{E}[\widehat{\Pi}'(\widehat{\Pi} + Pv)] - \mathbb{E}[\zeta^2] \\ &= \sigma_v^2\mathbb{E}[\|\widehat{\Pi}\|_2^2] + \sigma_v^4\mathbb{E}[d] - \mathbb{E}[\zeta^2] = 0.\end{aligned}$$

This proves the last identity in (E.9). Therefore,

$$\mathbb{E}[(v'\widehat{\Pi})^2] = \mathbb{E}[(\zeta + \sigma_v^2 d)^2] = \sigma_v^2\mathbb{E}[\|\widehat{\Pi}\|_2^2] + \sigma_v^4\mathbb{E}[d^2 + d].$$

The Gaussian error decomposition (A.5) and the fact that $\widehat{\Pi}$ is $\sigma(v)$-measurable yield

$$\mathbb{E}[(\varepsilon'\widehat{\Pi})^2] = \frac{\sigma_{\varepsilon v}^2}{\sigma_v^4}\mathbb{E}[(v'\widehat{\Pi})^2] + \left(\sigma_\varepsilon^2 - \frac{\sigma_{\varepsilon v}^2}{\sigma_v^2}\right)\mathbb{E}[\|\widehat{\Pi}\|_2^2].$$

Substitution proves (E.8). □

The rank bound below follows from the variance inequality of Theorem 2.2 of Bellec and Zhang (2021), applied to the LASSO fitted-value map.

Lemma E.4 (Variance of the Effective Dimension)**.** *Fix $\Pi \in \mathbb{R}^n$, $Z \in \mathbb{R}^{n\times p}$, and $\lambda > 0$, and let μ denote the LASSO fitted-value map defined by (2.4). For $v \sim N(0, \sigma_v^2 I_n)$ with $\sigma_v^2 > 0$, define*

$$d := \operatorname{rank}(Z_{E(\Pi+v)}), \qquad \bar{d} := \mathbb{E}[d].$$

Then $d = \operatorname{div}\mu(\Pi + v)$ $\mathbb{P}$-almost surely. There is a universal constant $C < \infty$ such that

$$\operatorname{Var}(d) \le C\left[1 + \bar{d}\log\left\{\frac{ep}{\bar{d}\vee 1}\right\}\right]. \tag{E.10}$$

Proof. Put

$$G := v/\sigma_v, \qquad F(u) := \mu(\Pi + \sigma_v u)/\sigma_v.$$

Then $G \sim N(0, I_n)$. By Lemma D.1, $DF(G) = P = P_{\operatorname{col}(Z_{E(\Pi+v)})}$ is an orthogonal projection with $\operatorname{tr}(P) = \operatorname{tr}(P^2) = d$ almost surely. The one-Lipschitz map F and $u'F(u)$ have the required weak derivatives and Gaussian integrability, so Theorem 2.2, equation (2.13), of Bellec and Zhang (2021) gives

$$\operatorname{Var}(d) \le \bar{d} + \mathbb{E}[\|PG\|_2^2]. \tag{E.11}$$

For $1 \le r \le \operatorname{rank}(Z)$, let

$$\mathscr{P}_r := \left\{P_{\operatorname{col}(Z_B)} : B \subseteq [p],\ |B| = \operatorname{rank}(Z_B) = r\right\}.$$

Since Z is fixed, this family is deterministic. On $\{d = r\}$, an independent subset of $E(\Pi + v)$ spans its column space; hence

$$|\mathscr{P}_r| \le \binom{p}{r} \le (ep/r)^r, \qquad P \in \mathscr{P}_r \quad \text{on } \{d = r\}. \tag{E.12}$$

For each fixed $P_0 \in \mathscr{P}_r$, $\|P_0 G\|_2^2 \sim \chi_r^2$. Apply Laurent and Massart (2000, equation (4.3)) and a union bound with $u = 2r\log(ep/r) + r + t$: its threshold $r + 2\sqrt{ru} + 2u$ is at most $11\{r\log(ep/r) + t\}$, and the resulting bound for each r is $|\mathscr{P}_r|e^{-u} \le e^{-r-t}$. Define $\psi_p(x) := x\log\{ep/(x \vee 1)\}$ for $0 \le x \le p$. On $\{d = 0\}$, $P = 0$. Summing over $r \ge 1$ gives

$$\mathbb{P}\Big\{\|PG\|_2^2 > 11\{\psi_p(d) + t\}\Big\} \le \frac{e^{-t}}{e-1}, \qquad t \ge 0. \tag{E.13}$$

Tail integration yields

$$\mathbb{E}[\|PG\|_2^2] \le 11\mathbb{E}[\psi_p(d)] + \frac{11}{e-1}. \tag{E.14}$$

The concave ψ_p dominates x on $[0, p]$. Jensen's inequality in (E.11)– (E.14) now gives $\mathrm{Var}(d) \le 12\{1 + \psi_p(\bar{d})\}$, proving (E.10). □

F Sparse First-Stage Bounds

Following Belloni et al. (2012), define the score event

$$\mathcal{G}_c := \left\{\big\|\mathbb{E}_n[z_{c,i}v_i]\big\|_\infty \le \frac{\lambda_c}{c_\lambda}\right\}. \tag{F.1}$$

Lemma F.1 (LASSO Rates and Recovery of the Strong Coordinates)**.** *Under Assumption 2, for all sufficiently large n, the following hold on the event $\mathcal{G}_c$ defined by (F.1). First, the prediction norm satisfies*

$$\mathbb{E}_n[\{z'_{c,i}(\widehat{\pi}_c - \pi_c^0)\}^2]^{1/2} \le 2\mathbb{E}_n[\xi_{c,i}^2]^{1/2} + \frac{2(1 + c_\lambda^{-1})\lambda_c\sqrt{s_c}}{\kappa_0}. \tag{F.2}$$

Next, for the measurable solution $\widehat{\pi}_c$ of Lemma D.2,

$$\|\widehat{\pi}_c - \pi_c^0\|_2 \le \frac{2}{\sqrt{\underline{\phi}}}\left\{\mathbb{E}_n[\xi_{c,i}^2]^{1/2} + \frac{(1 + c_\lambda^{-1})\lambda_c\sqrt{s_c}}{\kappa_0}\right\}. \tag{F.3}$$

The effective dimension d_c defined in (2.8) does not depend on which minimizer is taken, and satisfies

$$d_c < Ms_c + 1. \tag{F.4}$$

If Assumption 3 holds as well, then, again on $\mathcal{G}_c$,

$$k_c \le d_c, \tag{F.5}$$

and, for the measurable solution $\widehat{\pi}_c$ above,

$$A_c \subseteq \operatorname{supp}(\widehat{\pi}_c), \qquad \operatorname{sign}(\widehat{\pi}_{c,j}) = \operatorname{sign}(\pi^0_{c,j}), \quad j \in A_c. \tag{F.6}$$

Proof. Suppress the candidate subscript c throughout. Define $\Delta := \widehat{\pi} - \pi^0$, $a := \mathbb{E}_n[\xi_i^2]^{1/2}$, $t := \mathbb{E}_n[(z_i'\Delta)^2]^{1/2}$, and $\theta := c_\lambda^{-1}$. By Assumption 2(ii)–(iii),

$$\rho_n := \frac{a}{\lambda\sqrt{s}} \le \frac{C_a}{c_\lambda\sqrt{2}\sigma_v\{\log(2p) + \alpha\log\log(p \vee n)\}^{1/2}} = o(1). \tag{F.7}$$

The denominator diverges since $p \vee n \ge n$. Conditional on $\mathcal{G}$, the following argument is deterministic; the penalty lower bound is used here only to ensure $\rho_n \to 0$.

Prediction bound. The basic-inequality argument of Belloni et al. (2012, Lemma 6), with their penalty $2n\lambda$ and unit loadings under our normalization, compares the objective at $\widehat{\pi}$ and π^0. On $\mathcal{G}$, Hölder's and Cauchy–Schwarz inequalities and $\pi^0_{T^c} = 0$ give

$$\frac{1}{2}t^2 + (1-\theta)\lambda\|\Delta_{T^c}\|_1 \le (1+\theta)\lambda\|\Delta_T\|_1 + at. \tag{F.8}$$

If $t \le 2a$, (F.2) is immediate. Otherwise (F.8) gives the restricted-eigenvalue cone $\|\Delta_{T^c}\|_1 \le L_\lambda\|\Delta_T\|_1$, so $\|\Delta_T\|_1 \le \sqrt{s}\,t/\kappa_0$ by Assumption 2(iv). Substitution into the basic inequality gives

$$t \le 2a + \frac{2(1+\theta)\lambda\sqrt{s}}{\kappa_0},$$

which proves (F.2).

Rank bound. Put $E := E(x)$, $d := \operatorname{rank}(Z_E)$, and $\bar{s} := \lceil Ms \rceil$. The fitted vector and residual are unique by Lemma D.1, so E and d do not depend on the minimizer. If $d \ge \bar{s}$, choose $D \subseteq E$ with $|D| = \bar{s}$ and linearly independent columns. Since $\bar{s} \le d \le p$, Assumption 2(iv) applies at $\bar{s}$. Using $x - Z\widehat{\pi} = \xi + v - Z\Delta$, the KKT equalities on D, the score event, and that upper sparse-eigenvalue bound give

$$(1-\theta)\lambda\sqrt{\bar{s}} \le \sqrt{\overline{\phi}}\,(a+t) \le \sqrt{\overline{\phi}}\left\{3a + \frac{2(1+\theta)\lambda\sqrt{s}}{\kappa_0}\right\}.$$

Consequently,

$$\frac{\bar{s}}{s} \le \frac{\overline{\phi}}{(1-\theta)^2}\left\{3\rho_n + \frac{2(1+\theta)}{\kappa_0}\right\}^2 < M$$

for large n: the strict inequality follows from $\rho_n \to 0$ and $M > 4L_\lambda^2\overline{\phi}/\kappa_0^2$. This contradicts $\bar{s} \geq Ms$, so $d < \lceil Ms \rceil < Ms + 1$, proving (F.4).

Coefficient bound and strong core. Now use the measurable solution of Lemma D.2 and put $\widehat{T} := \operatorname{supp}(\widehat{\pi})$. By Lemmas D.1 and D.2,

$$|\widehat{T}| \leq d, \qquad |\operatorname{supp}(\Delta)| \leq (s+d) \wedge p \leq \lceil (M+1)s \rceil \wedge p.$$

The lower sparse-eigenvalue bound in Assumption 2(iv) and (F.2) therefore yield

$$\sqrt{\underline{\phi}}\,\|\Delta\|_2 \leq t \leq 2a + \frac{2(1+\theta)\lambda\sqrt{s}}{\kappa_0},$$

including the case $\Delta = 0$, proving (F.3). If Assumption 3 holds, Assumption 2(ii) and its beta-min threshold give

$$|\Delta_j| \leq \|\Delta\|_2 \leq \underline{b}_n/2 \leq |\pi_j^0|/2, \qquad j \in A.$$

Thus every $j \in A$ is selected with the correct sign, and $k \leq |\widehat{T}| \leq d$. This proves (F.6) and (F.5). □

Lemma F.2 (Consequences of the First-Stage Conditions)**.** *Suppose Assumptions 1 and 2 hold, with $J < \infty$ fixed. Let Φ denote the standard normal distribution function and define*

$$\eta_{n,c} := 2p_c\Phi\left(-\frac{\sqrt{n}\lambda_c}{c_\lambda \sigma_v}\right)$$

and

$$a_{n,c} := \mathbb{E}_n[\xi_{c,i}^2] + s_c\lambda_c^2 + \sqrt{\eta_{n,c}}.$$

Uniformly over candidates,

$$\eta_{n,c} \to 0, \qquad a_{n,c} \to 0,$$

and

$$\overline{\mathbb{E}}[(\widehat{\Pi}_{c,i} - \Pi_i)^2] = O(a_{n,c}), \tag{F.9}$$

$$h_c - H = O(\sqrt{a_{n,c}}), \qquad \overline{\mathbb{E}}[\widehat{\Pi}_{c,i}^2] - H = O(\sqrt{a_{n,c}}), \quad \mathcal{A}_c = O(a_{n,c}). \tag{F.10}$$

There are constants $\underline{h} > 0$ and $C < \infty$, independent of n and c, such that, for all sufficiently large n,

$$0 < \underline{h} \leq \min_c h_c, \qquad \max_c\{h_c + \overline{\mathbb{E}}[\widehat{\Pi}_{c,i}^2]\} \leq C. \tag{F.11}$$

If Assumption 3 also holds, then, for all sufficiently large n,

$$k_c(1-\eta_{n,c}) \le \bar{d}_c \le 2(Ms_c+1), \tag{F.12}$$

and, uniformly over candidates,

$$\bar{d}_c \asymp k_c, \bar{d}_c \ \to\infty, \frac{\mathrm{Var}(d_c)}{\bar{d}_c^2} \to 0, \frac{d_c}{\bar{d}_c} \ \to_p 1, \frac{R_{d,c}}{\bar{d}_c^2} \to 1, \frac{\bar{d}_c}{\sqrt{n}} \ \to 0. \tag{F.13}$$

Under the same additional Assumption 3, uniformly over candidates,

$$S_c \asymp \mathcal{A}_c + \frac{R_{d,c}}{n} \asymp \mathcal{A}_c + \frac{\bar{d}_c^2}{n}, \qquad \max_c S_c \to 0, \qquad \min_c nS_c \to \infty. \tag{F.14}$$

Moreover, for all sufficiently large n, uniformly over candidates,

$$\sqrt{\frac{\mathcal{A}_c}{n}} + \frac{\bar{d}_c}{n} \le C\sqrt{\frac{S_c}{n}}, \quad \sqrt{\frac{\mathcal{A}_c}{n}} + \frac{\bar{d}_c}{n} \le \frac{CS_c}{\bar{d}_c}, \quad \frac{\sqrt{\bar{d}_c}}{n} \le \frac{CS_c}{\bar{d}_c^{3/2}}. \tag{F.15}$$

Proof. Column normalization, the Gaussian bound $\Phi(-u) \le \exp(-u^2/2)$ for $u \ge 0$, and Assumption 2(iii) give

$$\mathbb{P}(\mathcal{G}_c^c) \le 2p_c\Phi\left(-\frac{\sqrt{n}\lambda_c}{c_\lambda\sigma_v}\right) = \eta_{n,c} \le \{\log(p_c \vee n)\}^{-\alpha} \to 0.$$

Parts (ii) and (v) of Assumption 2 then yield $a_{n,c} \to 0$. All limits below are uniform because J is fixed.

Prediction moments. Write $U_c(v) := \|\mu_c(\Pi+v) - \Pi\|_2/\sqrt{n}$, so $\mathbb{E}[U_c^2] = \overline{\mathbb{E}}[(\widehat{\Pi}_{c,i} - \Pi_i)^2]$. By Lemma D.1, U_c is $n^{-1/2}$-Lipschitz and $U_c \le 2\sqrt{H} + \|v\|_2/\sqrt{n}$; the latter bound ensures square integrability. The Gaussian Poincaré inequality Lemma E.1 therefore gives $\mathrm{Var}(U_c) \le \sigma_v^2/n$. On $\mathcal{G}_c$, Lemma F.1 and the triangle inequality imply

$$U_c \le B_c^{\mathrm{pred}} := 3\mathbb{E}_n[\xi_{c,i}^2]^{1/2} + \frac{2(1+c_\lambda^{-1})\lambda_c\sqrt{s_c}}{\kappa_0}. \tag{F.16}$$

Since $\mathbb{P}(\mathcal{G}_c) \ge 1/2$ eventually and, when $\mathbb{E}U_c > B_c^{\mathrm{pred}}$, the event $\mathcal{G}_c$ entails $|U_c - \mathbb{E}U_c| \ge \mathbb{E}U_c - B_c^{\mathrm{pred}}$,

$$\mathbb{P}(\mathcal{G}_c)(\mathbb{E}U_c - B_c^{\mathrm{pred}})_+^2 \le \mathrm{Var}(U_c).$$

Consequently $\mathbb{E}U_c \le B_c^{\mathrm{pred}} + \sqrt{2\,\mathrm{Var}(U_c)}$, whence

$$\mathbb{E}[U_c^2] = \mathrm{Var}(U_c) + (\mathbb{E}U_c)^2 \le 2(B_c^{\mathrm{pred}})^2 + 5\sigma_v^2/n = O(a_{n,c}).$$

Here $n^{-1} = O(s_c\lambda_c^2)$ follows from Assumption 2(iii) and $s_c \geq 1$. This proves (F.9) without a separate bad-event moment bound.

Fitted moments. By (3.5), Cauchy–Schwarz, and (F.9),

$$|h_c - H|^2 \leq H\mathbb{E}[U_c^2] = O(a_{n,c}), \qquad 0 \leq \mathcal{A}_c \leq \mathbb{E}[U_c^2] = O(a_{n,c}),$$

where the bound on $\mathcal{A}_c$ follows by evaluating the minimum in (3.6) at coefficient one. Moreover,

$$\overline{\mathbb{E}}[\widehat{\Pi}_{c,i}^2] - H = 2(h_c - H) + \mathbb{E}[U_c^2] = O(\sqrt{a_{n,c}}).$$

Since H is bounded and bounded away from zero by (2.2), these bounds prove (F.10) and (F.11).

Rank moments and ratios. Suppose Assumption 3 holds. By (F.5), $\bar{d}_c \geq k_c\mathbb{P}(\mathcal{G}_c) \geq k_c(1 - \eta_{n,c}) \geq k_c/2$ eventually. Using this lower bound in Lemma E.4 and (3.4) gives

$$V_c := \frac{\mathrm{Var}(d_c)}{\bar{d}_c^2} \leq C\left\{\frac{1}{k_c^2} + \frac{\log(2ep_c/k_c)}{k_c}\right\} \to 0.$$

Indeed, $d_c \leq p_c$ implies $\bar{d}_c \leq p_c$, while $\bar{d}_c \geq k_c/2$ bounds $\log(ep_c/\bar{d}_c) \leq \log(2ep_c/k_c)$. Put $b_c := Ms_c + 1$. Since $d_c \leq b_c$ on $\mathcal{G}_c$,

$$(1 - \eta_{n,c})\left(1 - \frac{b_c}{\bar{d}_c}\right)_+^2 \leq V_c.$$

If $\bar{d}_c > 2b_c$ along a subsequence, the left side is bounded away from zero, a contradiction. Hence $\bar{d}_c \leq 2b_c$ eventually, proving (F.12); with $s_c \leq C_s k_c$, it also gives $\bar{d}_c \asymp k_c \to \infty$. Chebyshev's inequality yields $d_c/\bar{d}_c \to_p 1$. Finally,

$$\frac{R_{d,c}}{\bar{d}_c^2} = 1 + V_c + \frac{1}{\bar{d}_c} \to 1, \qquad \frac{\bar{d}_c}{\sqrt{n}} \leq \frac{2(Ms_c + 1)}{\sqrt{n}} \to 0$$

by (A.2) and Assumption 2(v), proving (F.13).

Criterion rates. The closed form (3.7), the bounds (F.11), and fixed positive σ_ε^2 and $|\sigma_{\varepsilon v}|$ give

$$S_c \asymp \mathcal{A}_c + R_{d,c}/n \asymp \mathcal{A}_c + \bar{d}_c^2/n.$$

Here $R_{d,c}/\bar{d}_c^2 \to 1$ by (F.13); $\sigma_\varepsilon^2 > 0$ follows from Assumption 1(i), and $\sigma_{\varepsilon v} \neq 0$ from (2.2). Since $\mathcal{A}_c \to 0$, $\bar{d}_c/\sqrt{n} \to 0$, and $\bar{d}_c \to \infty$ by the preceding parts, these comparisons prove (F.14).

For (F.15), the same comparison gives $\mathcal{A}_c + \bar{d}_c^2/n \leq CS_c$, and

$$2\bar{d}_c\sqrt{\mathcal{A}_c/n} \leq \mathcal{A}_c + \bar{d}_c^2/n.$$

Taking square roots, or dividing these bounds by $\bar{d}_c$ and $\bar{d}_c^{3/2}$, respectively, gives (F.15). All comparisons are uniform because J is fixed. □

G Simulation Implementation and Additional Error Design

G.1 Implementation Details

We use the credit and export data for the intensive-margin sample of Paravisini et al. (2015), with 14,208 firm–product–destination export flows from 2,091 firms. We partial out product–destination fixed effects by demeaning credit growth and every instrument within the 5,996 product–destination groups. The Original dictionary contains the ten exposure measures used for LASSO by Paravisini et al. (2015): the share of a firm's 2006 credit from banks with more than 10% foreign funding, an indicator that this share is at least one half, and credit-weighted averages of the first eight powers of its lenders' foreign-funding shares. The Expanded dictionary adds pairwise products, alternative exposure measures, threshold and bin indicators, cubic B-splines, and interactions. An instrument-only screen removes two collinear or nearly collinear columns; the resulting dictionary has 114 columns and numerical rank 109. We center each column and scale it to unit empirical root mean square.

The main first-stage signal is a least-squares refit on 17 variables selected by LASSO from the Original instruments and their pairwise products (55 variables in all). Its penalty minimizes ten-fold cross-validation error averaged over five fold assignments, each keeping observations from the same firm together. The dense signal is the least-squares fit on all 55 variables, rescaled to match the main signal's H. Neither signal uses the export outcome. We divide both signals by the 55-variable fit's residual standard deviation, 1.975, to put them on the simulated unit-error scale. We use $H \in \{0.00605, 0.0121, 0.0242, 0.0484, 0.0968\}$, where 0.0484 is the main signal's value in the data. Both signals lie in the span of the Original instruments and their pairwise products, which has numerical rank 51. The iid-equivalent expected F statistic $1 + nH/51$ takes the reported values 2.7, 4.4, 7.7, 14.5, and 28.0; it differs from the smaller firm-clustered first-stage F statistic in the data. We set $\rho = c\widehat{\rho}$, where $\widehat{\rho} = -0.218$ is the correlation of the residuals from the just-identified IV regression of log export volume on credit growth, using exposed-bank share as the instrument, and the residuals from the main-signal fit. Section 4 reports $c = 0, 0.25, \ldots, 2.75$; the additional $c = 3$ cases remain in the design-level outputs. Each design uses 1,000 replications with paired underlying error draws across designs.

For each dictionary with p_c columns, the first-stage BCCH plug-in penalty and its 37-multiplier grid are

$$\lambda_c = \frac{1.1\,\varkappa_c}{\sqrt{n}}\Phi^{-1}\left(1-\frac{\gamma_c}{2p_c}\right), \qquad \gamma_c = \frac{0.1}{\log(\max\{p_c, n\})},$$
$$\varkappa_c = 2^a, \quad a = -15, -14.5, \ldots, 3.$$

Here Φ is the standard normal distribution function. The first-stage error standard deviation is known to be one in these simulations, but the structural-error moments in the feasible criterion are estimated using the pilot below. A rule's simulated risk is the replication average of $n(\widehat{\beta}-\beta)^2$. Its relative risk divides that risk by the lowest risk among the four feasible rules in the same design: the criterion, cross-validation, cross-fitting, and BCCH. Known Moments is excluded from this denominator and can have relative risk below one.

The cross-validation rule uses one prespecified, firm-grouped ten-fold assignment. Each training fit uses the full-sample penalty level and instruments rescaled to unit root mean square within the training sample; average prediction error over held-out observations selects a candidate, which is refit on the full sample. The cross-fitting rule uses a separate prespecified, firm-grouped ten-fold assignment. For each outer fold, it selects the dictionary and penalty by ten-fold cross-validation using only the other nine folds, again splitting by firm, computes plug-in penalties using the size of that outer training sample, and fits the selected candidate there. The held-out predictions serve as instruments, and the IV numerator and denominator are pooled across folds. Thus the selected candidate may differ across outer folds. Cross-fitting's effective dimension averages fold-specific dimensions with weights proportional to held-out fold size.

The pilot uses the cross-fitting rule's folds, with a BCCH-penalized first stage on the Original dictionary in each fold and a pooled IV numerator and denominator. If all ten BCCH fits are empty, it instead uses in every fold the candidate minimizing pooled ten-fold held-out prediction error. In this exceptional branch, the pooled choice can depend on each fold's held-out observations, unlike the training-only tuning of the cross-fitting rule. When the full-sample BCCH rule selects no instruments, its reported IV estimate is replaced by the pilot estimate. These fallbacks occur only rarely in the weakest identification designs. The design-level outputs retain the full-sample BCCH fallback frequencies.

Effective dimension is the rank of instruments with nonzero LASSO coefficients; in our checks it equals the rank of the equicorrelation set in (2.8). A candidate fit with a Karush–Kuhn–Tucker violation exceeding $10^{-3}\lambda_c$ is unavailable to every rule. When full-sample BCCH selects no instruments, its effective dimension is recorded as zero even though its IV estimate is replaced by the pilot.

The common-scale Laplace errors are defined in Section 4.2. The Gaussian designs use the same underlying draws a_i and b_i, and all other implementation choices are shared. Known Moments evaluates the same feasible criterion with the structural-error moments set to their population values, $\sigma_\varepsilon^2 = 1$ and $\sigma_{\varepsilon v} = \rho$, leaving its other estimated inputs unchanged. Design-level simulation outputs and the code used to generate the reported summaries are provided in the replication materials.

G.2 Smooth-Transformation Errors

For the smooth-transformation errors defined in Section 4.2, we keep the first-stage signals, candidate dictionaries, penalty grid, and implementation choices of Section 4.1 and appendix G.1 unchanged. We apply Q to the same Gaussian draws used in the Gaussian designs and use 1,000 replications for each of the 120 designs. As in the Gaussian designs, we retain the $\rho = 0$ exogenous benchmarks.

Section 4.2 and Table 2 summarize the criterion's risk and complexity results. Cross-validation retains an advantage near the calibrated correlation at $F = 7.7$ and $F = 14.5$, while cross-fitting remains competitive under stronger identification. The unnormalized risks of these two rules change by at most 6% relative to their Gaussian counterparts. BCCH exhibits larger differences at $F = 2.7$, where a few empty-selection replications invoke the pilot fallback.

Supplemental Appendix: Non-Gaussian Errors

Yukun Ma Manu Navjeevan Bohdan Salahub

October 22, 2026

S1 Non-Gaussian Extension of the Main Results

From the main text, we retain the model in (2.1), the candidate LASSO fits and IV estimators in (2.4)–(2.5), and the effective dimension in (2.8). For the geometric arguments, extend this last definition to any response w by $d_c(w) = \operatorname{rank}(Z_{c,E_c(w)})$, with $E_c(w)$ as in the main-text (2.7); then $d_c = d_c(x)$. As in the main text, there is one endogenous regressor, and all statements condition on the realized instruments. This appendix does not provide a separate treatment of exogenous controls. The expectations and probabilities below are taken under the non-Gaussian error distribution specified here. In particular, the maintained conditional-mean restrictions give $\mathbb{E}[\varepsilon_i] = \mathbb{E}[v_i] = 0$ and $\Pi_i = \mathbb{E}[x_i]$.

S1.1 Non-Gaussian Sampling Conditions

The following conditions replace the main-text Assumption 1. For $r \in \{1, 2\}$, write

$$\|X\|_{\psi_r} = \inf\{b > 0 : \mathbb{E}[\exp(|X|^r / b^r)] \le 2\}.$$

Assumption S1 (Non-Gaussian Sampling)**.** *Conditional on the instruments, the pairs (ε_i, v_i) are independent and identically distributed with a fixed distribution, and:*

(i) For fixed $a \neq 0$, $\sigma_u^2 > 0$, and $C_u < \infty$,

$$\varepsilon_i = a v_i + u_i, \qquad \mathbb{E}[u_i \mid v_i] = 0, \qquad \mathbb{E}[u_i^2 \mid v_i] = \sigma_u^2, \qquad \mathbb{E}[|u_i|^4] \le C_u. \tag{S1}$$

(ii) There is a three times continuously differentiable function $Q : \mathbb{R} \to \mathbb{R}$ such that

$$v_i = Q(e_i), \qquad e_i \sim N(0, 1), \tag{S2}$$

with $\mathbb{E}[Q(e_i)] = 0$. For fixed $0 < c_Q \le C_Q < \infty$ and $C_{\mathrm{sm}} < \infty$,

$$c_Q \le Q'(t) \le C_Q \quad (t \in \mathbb{R}), \qquad \|Q''(e_i)\|_{\psi_2} + \|Q'''(e_i)\|_{\psi_1} \le C_{\mathrm{sm}}. \tag{S3}$$

(iii) The strong-identification bound on $H = \mathbb{E}_n[\Pi_i^2]$ in the main-text (2.2) holds. For a fixed $K_\Pi < \infty$, additionally,

$$\max_{i\in[n]} |\Pi_i| \le K_\Pi. \tag{S4}$$

Part (i) gives a linear conditional mean and constant conditional variance for the structural error given v_i; it does not require independence of u_i and v_i. The fourth-moment bound supports estimation of the structural-error moments. Write $\sigma_v^2 = \mathbb{E}[v_i^2]$, $\sigma_{\varepsilon v} = a\sigma_v^2$, and $\sigma_\varepsilon^2 = a^2\sigma_v^2 + \sigma_u^2$. Thus the error covariance is nonsingular and $\sigma_{\varepsilon v} \neq 0$, as in the main text. Part (ii) permits asymmetric, non-Gaussian first-stage errors; the Gaussian first-stage case is $Q(t) = \sigma_v t$. Its derivative bounds yield the tail, Stein-kernel, and Poincaré bounds used below (Lemma S2). The additional bound in part (iii) controls projections weighted by that kernel.

The smooth-transformation construction in Section 4.2 provides an example satisfying Assumption S1(i)–(ii) when $0 < |\rho| < 1$. In that construction, $a = \rho$, $u_i = \sqrt{1-\rho^2}\, Q(B_i)$, and $\sigma_u^2 = 1 - \rho^2$. The transformation satisfies $7/(16\nu) \le Q'(t) \le 1/\nu$, with bounded second and third derivatives.

S1.2 Changes to the First-Stage Conditions

Use the approximate-sparsity representation (3.1), the restricted eigenvalue in (3.2), and the upper and lower sparse eigenvalues defined immediately after it in the main text. Use the constants of the main-text Assumption 2. Write $L_c = \log(p_c n)$. The next assumption records exactly which parts of that assumption are retained or replaced.

Assumption S2 (Non-Gaussian First-Stage Conditions)**.** *For every $c \in [J]$, with constants common to all candidates:*

(i) The column normalization in the main-text Assumption 2(i) holds, and, for a fixed $K_Z < \infty$, $\max_{i,j} |z_{c,ij}| \le K_Z$.

(ii) The approximate-sparsity bound in the main-text Assumption 2(ii) holds.

(iii) In place of the main-text Assumption 2(iii), the penalty satisfies

$$\lambda_c \ge c_\lambda C_Q \sqrt{\frac{2\{\log(2p_c) + 20\log n\}}{n}}. \tag{S5}$$

(iv) The restricted and sparse eigenvalue bounds in the main-text Assumption 2(iv) hold.

(v) In place of the main-text Assumption 2(v), require

$$\sqrt{s_c}\lambda_c \to 0, \qquad \frac{L_c}{s_c} \to 0, \qquad \frac{s_c^2 L_c}{n} \to 0. \tag{S6}$$

The entry bound in part (i) controls weighted Gram matrices. The penalty bound in part (iii) makes the score event hold with probability at least $1 - n^{-20}$, allowing integration of first-stage bounds over its complement. Part (v) controls terms from the nonconstant Stein kernel and implies the main-text rate $s_c/\sqrt{n} \to 0$. We also impose Assumption 3 exactly as stated in the main text, using the same $C_a, \kappa_0, \underline{\phi}$ and threshold, strong-coordinate set A_c, and size k_c; no support-recovery condition is added. When a candidate subscript is suppressed in the proofs, $T^0 = T_c$ denotes the support in (3.1), reserving T for the Stein-kernel matrix.

S1.3 Extension of the Main Results

Define h_c, $\mathcal{A}_c$, and S_c by (3.5), (3.6), and (3.7), respectively, with moments under Assumption S1; $R_{d,c} = \mathbb{E}[d_c^2 + d_c]$ is the shorthand in (A.2). Use the residual moment estimators, feasible criterion, selected candidate, and common centering B_n in (3.11), (3.12), and (3.15). Thus both criteria have exactly their main-text forms, with their population moments now evaluated under the non-Gaussian law. The positive-penalty candidates have finite exact MSE at each fixed n by the KKT denominator bound, the zero-fit convention, and $\mathbb{E}[\varepsilon_i^2] < \infty$ under Assumption S1. The decomposition and selection result concern AMSE rather than exact finite-sample MSE.

Theorem S1 (Non-Gaussian Extension)**.** *Suppose Assumptions S1 and S2 and the main-text Assumption 3 hold, with $J < \infty$ fixed.*

(i) Infeasible AMSE criterion. *For all sufficiently large n, use the random variables C_0, $\{Q_c\}_{c\in[J]}$, and $\{r_c\}_{c\in[J]}$ from (B.4). For every candidate,*

$$n(\widehat{\beta}_c - \beta)^2 = C_0 + Q_c + r_c, \tag{S7}$$

where C_0 is common to all candidates and $\mathbb{E}[C_0] = \sigma_\varepsilon^2/H + O(n^{-1})$. The expectation $\bar{Q}_c$ is taken under the non-Gaussian law, and the remainder satisfies

$$\max_{c\in[J]} \frac{|r_c|}{\bar{Q}_c} \to_p 0.$$

Uniformly over candidates, $\bar{Q}_c > 0$ eventually and

$$\bar{Q}_c = S_c\{1 + o(1)\}. \tag{S8}$$

(ii) Feasible selection. *If the pilot satisfies* $\check{\beta} - \beta = O_p(n^{-1/2})$, *then*

$$\max_{c\in[J]} \frac{|\widehat{S}_c - B_n - S_c|}{S_c} \to_p 0, \tag{S9}$$

$$\max_{c\in[J]} \frac{|\widehat{S}_c - B_n - \bar{Q}_c|}{\bar{Q}_c} \to_p 0, \tag{S10}$$

$$\frac{\bar{Q}_{\widehat{c}}}{\min_{c\in[J]} \bar{Q}_c} \to_p 1. \tag{S11}$$

The pilot may be the IV-LASSO estimate from a possibly data-dependent candidate in $[J]$ (Lemma S12). Proofs of both parts are in Appendix S3.

S2 Notation and Preliminary Reductions

We write C for a constant that depends only on the fixed constants in the assumptions and whose value may change between displays. Since J is fixed, a rate or limit established for each candidate holds uniformly over $c \in [J]$, and we use this without further comment.

Squared-error decomposition. We use Γ_c, Δ_c, L_n, q, Λ_c, C_0, Q_c, and r_c as defined in (B.1)–(B.4) in the proof of the main-text Theorem 1, with expectations now taken under Assumption S1. The decomposition in (S7) is therefore the same identity as in the main appendix.

Positive denominator event. By (S31), $h_c \to H$, and $\underline{H} \le H \le \overline{H}$ by the main-text bound (2.2), so that $\underline{H}/2 \le h_c \le 2\overline{H}$ for every candidate and all sufficiently large n. In place of the event used in the main appendix, we work on

$$\mathcal{H}_n := \left\{ \min_{c\in[J]} \widehat{h}_c \ge \underline{H}/4 \right\},$$

on which every sample denominator is bounded away from zero. Since $\widehat{h}_c = 0$ whenever $\widehat{\Pi}_c = 0$, every $\widehat{\beta}_c$ is the ratio in the main-text (2.5) on $\mathcal{H}_n$. Lemma S8 shows that $\mathbb{P}(\mathcal{H}_n) \to 1$.

Population projection residual. For a single candidate, with P as in (S16), write

$$m = \mu(\Pi + v) - \frac{h}{H}\Pi \tag{S12}$$

for the residual from the population projection in the main-text identity (3.6). This is the residual M_c of the main appendix, with the fit $\widehat{\Pi}_c = \mu_c(\Pi+v)$ written as μ, as in Lemma S3.

The definition of h, that identity, and (S16) give

$$\mathbb{E}[\|m\|_2^2] = n\mathcal{A}, \qquad Dm = P \quad \text{almost everywhere.} \tag{S13}$$

Let T and $D_\tau = \operatorname{tr}(TP)$ be the diagonal Stein-kernel matrix and the kernel-weighted rank in (S17). For Gaussian first-stage errors, $T = \sigma_v^2 I$ and $D_\tau = \sigma_v^2 d$, and the display below reduces to its Gaussian counterpart (A.4). Taking $b = (h/H)\Pi$ in (S27), and bounding the two terms with (S18) and $P^2 = P$, together with (S31) and (S13), we have

$$\mathbb{E}[(v'm - D_\tau)^2] = \mathbb{E}[m'Tm + \operatorname{tr}(TPTP)] \le C(n\mathcal{A} + s). \tag{S14}$$

S3 Proofs of the Main Results

S3.1 Proof of Theorem S1(i)

Proof. The proof follows the four steps of the proof of the main-text Theorem 1. Lemma S7 replaces (E.8) and Lemma B.2. Similarly, Lemmas S8 and S9 replace Lemmas B.1 and B.3, while Lemma S10 replaces the Gaussian moments of L_n and C_0.

Step 1: $\sqrt{n}(\widehat{\beta}_c - \beta) = \Lambda_c + o_p(S_c)$. Step 1 of the Gaussian proof uses $\Delta_c = O_p(n^{-1/2})$, $\mathbb{E}[(\Gamma_c - L_n)^2] \le CS_c$, $\mathbb{E}[L_n^2] = \sigma_\varepsilon^2/H$, $\max_c S_c \to 0$, $\min_c nS_c \to \infty$, and the fact that $1 + \Delta_c = \widehat{h}_c/h_c$ is bounded away from zero on $\mathcal{H}_n$. The first five hold by Lemmas S7, S8 and S10, and the last holds because $\widehat{h}_c \ge \underline{H}/4$ on $\mathcal{H}_n$, while $h_c \le 2\overline{H}$ for all sufficiently large n. The argument of that step therefore proves the claim and also shows that $\max_c |\Lambda_c| = O_p(1)$ and $\max_c |\sqrt{n}(\widehat{\beta}_c - \beta)| = O_p(1)$.

Step 2: $\mathbb{E}[\Gamma_c^2] = \sigma_\varepsilon^2/H + S_c + o(S_c)$. With the weighted Stein identities of Lemma S3 in place of the Gaussian ones, Lemma S7 shows that $\varepsilon'\widehat{\Pi}_c$ has the Gaussian second moment (E.8) up to additional kernel terms of order $o(n\mathcal{A}_c + s_c^2)$. Dividing (S40) by nh_c^2 and substituting $\mathbb{E}\|\widehat{\Pi}_c\|_2^2/n = \mathcal{A}_c + h_c^2/H$ from the main-text identity (3.6),

$$\mathbb{E}[\Gamma_c^2] = \frac{\sigma_\varepsilon^2}{H} + \frac{\sigma_\varepsilon^2 \mathcal{A}_c}{h_c^2} + \frac{\sigma_{\varepsilon v}^2 \mathbb{E}[d_c^2 + d_c]}{nh_c^2} + \frac{o(n\mathcal{A}_c + s_c^2)}{nh_c^2} = \frac{\sigma_\varepsilon^2}{H} + S_c + o(S_c),$$

where the last equality uses $S_c \asymp \mathcal{A}_c + s_c^2/n$ from Lemma S7 and $h_c \ge \underline{H}/2$.

Step 3: $\bar{Q}_c = S_c\{1 + o(1)\}$. Expanding Λ_c^2 as in Step 3 of the proof of the main-text Theorem 1,

$$\mathbb{E}[\Lambda_c^2] - \mathbb{E}[\Gamma_c^2] = -2\mathbb{E}[\Delta_c L_n^2] - 2\mathbb{E}[\Delta_c L_n(\Gamma_c - L_n)] + \mathbb{E}[\Delta_c^2 L_n^2].$$

By Lemma S9, $|\mathbb{E}[\Delta_c L_n^2]| \le C\{\sqrt{S_c/n} + 1/n\}$ and $\mathbb{E}[\Delta_c^2 L_n^2] \le C/n$. Cauchy–Schwarz and

Lemma S7 then bound the middle term by $C\sqrt{S_c/n}$. Meanwhile, $\mathbb{E}[C_0] = \sigma_\varepsilon^2/H + O(n^{-1})$ by Lemma S10. Combining these bounds with Step 2,

$$|\bar{Q}_c - S_c| \le \left|\mathbb{E}[\Lambda_c^2] - \mathbb{E}[\Gamma_c^2]\right| + \left|\mathbb{E}[\Gamma_c^2] - \frac{\sigma_\varepsilon^2}{H} - S_c\right| + \left|\frac{\sigma_\varepsilon^2}{H} - \mathbb{E}[C_0]\right|$$
$$\le C\left\{\sqrt{\frac{S_c}{n}} + \frac{1}{n}\right\} + o(S_c),$$

which is $o(S_c)$ because $\min_c nS_c \to \infty$. Dividing by S_c, we have $\max_c |\bar{Q}_c/S_c - 1| \to 0$, so that $\bar{Q}_c > 0$ for all sufficiently large n, which is (S8).

Step 4: $\max_c |r_c|/\bar{Q}_c \to_p 0$. This follows from Steps 1 and 3 by the argument in Step 4 of the proof of the main-text Theorem 1. □

S3.2 Proof of Theorem S1(ii)

Proof. The proof follows the four steps of the proof of the main-text Theorem 2, with Lemma S11, Lemma S12, Lemma S6, and Lemma S8 in place of the main-text Lemma C.1, Lemma C.2, Proposition 1, and Lemma B.1, respectively. We work on $\mathcal{H}_n$, whose probability tends to one by Lemma S8.

Step 1: $\widehat{\sigma}_\varepsilon^2 \mathbb{E}_n[\widehat{\Pi}_{c,i}^2]/\widehat{h}_c^2 = \sigma_\varepsilon^2 \mathcal{A}_c/h_c^2 + B_n + o_p(S_c)$. The decomposition (C.2) is an algebraic identity, and the rest of Step 1 of the Gaussian proof uses only the following facts. Lemma S11 shows that $\widehat{\mathcal{A}}_c - \mathcal{A}_c$ and $\widehat{q}_c - q$ are $o_p(S_c)$, while $\widehat{h}_c - h_c$ and $\widehat{\sigma}_\varepsilon^2 - \sigma_\varepsilon^2$ are $O_p(n^{-1/2})$ by Lemmas S8 and S12. By independence, $\mathbb{E}[q^2] = \sigma_v^2/(nH)$. Finally, $\mathcal{A}_c \le CS_c$ and $\min_c nS_c \to \infty$ by Lemma S7. With these facts, the Gaussian argument proves Step 1.

Step 2: $\widehat{\sigma}_{\varepsilon v}^2(d_c^2 + d_c)/(n\widehat{h}_c^2) = \sigma_{\varepsilon v}^2 \mathbb{E}[d_c^2 + d_c]/(nh_c^2) + o_p(S_c)$. Of the last three factors in (C.3), the first converges to one in probability by Lemma S12 and $\sigma_{\varepsilon v} = a\sigma_v^2 \neq 0$, the second by Lemma S6, and the third by Lemma S8. Since the leading factor is at most S_c, the claim follows.

Step 3: (S9) *and* (S10). Adding Steps 1 and 2, we obtain $\widehat{S}_c = B_n + S_c + o_p(S_c)$, which is (S9). Since $\max_c |\bar{Q}_c/S_c - 1| \to 0$ by (S8), adding and subtracting S_c in $\widehat{S}_c - B_n - \bar{Q}_c$ yields (S10).

Step 4: (S11). By (S8), every $\bar{Q}_c$ is positive for all sufficiently large n. Define ω_n as in the proof of the main-text Theorem 2, with $\bar{Q}_c$ now taken under Assumption S1. Then $\omega_n \to_p 0$ by Step 3, and on $\{\omega_n < 1\}$ the comparison leading to (C.4) gives

$$0 \le \frac{\bar{Q}_{\widehat{c}}}{\min_c \bar{Q}_c} - 1 \le \frac{2\omega_n}{1 - \omega_n}.$$

Since $\mathbb{P}(\omega_n < 1) \to 1$, the bound proves (S11). □

S4 Conditional Moments, Geometry, and Stein Identities

Lemma S1 (Conditional Moments of the Structural Error)**.** *Under Assumption S1, for any measurable vectors $b(v), g(v)$ for which the expectations exist,*

$$\mathbb{E}[\varepsilon' b \mid v] = av'b, \quad \mathbb{E}[(\varepsilon' b)(\varepsilon' g) \mid v] = a^2(v'b)(v'g) + \sigma_u^2 b'g. \tag{S15}$$

Proof. Independence of the pairs and (S1) imply $\mathbb{E}[u \mid v] = 0$ and $\mathbb{E}[uu' \mid v] = \sigma_u^2 I_n$. Thus $\mathbb{E}[\varepsilon \mid v] = av$ and $\mathbb{E}[\varepsilon\varepsilon' \mid v] = a^2 vv' + \sigma_u^2 I_n$. Multiplying by the v-measurable vectors b and g gives (S15). □

LASSO geometry. We use the deterministic fit properties in Lemma D.1 and the Borel coefficient selection in Lemma D.2. For a fixed candidate, define $P(w) := P_{\mathrm{col}(Z_{E(w)})}$ for every w; this map is Borel because the continuous residual determines $E(w)$ among finitely many subsets. Suppressing the candidate subscript, write $P := P(\Pi + v)$, $d := d(\Pi + v)$, and $\mu := \mu(\Pi + v)$ when no argument is shown. Under Assumption S1, $\Pi + v$ has a Lebesgue density, so (D.5) gives, almost surely,

$$D\mu(\Pi + v) = P, \qquad P = P' = P^2, \qquad P\mu = \mu, \qquad \operatorname{tr} P = d. \tag{S16}$$

Whenever coefficients are needed, $\widehat{\pi}(w)$ denotes the selection in Lemma D.2. Its support is contained in $E(w)$ by (D.1), and its selected columns are linearly independent by that lemma. Thus its support size is at most $\operatorname{rank}(Z_{E(w)}) = d(w)$.

Let p denote the density of v_i, which exists and is positive because $Q' > 0$. Define the canonical Stein kernel and its diagonal matrix by

$$\tau(t) = \frac{\int_t^\infty z p(z)\, dz}{p(t)}, \quad T = \operatorname{diag}\{\tau(v_i)\}, \quad W = T - \sigma_v^2 I, \quad D_\tau = \operatorname{tr}(TP). \tag{S17}$$

With subscripts restored, write $P_c := P_{\mathrm{col}(Z_{c,E_c(x)})}$ and $D_{\tau,c} := \operatorname{tr}(TP_c)$; by (S16), $D\mu_c(x) = P_c$ almost surely. The support T_c in (3.1) is distinct from the diagonal kernel matrix T in (S17).

Lemma S2 (Properties of the First-Stage Errors)**.** *Under Assumption S1, p is positive and twice continuously differentiable, and τ is continuously differentiable. The kernel satisfies*

$$c_Q^2 \le \tau(t) \le C_Q^2, \quad \mathbb{E}[\tau(v_i)] = \sigma_v^2, \quad (\tau p)'(t) = -t\, p(t). \tag{S18}$$

For every $b \in \mathbb{R}^n$ and $r > 0$,

$$\mathbb{E}e^{b'v} \le e^{C_Q^2 \|b\|_2^2/2}, \quad \mathbb{P}(|b'v| > r) \le 2e^{-r^2/(2C_Q^2\|b\|_2^2)} \quad (b \neq 0). \tag{S19}$$

For locally Lipschitz f with square-integrable weak gradient,

$$\mathrm{Var}(f(v)) \le \mathbb{E}[\nabla f(v)' T \nabla f(v)] \le C_Q^2 \mathbb{E}[\|\nabla f(v)\|_2^2]. \tag{S20}$$

Finally, if $f : \mathbb{R} \to \mathbb{R}$ is locally absolutely continuous and $\mathbb{E}|v_i f(v_i)| + \mathbb{E}|\tau(v_i) f'(v_i)| < \infty$, then

$$\mathbb{E}[v_i f(v_i)] = \mathbb{E}[\tau(v_i) f'(v_i)]. \tag{S21}$$

Proof. Step 1: Density and kernel bounds.

Let φ denote the standard normal density and put $q_0 = Q(0)$. Since $c_Q \le Q' \le C_Q$,

$$c_Q|z| \le |Q(z) - q_0| \le C_Q|z|, \qquad Q(\mathbb{R}) = \mathbb{R}.$$

The change-of-variable formula gives

$$p(Q(z)) = \frac{\varphi(z)}{Q'(z)}. \tag{S22}$$

Thus p is positive and twice continuously differentiable, and

$$\frac{e^{-(t-q_0)^2/(2c_Q^2)}}{C_Q\sqrt{2\pi}} \le p(t) \le \frac{e^{-(t-q_0)^2/(2C_Q^2)}}{c_Q\sqrt{2\pi}}. \tag{S23}$$

In particular, $\int (1+|t|^j) e^{at} p(t)\, dt < \infty$ for all fixed $j \ge 0$ and $a \in \mathbb{R}$.

For independent standard Gaussian $e, \widetilde{e}$, and nondecreasing F, G with the following expectations finite,

$$\mathrm{Cov}(F(e), G(e)) = \tfrac{1}{2}\mathbb{E}[(F(e) - F(\widetilde{e}))(G(e) - G(\widetilde{e}))] \ge 0.$$

Apply this identity with $G(e) = \mathbb{1}_{\{e>z\}}$ and, in turn, $F(e) = Q(e) - c_Q e$ and $F(e) = C_Q e - Q(e)$. Both choices of F are centered and nondecreasing. Since $\mathbb{E}[e\mathbb{1}_{\{e>z\}}] = \varphi(z)$,

for every $z \in \mathbb{R}$,

$$c_Q\varphi(z) \le \int_z^\infty Q(t)\varphi(t)\,dt \le C_Q\varphi(z), \quad \tau(Q(z)) = \frac{Q'(z)}{\varphi(z)} \int_z^\infty Q(t)\varphi(t)\,dt.$$

Hence $c_Q^2 \le \tau(Q(z)) \le C_Q^2$ for all z, as required. The definition of τ gives $(\tau p)'(t) = -t\,p(t)$. Since p is positive and twice continuously differentiable, τ is continuously differentiable.

Step 2: First-order identity and tails.

For compactly supported, locally absolutely continuous f,

$$\mathbb{E}[v_i f(v_i)] = -\int_{\mathbb{R}} f(t)(\tau p)'(t)\,dt = \int_{\mathbb{R}} f'(t)\tau(t)p(t)\,dt.$$

For general f satisfying the stated integrability conditions, choose smooth χ_R equal to one on $[-R, R]$, zero outside $[-2R, 2R]$, and with $|\chi_R'| \le C/R$. Applying the preceding identity to $f\chi_R$ gives

$$\mathbb{E}[v_i f(v_i)\chi_R(v_i)] = \mathbb{E}[\tau(v_i)f'(v_i)\chi_R(v_i)] + \mathbb{E}[\tau(v_i)f(v_i)\chi_R'(v_i)].$$

The last term satisfies

$$|\mathbb{E}[\tau(v_i)f(v_i)\chi_R'(v_i)]| \le \frac{C}{R^2}\mathbb{E}[|v_i f(v_i)|\mathbb{1}_{\{|v_i|\ge R\}}] \longrightarrow 0.$$

Dominated convergence proves (S21). At $f(t) = t$, it gives $\mathbb{E}\tau(v_i) = \sigma_v^2$. The density envelope (S23) ensures all exponential moments exist. Since $\mathbb{E}[v_i] = 0$ and $\|\tau\|_\infty \le C_Q^2$, the Laplace-transform bound in the proof of Theorem 14 of Saumard (2018), applied to the one-Lipschitz functions $t \mapsto t$ and $t \mapsto -t$, gives

$$\mathbb{E}e^{tv_i} \le e^{C_Q^2 t^2/2}, \qquad t \in \mathbb{R}.$$

Independence yields $\mathbb{E}e^{b'v} = \prod_i \mathbb{E}e^{b_i v_i} \le e^{C_Q^2\|b\|_2^2/2}$. The two-sided Chernoff bound gives (S19) for $b \ne 0$; for $b = 0$, the tail probability is zero.

Step 3: Weighted Poincaré inequality.

The density is positive on $\mathbb{R}$ and $\mathbb{E}|v_i| < \infty$. Thus Theorem 6 of Saumard (2018) gives

$$\mathrm{Var}(f(v_i)) \le \mathbb{E}[\tau(v_i)f'(v_i)^2], \qquad f \in L^2(p) \text{ absolutely continuous.}$$

For a smooth compactly supported function of n variables, let $v_{-1} = (v_2, \ldots, v_n)$. Condi-

tional variance, induction, and Jensen's inequality give

$$\begin{aligned}\operatorname{Var}(f(v)) &= \mathbb{E}[\operatorname{Var}(f(v) \mid v_{-1})] + \operatorname{Var}\{\mathbb{E}[f(v) \mid v_{-1}]\} \\ &\le \mathbb{E}[\tau(v_1)(\partial_1 f)^2] + \sum_{j=2}^{n} \mathbb{E}[\tau(v_j)\{\mathbb{E}[\partial_j f \mid v_{-1}]\}^2] \\ &\le \sum_{j=1}^{n} \mathbb{E}[\tau(v_j)(\partial_j f)^2] \le C_Q^2 \mathbb{E}[\|\nabla f\|_2^2].\end{aligned}$$

For locally Lipschitz f with square-integrable weak gradient, put $f_N = (-N) \vee (f \wedge N)$. Spatial cutoffs and local convolution extend the inequality to each bounded f_N, with $\|\nabla f_N\|_2 \le \|\nabla f\|_2$ almost everywhere. Thus $\sup_N \operatorname{Var}(f_N) \le C_Q^2 \mathbb{E}\|\nabla f\|_2^2$. On a fixed ball A with positive probability, $|f_N| \le M_A$, so

$$\mathbb{P}(A)(|\mathbb{E} f_N| - M_A)_+^2 \le \operatorname{Var}(f_N).$$

Hence $\sup_N \mathbb{E}[f_N^2] < \infty$, and Fatou's lemma gives $f \in L^2$. Passing to the limit in the truncations proves (S20). □

For a locally Lipschitz vector field m, set

$$\mathfrak{D}(m) = v'm - \sum_i \tau(v_i)\partial_i m_i. \tag{S24}$$

Derivatives refer to weak derivatives, defined almost everywhere.

Lemma S3 (Weighted Stein Identities)**.** *Under Assumption S1, let m, g and their first derivatives have polynomial growth, with $m : \mathbb{R}^n \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}$ locally Lipschitz. Then*

$$\begin{aligned}\mathbb{E}[\mathfrak{D}(m)] &= 0, \\ \mathbb{E}[\mathfrak{D}(m)g] &= \mathbb{E}[m'T\nabla g], \qquad \text{(S25)} \\ \mathbb{E}[\mathfrak{D}(m)^2] &= \mathbb{E}\{m'Tm + \operatorname{tr}(TDm\,TDm)\}. \qquad \text{(S26)}\end{aligned}$$

In particular, for $m(v) = \mu(\Pi + v) - b$ and nonrandom b,

$$\mathbb{E}[v'm] = \mathbb{E}[D_\tau], \quad \mathbb{E}[(v'm - D_\tau)^2] = \mathbb{E}[m'Tm + \operatorname{tr}(TPTP)]. \tag{S27}$$

Proof. Step 1: Coordinate and covariance identities.

Equation (S21), conditional on the other coordinates, gives

$$\mathbb{E}[v_i f(v)] = \mathbb{E}[\tau(v_i)\partial_i f(v)].$$

This holds for locally Lipschitz f whenever both sides are absolutely integrable, by Fubini's theorem. For smooth compactly supported m and continuously differentiable compactly supported g, the product rule gives

$$\begin{aligned}\mathbb{E}[\mathfrak{D}(m)g] &= \sum_i \mathbb{E}[v_i m_i g - \tau(v_i)(\partial_i m_i)g] \\ &= \sum_i \mathbb{E}[\tau(v_i)\partial_i(m_i g) - \tau(v_i)(\partial_i m_i)g] \\ &= \sum_i \mathbb{E}[\tau(v_i) m_i \partial_i g] = \mathbb{E}[m' T \nabla g].\end{aligned}$$

Applying the coordinate identity to each m_i also gives $\mathbb{E}\mathfrak{D}(m) = 0$.

Step 2: Second moments for smooth fields.

Take m smooth and compactly supported. Since τ is continuously differentiable, all terms involving τ' are integrable on this support. Moreover, $\mathfrak{D}(m)$ is continuously differentiable and compactly supported, so Step 1 applies with $g = \mathfrak{D}(m)$. Write $m_{j,i} = \partial_i m_j$, $\tau_i = \tau(v_i)$, and $\tau_i' = \tau'(v_i)$. By Step 1,

$$\begin{aligned}\mathbb{E}[\mathfrak{D}(m)^2] &= \mathbb{E}\sum_i \tau_i m_i \partial_i \left[\sum_j \{v_j m_j - \tau_j m_{j,j}\}\right] \\ &= \mathbb{E}\sum_i \tau_i m_i^2 + \mathbb{E}\sum_{i,j} \tau_i m_i v_j m_{j,i} \\ &\quad - \mathbb{E}\sum_i \tau_i \tau_i' m_i m_{i,i} - \mathbb{E}\sum_{i,j} \tau_i \tau_j m_i m_{j,ji}.\end{aligned}$$

Integration by parts in coordinate j gives

$$\begin{aligned}\mathbb{E}\sum_{i,j} \tau_i m_i v_j m_{j,i} &= \mathbb{E}\sum_{i,j} \tau_j \partial_j(\tau_i m_i m_{j,i}) \\ &= \mathbb{E}\sum_{i,j} \tau_i \tau_j m_{i,j} m_{j,i} + \mathbb{E}\sum_i \tau_i \tau_i' m_i m_{i,i} \\ &\quad + \mathbb{E}\sum_{i,j} \tau_i \tau_j m_i m_{j,ij}.\end{aligned}$$

Since $m_{j,ij} = m_{j,ji}$, substitution yields

$$\mathbb{E}[\mathfrak{D}(m)^2] = \mathbb{E}\sum_i \tau_i m_i^2 + \mathbb{E}\sum_{i,j} \tau_i\tau_j m_{i,j} m_{j,i} = \mathbb{E}\{m'Tm + \operatorname{tr}(TDm\,TDm)\}.$$

Step 3: Extension to locally Lipschitz fields.

Fix n, write $q_n(v) = \prod_i p(v_i)$, and put

$$w_0(v) = 1 + \|v\|_2^2.$$

For every fixed $k \geq 0$, (S19) gives

$$\mathbb{E}[w_0(v)(1 + \|v\|_2^k)] < \infty. \tag{S28}$$

Choose smooth cutoffs χ_R equal to one on $\{\|v\|_2 \leq R\}$, zero on $\{\|v\|_2 \geq 2R\}$, and with $\|\nabla\chi_R\|_2 \leq C/R$. For a smooth compactly supported approximate identity φ_δ, set

$$m_R = \chi_R m, \qquad m_{R,\delta} = \varphi_\delta * m_R,$$

and define $g_R, g_{R,\delta}$ in the same way. On compact sets, w_0 is bounded and q_n is bounded above and below by positive constants. Hence, for fixed R,

$$\int \Big\{ w_0 \|m_{R,\delta} - m_R\|_2^2 + \|Dm_{R,\delta} - Dm_R\|_F^2 \Big\}\, q_n\, dv \longrightarrow 0 \text{ as } \delta \downarrow 0.$$

Polynomial growth and (S28) give, by dominated convergence, including the cutoff-gradient term $m(\nabla\chi_R)'$,

$$\int \Big\{ w_0 \|m_R - m\|_2^2 + \|Dm_R - Dm\|_F^2 \Big\}\, q_n\, dv \longrightarrow 0 \text{ as } R \to \infty.$$

The same limits hold for g and its gradient. Choose a diagonal sequence m_k, g_k of these smooth approximations. Write $\|h\|_{L^2(w_0)}^2 = \mathbb{E}[w_0\|h\|_2^2]$, and omit w_0 for the ordinary L^2 norm; use Frobenius norms for matrices. Then

$$\|\mathfrak{D}(m_k) - \mathfrak{D}(m)\|_{L^2} \leq \|m_k - m\|_{L^2(w_0)} + C_Q^2\sqrt{n}\|Dm_k - Dm\|_{L^2} \longrightarrow 0.$$

Boundedness of T and Cauchy–Schwarz also give

$$\begin{aligned}\mathbb{E}\,\big|\operatorname{tr}(TDm_k\,TDm_k) - \operatorname{tr}(TDm\,TDm)\big| \\ \leq C\left(\|Dm_k\|_{L^2} + \|Dm\|_{L^2}\right)\|Dm_k - Dm\|_{L^2} \longrightarrow 0,\end{aligned}$$

$$\mathbb{E}[m_k' T m_k] \to \mathbb{E}[m' T m], \quad \mathbb{E}[\mathfrak{D}(m_k) g_k] \to \mathbb{E}[\mathfrak{D}(m) g], \quad \mathbb{E}[m_k' T \nabla g_k] \to \mathbb{E}[m' T \nabla g].$$

These limits prove the weighted identities.

For the offset fit, Lemma D.1 and (S16) give

$$\|m\|_2 \le \|\Pi + v\|_2 + \|b\|_2, \qquad Dm = P \quad \text{almost everywhere}, \qquad \|Dm\|_{\mathrm{op}} \le 1.$$

The product density assigns zero probability to the exceptional Lebesgue-null set in Lemma D.1. Substitution in (S25) and (S26) proves (S27). □

S5 Sparse Fits, Uniform Projections, and Effective Dimension

Let

$$\mathcal{G}_c = \{\|Z_c' v/n\|_\infty \le \lambda_c / c_\lambda\}. \tag{S29}$$

Lemma S4 (LASSO Rates under Non-Gaussian Errors)**.** *Under Assumptions S1 and S2,* $\mathbb{P}(\mathcal{G}_c^c) \le n^{-20}$*. On* $\mathcal{G}_c$*, eventually,*

$$\begin{aligned} \|Z_c(\widehat{\pi}_c - \pi_c^0)\|_2/\sqrt{n} &\le 2C_a\sqrt{s_c/n} + 2(1 + c_\lambda^{-1})\lambda_c\sqrt{s_c}/\kappa_0, \\ d_c < \lceil M s_c \rceil, \qquad \|\widehat{\pi}_c - \pi_c^0\|_2 &\le \underline{b}_{n,c}/2. \end{aligned} \tag{S30}$$

The coefficient bound uses the Borel solution in Lemma D.2. Under Assumption 3, $A_c \subseteq \mathrm{supp}(\widehat{\pi}_c)$ *with correct signs, and* $k_c \le d_c$ *on this event. Furthermore, with* $a_{n,c} = s_c\lambda_c^2 + s_c/n + n^{-10}$*,*

$$\begin{aligned} \mathbb{E}[\|\widehat{\Pi}_c - \Pi\|_2^2]/n \le C a_{n,c}, \quad |h_c - H| \le C\sqrt{a_{n,c}}, \\ 0 \le \mathcal{A}_c \le C a_{n,c} = o(1), \quad \mathbb{E}[d_c^r] \le C s_c^r \quad (1 \le r \le 8). \end{aligned} \tag{S31}$$

In particular, h_c *is bounded away from zero eventually and* $\mathbb{E}[d_c] \asymp s_c$ *under Assumption 3.*

Proof. Suppress the candidate subscript. By normalization, (S19), and (S5),

$$\mathbb{P}(\mathcal{G}^c) \le 2p \exp\{-n\lambda^2/(2c_\lambda^2 C_Q^2)\} \le n^{-20}.$$

Put $e = \|\xi\|_2/\sqrt{n}$ and $\theta = c_\lambda^{-1}$. Assumption S2(ii) and (S5) give

$$\frac{e}{\lambda\sqrt{s}} \le \frac{C_a}{c_\lambda C_Q \sqrt{2\{\log(2p) + 20\log n\}}} \to 0. \tag{S32}$$

The basic-inequality, rank, and coefficient arguments in the proof of Lemma F.1 are de-

terministic on $\mathcal{G}$: their approximation, restricted eigenvalue, and sparse eigenvalue inputs hold here by Assumption S2(ii),(iv) and the preceding ratio bound. They yield (S30). The coefficient bound uses the Borel solution in Lemma D.2; under Assumption 3, the same argument gives the asserted inclusion, signs, and $k \le d$.

For the moments, let $U(v) = \|\mu(\Pi + v) - \Pi\|_2/\sqrt{n}$. By Lemma D.1, U is $n^{-1/2}$-Lipschitz and $U(v) \le 2\sqrt{H} + \|v\|_2/\sqrt{n}$, so it is square-integrable. The non-Gaussian Poincaré bound (S20) gives $\mathrm{Var}(U) \le C_Q^2/n$. On $\mathcal{G}$, the deterministic prediction bound above gives $U \le B := 3e + 2(1+\theta)\lambda\sqrt{s}/\kappa_0$. Thus

$$\mathbb{P}(\mathcal{G})(\mathbb{E}U - B)_+^2 \le \mathrm{Var}(U), \qquad \mathbb{E}[U^2] \le 2B^2 + 5C_Q^2/n \le C\{s/n + s\lambda^2 + n^{-1}\} \le Ca_n,$$

where $\mathbb{P}(\mathcal{G}) \ge 1/2$ eventually and $s \ge 1$. By the definitions of $h, \mathcal{A}$ and Cauchy–Schwarz,

$$|h - H|^2 \le H\mathbb{E}[U^2] \le Ca_n, \qquad 0 \le \mathcal{A} \le \mathbb{E}[U^2] \le Ca_n = o(1),$$

where the limit follows from (S6).

The exceptional-event bound remains useful below. By (S19), $\mathbb{E}[(n + \|v\|_2^2)^{2r}] \le C_r n^{2r}$ for $1 \le r \le 4$; hence Cauchy–Schwarz gives, whenever $\mathbb{P}(\mathcal{B}) \le Cn^{-20}$,

$$\mathbb{E}[(n + \|v\|_2^2)^r \mathbb{1}_{\mathcal{B}}] \le C_r n^{r-10}. \tag{S33}$$

Finally, $d = \mathrm{rank}(Z_E) \le n$, so for $1 \le r \le 8$,

$$\mathbb{E}[d^r] \le (Ms+1)^r + n^r \mathbb{P}(\mathcal{G}^c) \le Cs^r.$$

Under Assumption 3,

$$k(1 - n^{-20}) \le \mathbb{E}d \le Cs, \qquad k \le s \le C_s k,$$

so $\mathbb{E}d \asymp s$. Also $h \ge H - C\sqrt{a_n}$ is bounded away from zero eventually. □

For a deterministic subset $B \subseteq [p_c]$, let $P_{c,B}$ be projection onto $\mathrm{col}(Z_{c,B})$, whether or not its columns are independent.

Lemma S5 (Projections onto Sparse Sets of Columns)**.** *Under Assumptions S1 and S2, set* $m_c^* = \lceil (M+1)s_c \rceil \wedge p_c$. *For* $z = v$ *and for* $z = W\Pi$, *and every fixed* $r \in [1, 8]$,

$$\mathbb{E}\left[\max_{|B| \le m_c^*} \|P_{c,B} z\|_2^r\right] \le C_r (s_c L_c)^{r/2}. \tag{S34}$$

There is an event $\mathcal{W}_c$, with $\mathbb{P}(\mathcal{W}_c^c) \le n^{-20}$, on which, for every $|B| \le \lceil (M+1)s_c \rceil \wedge p_c$,

$$\|P_{c,B} W P_{c,B}\|_{\mathrm{op}} \le \eta_c, \qquad \eta_c = C s_c \sqrt{L_c/n} = o(1). \tag{S35}$$

Consequently, for the random projection $P = P_c$,

$$\mathbb{E}[\|Pv\|_2^r] + \mathbb{E}[\|PW\Pi\|_2^r] \le C_r (s_c L_c)^{r/2}, \qquad 1 \le r \le 8, \tag{S36}$$

$$\mathbb{E}[(D_\tau - \sigma_v^2 d_c)^2] \le C\eta_c^2 s_c^2 + Cn^{-18}. \tag{S37}$$

Proof. Step 1: Uniform projection moments. For $z = v$, (S19) gives a centered sub-Gaussian moment-generating-function bound. The coordinates of $z = W\Pi$ are also independent and centered by (S18); they are bounded by Assumption S1(iii), so Hoeffding's lemma gives the same bound with a possibly larger fixed constant. Thus, for both vectors and every $b \in \mathbb{R}^n$,

$$\mathbb{E}e^{b'z} \le \exp\{K_{\mathrm{sg}}^2 \|b\|_2^2/2\}, \qquad K_{\mathrm{sg}} < \infty.$$

For deterministic B with $r_B := \mathrm{rank}(P_B) > 0$, apply Hsu et al. (2012, Theorem 2.1) with $A = P_B$, $\Sigma = A'A = P_B$, and zero mean:

$$\mathbb{P}\Big\{ \|P_B z\|_2^2 > K_{\mathrm{sg}}^2 \{ r_B + 2\sqrt{r_B t} + 2t \} \Big\} \le e^{-t}, \qquad t > 0.$$

The rank-zero projection is identically zero. Since $r_B \le |B|$ and $\sum_{j=0}^m \binom{p}{j} \le (p+1)^m$, a union bound with $t = m\log(p+1) + u$ gives, for $1 \le m \le p$,

$$\mathbb{P}\left\{ \max_{|B| \le m} \|P_B z\|_2 > C\sqrt{m\log(2p) + u} \right\} \le e^{-u}, \qquad u \ge 0.$$

Integrating this tail for fixed $r \in [1, 8]$, then using $m = m^* \le Cs$ and $\log(2p) \le CL$, proves (S34).

Step 2: Restricted weighted Gram matrix. The variables $z_{ij} z_{ik} W_{ii}$ are independent, centered, and uniformly bounded by Assumption S2(i) and (S18). Hoeffding's inequality and a union bound give an event $\mathcal{W}$, with $\mathbb{P}(\mathcal{W}^c) \le n^{-20}$, on which

$$\max_{j,k} |(Z'WZ/n)_{jk}| \le C\sqrt{L/n}.$$

For $g = Z_B b$ with $|B| \le m^*$, the lower sparse-eigenvalue bound in Assumption S2(iv) therefore yields

$$|g'Wg| \le n \max_{j,k} |(Z'WZ/n)_{jk}| \|b\|_1^2 \le C|B|\sqrt{L/n}\, \|g\|_2^2.$$

Symmetry of W implies $\|P_B W P_B\|_{\mathrm{op}} \le \eta$, including $B = \varnothing$, and $\eta^2 \le Cs^2L/n \to 0$ by (S6). This proves (S35).

Step 3: Selected projections. On $\mathcal{G}$, an independent basis $B \subseteq E(x)$ has $|B| = d < \lceil Ms \rceil$ and $|B| \le m^*$, so $P = P_B$. On $\mathcal{G}^c$, use $\|Pv\|_2 \le \|v\|_2$ and $\|PW\Pi\|_2 \le C\sqrt{n}$. Cauchy–Schwarz and (S19) then give, for either $z = v, W\Pi$,

$$\mathbb{E}[\|Pz\|_2^r] \le \mathbb{E} \max_{|B| \le m^*} \|P_B z\|_2^r + O(n^{r/2-10}) \le C_r (sL)^{r/2}, \quad 1 \le r \le 8.$$

On $\mathcal{G} \cap \mathcal{W}$, $|D_\tau - \sigma_v^2 d| = |\operatorname{tr}(PWP)| \le \eta d \le Cs\eta$. Since $D_\tau + d \le Cn$ everywhere,

$$\mathbb{E}[(D_\tau - \sigma_v^2 d)^2] \le C\eta^2 s^2 + Cn^2\{\mathbb{P}(\mathcal{G}^c) + \mathbb{P}(\mathcal{W}^c)\} \le C\eta^2 s^2 + Cn^{-18}.$$

These are (S36)–(S37). □

Lemma S6 (Concentration of the Effective Dimension)**.** *Under the hypotheses of Theorem S1(i),*

$$\begin{aligned} \operatorname{Var}(d_c) &\le C\{s_c L_c + \eta_c^2 s_c^2 + n^{-18}\} = o(s_c^2), \\ R_{d,c} &= \mathbb{E}[d_c]^2\{1 + o(1)\} \asymp s_c^2. \end{aligned} \tag{S38}$$

Consequently,

$$\max_c \left| \frac{d_c}{\mathbb{E}[d_c]} - 1 \right| \to_p 0, \quad \max_c \left| \frac{d_c^2 + d_c}{R_{d,c}} - 1 \right| \to_p 0. \tag{S39}$$

Proof. Let $m_0(v) = \mu(\Pi + v) - \mathbb{E}[\mu(\Pi + v)]$. Apply (S20) coordinatewise. By (S16) and Lemma S4,

$$\mathbb{E}\|m_0\|_2^2 \le \mathbb{E}[\operatorname{tr}(PTP)] = \mathbb{E}[D_\tau] \le C\mathbb{E}d \le Cs.$$

For $F_0 = v'm_0$, $\nabla F_0 = m_0 + Pv$ almost everywhere. Equations (S20), (S36), and (S27) give

$$\operatorname{Var}(F_0) \le CsL, \qquad \mathbb{E}[(F_0 - D_\tau)^2] = \mathbb{E}[m_0' T m_0 + \operatorname{tr}(TPTP)] \le Cs.$$

Hence $\operatorname{Var}(D_\tau) \le CsL$, and (S37) yields

$$\sigma_v^4 \operatorname{Var}(d) \le 2\operatorname{Var}(D_\tau) + 2\mathbb{E}[(D_\tau - \sigma_v^2 d)^2] \le C\{sL + \eta^2 s^2 + n^{-18}\}.$$

Since $\mathbb{E}d \asymp s$ by Lemma S4, $L/s \to 0$, and $\eta \to 0$, this is (S38). Chebyshev's inequality gives $d/\mathbb{E}d \to_p 1$. Moreover,

$$\frac{R_d}{(\mathbb{E}d)^2} = 1 + \frac{\operatorname{Var}(d)}{(\mathbb{E}d)^2} + \frac{1}{\mathbb{E}d} \to 1,$$

and $(d^2+d)/(\mathbb{E}d)^2 \to_p 1$. The fixed candidate list makes these two ratios uniform, proving (S39). □

S6 The Second Moment of the Numerator

Lemma S7 (Second Moment of the Numerator)**.** *Under the hypotheses of Theorem S1(i), for all sufficiently large n,*

$$\mathbb{E}[(\varepsilon'\widehat{\Pi}_c)^2] = \sigma_\varepsilon^2 \mathbb{E}[\|\widehat{\Pi}_c\|_2^2] + \sigma_{\varepsilon v}^2 \mathbb{E}[d_c^2 + d_c] + o(n\mathcal{A}_c + s_c^2), \tag{S40}$$

$$\mathbb{E}[(\Gamma_c - L_n)^2] \le C S_c. \tag{S41}$$

Also, $S_c \asymp \mathcal{A}_c + s_c^2/n = o(1)$ and $nS_c \to \infty$.

Proof. Fix a candidate and suppress its subscript. By Lemmas S4 and S6, the main-text (3.7), $a \neq 0$, and $\underline{H}/2 \le h \le 2\overline{H}$,

$$\mathcal{R} := n\mathcal{A} + s^2 \asymp nS, \qquad S = o(1), \qquad s^2 \le CnS.$$

Since $L = \log(pn) \ge \log n$, (S6) forces $s \to \infty$, so that $nS \ge s^2/C \to \infty$. With m as in (S12), replacing Π by its sparse approximation $Z\pi^0$ defines the auxiliary fit $g = m + (h/H)\xi = \mu(\Pi + v) - (h/H)Z\pi^0$. By Assumption S2(ii) and (S13),

$$\mathbb{E}\|g\|_2^2 \le 2\mathbb{E}\|m\|_2^2 + 2\frac{h^2}{H^2}\|\xi\|_2^2 \le C(n\mathcal{A} + s). \tag{S42}$$

Step 1: the decomposition (S43). Put $F = v'\mu$, $D = D_\tau$, and $\delta = F - D$. The weighted covariance and second-moment identities (S25) and (S26), together with $\nabla F = \mu + Pv$, give

$$\mathbb{E}[\delta F] = \mathbb{E}[\mu' T\mu + \mu' TPv], \qquad \mathbb{E}[\delta^2] = \mathbb{E}[\mu' T\mu + \operatorname{tr}(TPTP)].$$

Since $F^2 = D^2 + 2\delta F - \delta^2$, $P\mu = \mu$, and $\mathbb{E}[\mu' v] = \mathbb{E}D$ by (S27),

$$\begin{aligned}\mathbb{E}[F^2] - \sigma_v^2\mathbb{E}[\|\mu\|_2^2] - \sigma_v^4\mathbb{E}[d^2+d] = {} & \mathbb{E}[\mu' W\mu] + \mathbb{E}[D^2 - \sigma_v^4 d^2] \\ & - \mathbb{E}[\operatorname{tr}(TPTP) - \sigma_v^4 d] + 2\sigma_v^2\mathbb{E}[D - \sigma_v^2 d] \\ & + 2\mathbb{E}[\mu' WPv]. \end{aligned} \tag{S43}$$

Step 2: the bounds (S44). Because $\mathbb{E}W = 0$ and Π, ξ, and h/H are nonrandom,

$$\mathbb{E}[\mu' W\mu] = 2\frac{h}{H}\mathbb{E}[\Pi' Wg] + \mathbb{E}[m' Wm].$$

For the independent-support solution in Lemma D.2, whose support lies in E by (D.1), set $B = \mathrm{supp}(\widehat{\pi}) \cup T^0$. On $\mathcal{G}$, (S30) gives $|B| \le m^*$ and $g = P_B g$. We can therefore bound $|\Pi' W g| \le \|P_B W \Pi\|_2 \|g\|_2$, so that (S34), (S42), and Cauchy–Schwarz control the first term on that event. For the second, we write $m = g - (h/H)\xi$ and use $\|P_B W P_B\|_{\mathrm{op}} \le \eta$ on $\mathcal{G} \cap \mathcal{W}$ together with $\|\xi\|_2^2 \le Cs$. The complements are controlled by (S33). Indeed, $\|\mu\|_2 \le \|\Pi + v\|_2$, $\|\Pi\|_2 = O(\sqrt{n})$, $\|\xi\|_2 = O(\sqrt{s})$, and $h/H = O(1)$, so $\|m\|_2 + \|g\|_2 \le C(\sqrt{n} + \|v\|_2)$ and each quadratic product is bounded by $C(n + \|v\|_2^2)$. These arguments show that

$$\begin{aligned} |\mathbb{E}[\Pi' W g]| &\le C\sqrt{sL(n\mathcal{A} + s)} + Cn^{-8}, \\ |\mathbb{E}[m' W m]| &\le C\{\eta(n\mathcal{A} + s) + \sqrt{s(n\mathcal{A} + s)} + s + n^{-8}\}. \end{aligned} \tag{S44}$$

Step 3: the bounds (S45) *and* $|\mathbb{E}[\mu' W P v]| \le C\{sL + \sqrt{n\mathcal{A}\, sL}\}$. On $\mathcal{G} \cap \mathcal{W}$, $|D - \sigma_v^2 d| \le \eta d$ by (S35). Moreover, $0 \le D \le Cd$ and $0 \le \mathrm{tr}(TPTP) \le Cd$ by (S18) and $P^2 = P$. Since $\mathbb{E}[d^j] \le Cs^j$ for $j = 1, 2$, these bounds control the expectations over $\mathcal{G} \cap \mathcal{W}$. On its complement, $D + d \le Cn$ and $\mathbb{P}((\mathcal{G} \cap \mathcal{W})^c) \le 2n^{-20}$, so that the quadratic and linear terms contribute $O(n^{-18})$ and $O(n^{-19})$, respectively. We therefore have

$$\begin{aligned} |\mathbb{E}[D^2 - \sigma_v^4 d^2]| &\le C\eta s^2 + Cn^{-18}, \\ |\mathbb{E}[D - \sigma_v^2 d]| &\le C\eta s + Cn^{-19}, \\ |\mathbb{E}[\mathrm{tr}(TPTP) - \sigma_v^4 d]| &\le Cs. \end{aligned} \tag{S45}$$

Finally, write $\mu = (h/H)\Pi + m$. Since P is an orthogonal projection and $\|W\|_{\mathrm{op}} \le C$, the adaptive projection moments in (S36) give

$$|\mathbb{E}[\mu' W P v]| \le C\mathbb{E}[\|PW\Pi\|_2 \|Pv\|_2] + C\mathbb{E}[\|m\|_2 \|Pv\|_2] \le C\{sL + \sqrt{n\mathcal{A}\, sL}\}.$$

Step 4: (S40) *and* (S41). The five terms in (S43), bounded as above, have total absolute value at most

$$C\big\{\eta\mathcal{R} + \sqrt{sL(n\mathcal{A} + s)} + \sqrt{s(n\mathcal{A} + s)} + sL + s + n^{-8}\big\}.$$

Since $s \ge 1$, we have $n\mathcal{A} + s \le \mathcal{R}$, and $\mathcal{R} \ge s^2$ by definition. Relative to $\mathcal{R}$, this total is therefore at most

$$C\left\{\eta + \sqrt{L/s} + s^{-1/2} + L/s + s^{-1} + n^{-8}/\mathcal{R}\right\} \longrightarrow 0$$

by (S6) and $s \to \infty$. The conditional structural moments (S15) now imply

$$\mathbb{E}[(\varepsilon'\widehat{\Pi})^2] = a^2\mathbb{E}[F^2] + \sigma_u^2\mathbb{E}[\|\mu\|_2^2] = \sigma_\varepsilon^2\mathbb{E}[\|\mu\|_2^2] + \sigma_{\varepsilon v}^2\mathbb{E}[d^2 + d] + o(\mathcal{R}),$$

which is (S40).

Lastly, (S14) and $D \le Cd$ yield

$$\mathbb{E}[(v'm)^2] \le 2\mathbb{E}[(v'm - D)^2] + 2\mathbb{E}[D^2] \le C\{n\mathcal{A} + s + \mathbb{E}d^2\} \le C\mathcal{R}.$$

By (S15) and $\Gamma - L_n = \varepsilon' m/(\sqrt{n}\, h)$, we then have

$$\mathbb{E}[(\Gamma - L_n)^2] = \frac{a^2\mathbb{E}[(v'm)^2] + \sigma_u^2\mathbb{E}\|m\|_2^2}{nh^2} \le C\mathcal{R}/n \le CS.$$

□

S7 The Denominator and the Optimal Instrument

Lemma S8 (The Sample Denominator)**.** *Under the hypotheses of Theorem S1(i),*

$$\mathbb{E}[\widehat{h}_c] = h_c + \mathbb{E}[D_{\tau,c}]/n, \qquad \mathbb{E}[(\widehat{h}_c - h_c)^2] \le C/n, \tag{S46}$$

$$\max_c |\widehat{h}_c - h_c| = O_p(n^{-1/2}), \qquad \mathbb{P}(\mathcal{H}_n) \to 1. \tag{S47}$$

Consequently, $\max_c |\Delta_c| = O_p(n^{-1/2})$.

Proof. Fix a candidate and suppress its subscript. By the weighted first-moment identity (S27) with $b = 0$, $\mathbb{E}[v'\mu] = \mathbb{E}[D_\tau]$, so that $\mathbb{E}[\widehat{h}] - h = \mathbb{E}[D_\tau]/n$, and $\mathbb{E}[D_\tau] \le C\mathbb{E}[d] \le Cs$ by (S18) and (S31). For the variance, the gradient of $x'\mu$ with respect to v is $\mu + Px$ almost everywhere. Since $\|\mu\|_2 \le \|x\|_2$ by Lemma D.1 and P is a projection, the weighted Poincaré inequality (S20) yields

$$\mathrm{Var}(\widehat{h}) \le \frac{C}{n^2}\mathbb{E}\|\mu + Px\|_2^2 \le \frac{C}{n^2}\mathbb{E}\|x\|_2^2 \le \frac{C}{n}.$$

By (S6), $s = o(\sqrt{n})$, so that $\mathbb{E}[(\widehat{h} - h)^2] = \mathrm{Var}(\widehat{h}) + \mathbb{E}[D_\tau]^2/n^2 \le C/n + Cs^2/n^2 \le C/n$, which proves (S46). Since $\min_c h_c \ge \underline{H}/2$ for all sufficiently large n, the first bound in (S47), the bound on Δ_c, and $\mathbb{P}(\mathcal{H}_n^c) \le \mathbb{P}(\max_c |\widehat{h}_c - h_c| > \underline{H}/4) \le CJ/n \to 0$ then follow from Chebyshev's inequality. □

Lemma S9 (Moments of the Denominator Correction)**.** *Under the hypotheses of Theorem S1(i), with Δ_c as in (B.1), there is a constant $C < \infty$ such that, for all sufficiently large n and every candidate,*

$$\mathbb{E}[\Delta_c^2 L_n^2] \le C/n, \tag{S48}$$

$$|\mathbb{E}[\Delta_c L_n^2]| \le C\{\sqrt{S_c/n} + 1/n\}. \tag{S49}$$

Proof. Fix a candidate and suppress its subscript. We follow the proof of Lemma B.3 in the main appendix, using the weighted Stein identity

$$\mathbb{E}[(\Pi' v) f(v)] = \mathbb{E}[(T\Pi)' \nabla f(v)] \tag{S50}$$

in place of (B.11). This identity is (S25) applied with the vector Π in place of the field there, for which the weighted divergence in (S24) is $\Pi' v$. The fit is Lipschitz with at most linear growth, so every f used below has polynomially bounded value and weak gradient, and the required integrability follows from (S19).

As in the Gaussian case, we first collect three facts. The gradient of Δ is $\nabla\Delta = (\mu + Px)/(nh)$, and since $\|\mu\|_2 \le \|x\|_2$ by Lemma D.1, $\|Px\|_2 \le \|x\|_2$, and $\mathbb{E}\|x\|_2^2 = n(H + \sigma_v^2)$, we have $\mathbb{E}\|\nabla\Delta\|_2^2 \le C/n$. By Lemma S8, $\mathbb{E}[\Delta] = \mathbb{E}[D_\tau]/(nh) = O(s/n)$ and $\mathbb{E}[\Delta^2] = \mathbb{E}[(\widehat{h} - h)^2]/h^2 \le C/n$. Finally, by the conditional structural moments (S15),

$$\mathbb{E}[L_n^2 \mid v] = \frac{\sigma_u^2}{H} + \frac{a^2}{nH^2}(\Pi' v)^2. \tag{S51}$$

Since Δ depends only on v, we may multiply (S51) by functions of Δ and take expectations.

Step 1: $\mathbb{E}[\Delta^2 L_n^2] \le C/n$. By (S51), it suffices to show that $\mathbb{E}[(\Pi' v)^2 \Delta^2] \le C$. Apply (S50) with $f = (\Pi' v)\Delta^2$. Because $\nabla(\Pi' v) = \Pi$, the product rule gives

$$\begin{aligned}\mathbb{E}[(\Pi' v)^2 \Delta^2] &= \mathbb{E}[(\Pi' T\Pi)\Delta^2] + 2\mathbb{E}[(\Pi' v)\Delta \cdot (T\Pi)' \nabla\Delta] \\ &\le Cn\mathbb{E}[\Delta^2] + \tfrac{1}{2}\mathbb{E}[(\Pi' v)^2 \Delta^2] + 2\mathbb{E}[((T\Pi)' \nabla\Delta)^2],\end{aligned}$$

where the inequality uses $\|T\|_{\mathrm{op}} \le C$ and Young's inequality, as in Step 1 of the proof of Lemma B.3. The last term is at most $Cn\mathbb{E}\|\nabla\Delta\|_2^2 \le C$ because $\|T\Pi\|_2^2 \le Cn$. Moving the finite term $\frac{1}{2}\mathbb{E}[(\Pi' v)^2\Delta^2]$ to the left-hand side, we obtain $\mathbb{E}[(\Pi' v)^2 \Delta^2] \le C$.

Step 2: $|\mathbb{E}[\Delta L_n^2]| \le C\{\sqrt{S/n} + 1/n\}$. Multiply (S51) by Δ, take expectations, and apply (S50) with $f = (\Pi' v)\Delta$. Since $T = \sigma_v^2 I + W$ and $\sigma_\varepsilon^2 = \sigma_u^2 + a^2\sigma_v^2$, we obtain

$$\mathbb{E}[\Delta L_n^2] = \frac{\sigma_\varepsilon^2}{H}\mathbb{E}[\Delta] + \frac{a^2}{nH^2}\left\{\mathbb{E}[(\Pi' W\Pi)\Delta] + \mathbb{E}[(\Pi' v)(T\Pi)' \nabla\Delta]\right\}. \tag{S52}$$

The first term is $O(s/n)$, while the term $\mathbb{E}[(\Pi' W \Pi)\Delta]$ does not have a Gaussian counterpart. Since W is diagonal with independent, mean-zero entries and $|\Pi_i| \le K_\Pi$ by (S4),

$$\mathbb{E}[(\Pi' W \Pi)^2] = \mathrm{Var}(\tau(v_i)) \sum_{i=1}^{n} \Pi_i^4 \le Cn,$$

so that $|\mathbb{E}[(\Pi' W \Pi)\Delta]| \le C$ by Cauchy–Schwarz.

For the last term in (S52), split $(T\Pi)'\nabla\Delta = \sigma_v^2 \Pi'\nabla\Delta + (W\Pi)'\nabla\Delta$. The first part satisfies $|\mathbb{E}[(\Pi' v)\Pi'\nabla\Delta]| \le C(1 + \sqrt{n\mathcal{A}})$ by the argument in Step 2 of the proof of Lemma B.3, with two changes. First, (S50) gives $\mathbb{E}[(\Pi' v)\Pi'\mu] = \mathbb{E}[(T\Pi)' P\Pi]$, which is at most Cn in absolute value. Second, the bound $\mathbb{E}[(\Pi' v)^4] + \mathbb{E}\|x\|_2^4 \le Cn^2$ now follows from (S19). The bound $\mathbb{E}\|(I - P)\Pi\|_2^2 \le (H/h)^2 n\mathcal{A}$ uses $P\mu = \mu$ and (S13), neither of which depends on Gaussian errors. For the second part, $nh\,(W\Pi)'\nabla\Delta = (W\Pi)'(\mu + Px) = (PW\Pi)'(\mu + x)$, since $P\mu = \mu$. Applying Cauchy–Schwarz twice, with $\|\mu + x\|_2 \le 2\|x\|_2$, and using (S36), we obtain

$$\begin{aligned} |\mathbb{E}[(\Pi' v)(PW\Pi)'(\mu + x)]| &\le \left\{\mathbb{E}[(\Pi' v)^4]\right\}^{1/4} \left\{\mathbb{E}\|\mu + x\|_2^4\right\}^{1/4} \left\{\mathbb{E}\|PW\Pi\|_2^2\right\}^{1/2} \\ &\le Cn\sqrt{s\log(pn)}. \end{aligned}$$

Dividing the second part by nh and combining it with the first, $|\mathbb{E}[(\Pi' v)(T\Pi)'\nabla\Delta]| \le C\{1 + \sqrt{n\mathcal{A}} + \sqrt{s\log(pn)}\}$, so that by (S52),

$$|\mathbb{E}[\Delta L_n^2]| \le C\left\{\frac{s}{n} + \sqrt{\frac{\mathcal{A}}{n}} + \frac{\sqrt{s\log(pn)}}{n} + \frac{1}{n}\right\}.$$

Finally, $\mathcal{A} \le CS$ and $s/n \le C\sqrt{S/n}$ by Lemma S7, while

$$\frac{\sqrt{s\log(pn)}}{n} = \frac{s}{n}\sqrt{\frac{\log(pn)}{s}} = o\left(\frac{s}{n}\right)$$

by (S6). This proves (S49). □

Lemma S10 (Moments for the Optimal Instrument)**.** *Under Assumption S1,*

$$\mathbb{E}[L_n^2] = \sigma_\varepsilon^2/H, \quad |\mathbb{E}[qL_n^2]| + \mathbb{E}[q^2 L_n^2] \le C/n. \tag{S53}$$

Consequently, $C_0 = (1 - q)^2 L_n^2$ *satisfies*

$$\mathbb{E}[C_0] = \sigma_\varepsilon^2/H + O(n^{-1}). \tag{S54}$$

Proof. By independence and centering, $\mathbb{E}[L_n^2] = \sigma_\varepsilon^2/H$. Although (S51) appears in the proof of Lemma S9, it uses only (S15) and so holds under Assumption S1. Since $q = \Pi' v/(nH)$ depends only on v, multiplying (S51) by q and using independence and $\mathbb{E}v_i = 0$, we have

$$\mathbb{E}[qL_n^2] = \frac{a^2\mathbb{E}[(\Pi' v)^3]}{n^2H^3} = \frac{a^2\mathbb{E}[v_i^3]\sum_i \Pi_i^3}{n^2H^3} = O(n^{-1}),$$

where $\sum_i |\Pi_i|^3 \le K_\Pi nH$ by (S4) and $\mathbb{E}|v_i|^3 \le C$ by (S19). By the same tail bound, $\mathbb{E}[(\Pi' v)^4] \le Cn^2$, and by independence $\mathbb{E}[q^2] = \sigma_v^2/(nH)$. Therefore

$$\mathbb{E}[q^2 L_n^2] = \frac{\sigma_u^2}{H}\mathbb{E}[q^2] + \frac{a^2\mathbb{E}[(\Pi' v)^4]}{n^3H^4} = O(n^{-1}).$$

Expanding $C_0 = (1 - 2q + q^2)L_n^2$ proves (S54). □

S8 Supporting Results for Theorem S1(ii)

Lemma S11 (Sample Counterparts of $\mathcal{A}_c$ and q)**.** *Under the hypotheses of Theorem S1(i), let* $\widehat{\mathcal{A}}_c = \min_{b\in\mathbb{R}} \mathbb{E}_n[(\widehat{\Pi}_{c,i} - b\Pi_i)^2]$*, and let* q *and* $\widehat{q}_c$ *be as in the proofs of the main-text Theorems 1 and 2. Then*

$$\widehat{\mathcal{A}}_c - \mathcal{A}_c = O_p\Big(\sqrt{\mathcal{A}_c/n} + n^{-1}\Big) = o_p(S_c), \tag{S55}$$

$$\widehat{q}_c - q = O_p\Big(\sqrt{\mathcal{A}_c/n} + s_c/n + n^{-1}\Big) = o_p(S_c), \tag{S56}$$

where the second assertion holds on $\mathcal{H}_n$*.*

Proof. Fix a candidate and suppress its subscript.

Step 1: $\widehat{\mathcal{A}} - \mathcal{A} = O_p(\sqrt{\mathcal{A}/n} + 1/n)$. Step 1 of the proof of Lemma C.1 uses Gaussian errors only through the Poincaré inequality, which it applies to $\mathbb{E}_n[\Pi_i\widehat{\Pi}_i]$ and $\widehat{\mathcal{A}}$. By (S16), their gradients are again $P\Pi/n$ and $(2/n)P\{I - \Pi\Pi'/(nH)\}\widehat{\Pi}$. With the weighted Poincaré inequality (S20) in place of (E.1), the same argument shows that $0 \le \mathcal{A} - \mathbb{E}[\widehat{\mathcal{A}}] \le C/n$ and $\mathrm{Var}(\widehat{\mathcal{A}}) \le C\mathcal{A}/n$, and the claim follows from Chebyshev's inequality.

Step 2: $\widehat{q} - q = O_p(\sqrt{\mathcal{A}/n} + s/n + 1/n)$ *on* $\mathcal{H}_n$. Since $\widehat{\Pi} = (h/H)\Pi + m$ by (S12), we have $\mathbb{E}_n[v_i\widehat{\Pi}_i] = v'm/n + hq$, and $\mathbb{E}[(v'm - D_\tau)^2] \le C(n\mathcal{A} + s)$ by (S14). Also, $0 \le D_\tau \le Cd = O_p(s)$ by Lemmas S2 and S4. Since $s \ge 1$, it follows that $v'm/n = O_p(\sqrt{\mathcal{A}/n} + s/n)$. By independence, $\mathbb{E}[q^2] = \sigma_v^2/(nH)$, while $\widehat{h} - h = O_p(n^{-1/2})$ by Lemma S8. On $\mathcal{H}_n$, where $\widehat{h} \ge \underline{H}/4$,

$$\widehat{q} - q = \frac{v'm/n - q(\widehat{h} - h)}{\widehat{h}} = O_p\Big(\sqrt{\mathcal{A}/n} + s/n + n^{-1}\Big).$$

Step 3: Both rates are $o_p(S)$. By Lemma S7, $S \asymp \mathcal{A} + s^2/n$. The inequality $2s\sqrt{\mathcal{A}/n} \leq \mathcal{A} + s^2/n$ and $s^2/n \leq CS$ imply

$$\frac{\sqrt{\mathcal{A}/n} + s/n + n^{-1}}{S} \leq C(s^{-1} + s^{-2}) \longrightarrow 0,$$

since $s \to \infty$ by (S6). □

Lemma S12 (Estimating the Error Moments)**.** *Suppose the hypotheses of Theorem S1(i) hold. For any (possibly data-dependent) $[J]$-valued index $\check{c}$, $\widehat{\beta}_{\check{c}} - \beta = O_p(n^{-1/2})$. If the pilot satisfies $\check{\beta} - \beta = O_p(n^{-1/2})$, the moment estimators in the main-text (3.11) satisfy*

$$\widehat{\sigma}_\varepsilon^2 - \sigma_\varepsilon^2 = O_p(n^{-1/2}), \quad \widehat{\sigma}_{\varepsilon v} - \sigma_{\varepsilon v} = O_p(n^{-1/2}). \tag{S57}$$

Proof. Step 1 of the proof of Theorem S1(i) shows that $\max_c |\sqrt{n}(\widehat{\beta}_c - \beta)| = O_p(1)$, so that

$$|\sqrt{n}(\widehat{\beta}_{\check{c}} - \beta)| \leq \max_c |\sqrt{n}(\widehat{\beta}_c - \beta)| = O_p(1)$$

for every possibly data-dependent $[J]$-valued index. By the moment conditions (S1), (S19), and (S4), $\sup_i \mathbb{E}[\varepsilon_i^4 + x_i^4] < \infty$, and independence and Chebyshev's inequality then give

$$\mathbb{E}_n[\varepsilon_i^2] = \sigma_\varepsilon^2 + O_p(n^{-1/2}), \qquad \mathbb{E}_n[\varepsilon_i x_i] = \sigma_{\varepsilon v} + O_p(n^{-1/2}).$$

Also, $\overline{\mathbb{E}}[x_i^2] = H + \sigma_v^2 = O(1)$, so $\mathbb{E}_n[x_i^2] = O_p(1)$ by Markov's inequality. For any pilot with $\check{\beta} - \beta = O_p(n^{-1/2})$, substituting these bounds into the identities (C.11), which do not use Gaussian errors, proves (S57). □

References

Hsu, D., S. M. Kakade, and T. Zhang (2012). A tail inequality for quadratic forms of subgaussian random vectors. *Electronic Communications in Probability 17*(52), 1–6.

Saumard, A. (2018). Weighted Poincaré inequalities, concentration inequalities and tail bounds related to the Stein kernels in dimension one. arXiv:1804.03926v3.